Terpene diversity, biosynthetic evolution, and antiparasitic evidence across *Artemisia* species

Abstract

The genus *Artemisia* contains a chemically diverse collection of volatile terpenes, nonvolatile terpenoids, and sesquiterpene lactones. This review develops a source-backed framework for comparing that chemistry across species while avoiding a common error: treating an essential-oil profile, a pathway gene, or an in-vitro parasite assay as a genus-wide pharmacological conclusion. Across 516 registered sources and 519 source-linked evidence records, we synthesize monoterpenes, sesquiterpenes, diterpenes, triterpenes, and sesquiterpene lactones; evaluate biosynthetic compartmentalization and gene-family evolution; and connect candidate pathway changes to chemotype and antiparasitic evidence. The best-resolved pathway is artemisinin biosynthesis in *A. annua*, but recent *A. argyi* results illustrate why gene presence/absence claims require assembly-, ploidy-, and annotation-aware reconciliation. Malaria and artemisinin form the primary validated case study, while wormwood (*A. absinthium*) provides a separate, preparation-specific veterinary vermifuge/anthelmintic case. Essential-oil, extract, whole-leaf, and purified-compound assays provide phenotype anchors across protozoa, helminths, and vectors, yet constituent attribution remains unresolved in most mixture studies. We propose a reproducible, phylogeny-aware research program combining voucher-linked metabolomics, secretory-tissue transcriptomics, comparative genomics, enzyme assays, and fractionated parasite testing.

Keywords: *Artemisia*; terpenes; sesquiterpene lactones; artemisinin; chemotype; comparative genomics; essential oil; antiparasitic activity

1. Introduction

Artemisia is a large genus with substantial ecological, morphological, ploidy, and phytochemical diversity. Its volatile chemistry is concentrated especially in leaves and flowers, but reported profiles vary with genotype, chemotype, locality, tissue, ontogeny, harvest, drying, and extraction. Consequently, "the terpene profile of *Artemisia*" is not a single biological object; the appropriate unit of comparison is a documented specimen or accession with analytical context.

This review has two aims. First, it organizes the major terpene classes reported from *Artemisia* and summarizes the biosynthetic logic connecting precursor supply, synthase diversification, oxidation, lactonization, storage, and regulation. Second, it evaluates whether comparative evolutionary genomics can connect that diversity to antiparasitic phenotypes without overstating causality.

The evidence extraction unit is a specimen, accession, or explicitly bounded experimental preparation. Representative records are provided in `chemotype-table.csv`; they are not species means. The review follows the protocol in `review-protocol.md` and the source identifiers in `sources.json`.

1.1 What previous reviews establish-and what remains open

Yes, *Artemisia* terpene and terpenoid reviews have been published. The principal precedents include a genus-level essential-oil review, a 2012-2017 essential-oil update, a dedicated sesquiterpene-lactone analysis and methods review, a review of edible species and selected sesquiterpene lactones, a 2024 genus-wide phytochemistry, pharmacology, and toxicology review, a 2024 synthesis of STL biosynthesis/regulation/genomics, a 2024 evolution-focused STL review, a recent review focused on *A. herba-alba* and *A. judaica* chemistry and omics, a 1990 review of *A. annua* constituents and antimalarial pharmacology, a 2022 review of *A. annua* artemisinin biotechnology, a 2023 malaria-focused review of isolated *A. annua* and *A. afra* metabolites and PK/PD, a 2025 medicinal-chemistry review of *Artemisia* antimalarials, and an August 2026 systematic review of whole-plant *A. annua* tea (Abad et al. 2012; Pandey and Singh 2017; Ivanescu et al. 2015; Trendafilova et al. 2021; Hussain et al. 2024; Frey et al. 2024; Agatha et al. 2024; Al-Khayri et al. 2022; Woerdenbag et al. 1990; Shinyuy et al. 2023; Diab et al. 2026; Rauf et al. 2025; Munyangi Wa Nkola et al. 2026). These reviews establish the breadth of reported chemistry and biological claims, but they generally do not provide one specimen-level, class-balanced, phylogeny-aware evidence model connecting terpene chemistry, evolutionary genomics, tissue context, antiparasitic stage, and safety. The present package is designed as an auditable update and integration, not as a claim that prior reviews were absent or noncomprehensive. Two adjacent reviews are particularly useful for scoping the parasite sections. Sharma's 2025 review surveys *Artemisia*, artemisinin, semisynthetic derivatives, and derived oligomers across protozoal, helminth, bacterial, viral, and fungal contexts, while Jamil and colleagues' 2025/2026 review situates *A. absinthium* within botanical anthelmintics for ruminants (Sharma 2025; Jamil et al. 2025/2026). We use both as discovery and framing sources. Their broad mechanism and efficacy summaries are not substituted for specimen-level chemistry, parasite-stage assays, host counter-screens, or primary clinical and veterinary evidence. A species-focused wormwood review provides a complementary synthesis of *A. absinthium* constituents, pharmacological reports, anthelmintic context, pharmacokinetics, and toxicity (Batiha et al. 2020). A veterinary network meta-analysis likewise places *A. absinthium* among botanical anthelmintic candidates, but its cross-plant and heterogeneous evidence base requires caution when interpreting any species-specific or terpene-specific effect (Calzetta et al. 2020). These reviews strengthen the rationale for the wormwood section while leaving preparation, constituent attribution, and human translation as primary-study questions.

1.2 Literature-search and evidence strategy

The literature review was organized around five linked evidence domains: (i) compound occurrence and analytical identity; (ii) enzyme, pathway, and compartment evidence; (iii) genotype, population, tissue, developmental, and environmental variation; (iv) comparative genomics and gene-family evolution; and (v) antiparasitic phenotype, target, and safety evidence. Searches combined *Artemisia* taxon terms with terpene-class, analytical, biosynthetic, genomic, and parasite terms. The reproducible search strings, dates, database results, de-duplication outputs, and title/abstract decisions are archived in `search-log.md`, `search-results.json`, and the screening CSV files. The working corpus contains both primary studies and prior reviews, but mechanistic and quantitative statements are preferentially anchored to primary studies.

Evidence was graded by what an experiment actually established. Isolation with NMR or a matched analytical standard supports compound identity in the tested material; GC-MS library matching supports a lower-confidence assignment unless retention-index and standard data are also available. Heterologous enzyme assays support catalytic capacity under the assay conditions, not native flux. Transcript abundance and co-expression prioritize candidates but do not prove product identity. Parasite growth inhibition supports a preparation-specific phenotype; causal target claims require biochemical, structural, genetic, or activity-based target-engagement evidence. This hierarchy allows broad coverage without converting association into mechanism.

The review is intentionally class-balanced. Essential-oil surveys dominate the searchable literature, but the synthesis separately tracks nonvolatile sesquiterpene lactones, diterpenes, triterpenes, and cuticular metabolites so that analytical convenience does not become a biological conclusion. Likewise, *A. annua* is treated as the best-resolved reference rather than a proxy for the genus. Evidence from *A. argyi*, *A. absinthium*, *A. herba-alba*, *A. afra*, *A. campestris*, and other species is retained with its own voucher, population, tissue, and preparation boundaries wherever those data are reported.

2. Chemical scope and terminology

Terpenes are assembled from C5 isoprenoid units. Monoterpenes are principally C10, sesquiterpenes C15, diterpenes C20, and triterpenes C30; oxygenated and rearranged products are more precisely terpenoids. Sesquiterpene lactones are C15 terpenoids containing a lactone ring and are generally treated as extractable specialized metabolites rather than ordinary essential-oil constituents.

Representative volatile classes include pinene, limonene, myrcene, camphor, 1,8-cineole, borneol, thujones, germacrene D, β-caryophyllene, α-humulene, davanone, spathulenol, and caryophyllene oxide. Important nonvolatile or less volatile chemistry includes artemisinin and related compounds, guaianolide and germacranolide sesquiterpene lactones, and triterpenoids associated with plant surfaces and defense.

2.1 Evidence map by class

Table 1. Major terpene classes, analytical contexts, and interpretation limits in *Artemisia*.

Class	Representative *Artemisia* chemistry	Main analytical context	Review interpretation
Monoterpenes and oxygenated monoterpenes	α-/β-pinene, limonene, myrcene, camphor, 1,8-cineole, borneol, thujones, artemisia ketone	GC-MS/GC-FID of volatile oils and headspace fractions	Often abundant, but highly sensitive to chemotype, tissue, stage, drying, and distillation
Sesquiterpene hydrocarbons and oxygenated sesquiterpenes	germacrene D, β-caryophyllene, α-humulene, davanone, spathulenol, caryophyllene oxide	GC-MS of oils; targeted volatile profiling	Useful markers of population chemistry, not universal species signatures
Sesquiterpene lactones	artemisinin, arteannuin B, artemisinic acid, santonin, guaianolides, germacranolides, absinthin-related compounds	LC-MS/HPLC, NMR, isolation; GC is often unsuitable without derivatization	Usually nonvolatile; structural and stereochemical confirmation is critical
Diterpenes	species-specific diterpenoid constituents and chlorophyll-associated phytol derivatives	Solvent extraction, LC-MS, NMR; often reported in broad phytochemical surveys	Less comprehensively sampled than C10/C15 chemistry; presence should be linked to isolation or validated MS evidence

Class	Representative *Artemisia* chemistry	Main analytical context	Review interpretation
Triterpenes and sterols	β-amyrin/α-amyrin-, friedelin-, cycloartane- and sterol-associated reports	Solvent extraction, LC-MS, GC-MS after derivatization, NMR	Surface and storage chemistry; not part of ordinary essential-oil comparisons

The unevenness of this evidence map is itself a result: volatile mono- and sesquiterpenes are overrepresented because they are accessible to routine GC analysis, whereas nonvolatile diterpenes, triterpenes, and lactones require extraction workflows that are less directly comparable across studies.

Recent structure-confirmed work expands the under-sampled C10 and C15 space without supplying a shortcut to pharmacology. Aerial parts of *A. persica* yielded ten undescribed monoterpenoids, including four enantiomeric pairs, characterized by NMR, high-resolution mass spectrometry, and ECD; none exceeded 50% activity at the tested concentrations (Umaraliev et al. 2026). The same species also yielded eleven undescribed bisabolane-type sesquiterpenoids, with only limited anti-inflammatory activity reported for two compounds (Umaraliev et al. 2025). These records add authenticated chemical space and a useful negative/limited-activity control, but neither study establishes antiparasitic activity, native biosynthetic genes, or species-wide abundance. They support a core rule for the comparative dataset: a newly named terpene is an occurrence record until its pathway, ecological context, and parasite phenotype are independently linked.

Postharvest handling can create an apparent chemotype shift even when genotype is unchanged. Five weeks of extended natural shade drying reduced *A. monosperma* essential-oil yield by 23.7% and altered its constituent profile, although beta-pinene and selected major constituents were comparatively stable (Mohammed and Alfazi 2026). In three *A. argyi* cultivars, drying method also changed the volatile balance: 80 °C hot-air drying/fixation reduced oxygenated monoterpenes such as eucalyptol and thujone while increasing selected sesquiterpenes including caryophyllene and neointermedeol (Ming et al. 2025). Any parasite assay using dried plant material should therefore record drying duration, temperature, cultivar, oil yield, and full chromatographic composition before attributing activity to a terpene or comparing studies.

An additional volatile comparator is *A. serotina*, or winter wormwood. Its aerial-part oil was dominated by the irregular monoterpene santolina alcohol (34.6%) and included santolinane-, artemisane-, and lavandulane-type constituents; the oil showed fungicidal activity against *Candida albicans* in the reported assay (Kadyrbay et al. 2025). This record broadens the irregular-monoterpene space but does not establish malaria, helminth, or vermifuge activity. It also illustrates why common names such as "wormwood" should not be used as chemical surrogates for *A. absinthium* or for a particular thujone exposure.

Wormwood stereochemistry and postharvest history add two distinct sources of chemical uncertainty. VCD/DFT and NMR assigned the absolute configurations of two naturally occurring ocimene monoterpenoids from domesticated *A. absinthium* oils, converting tentative C10 identities into stereochemically resolved records (Julio et al. 2017). In a seasonal Iranian collection, 44 oil constituents and total yield changed across preflowering, flowering, and post-flowering stages, with shifts among alpha-phellandrene, beta-pinene, chamazulene, and related volatiles (Mohammadi et al. 2014). A separate storage model showed that drying and storage alter wormwood profiles through evaporation and parent-to-daughter chemical transformations (Blagojević et al. 2015). These findings make stereochemical confirmation, harvest stage, drying, and storage mandatory fields when comparing wormwood vermifuge preparations.

Primary studies show that the under-sampled classes are not merely theoretical. In *A. annua*, seed isolation recovered fourteen sesquiterpenes, three monoterpenes, and one diterpene (Brown et al. 2003). Separate trichome transcriptome and heterologous-expression work functionally characterized the oxidosqualene cyclase OSC2 and P450 CYP716A14v2 in aerial-cuticle triterpenoid production (Moses et al. 2015), while a diterpene-synthase study identified ten *A. annua* diTPS genes in a glandular-trichome and stress-resilience context (Chen et al. 2021). These are class-specific anchors, not evidence that all *Artemisia* species share the same diterpene or triterpene repertoire.

2.2 Monoterpenes and volatile sesquiterpenes

Chiral stereochemistry is an additional authentication variable for volatile antiparasitic candidates. Four *A. herba-alba* oils were modeled with infrared/VCD spectra and compared with chiral HPLC, resolving variable proportions and prevailing enantiomers of (-)-alpha-thujone, (+)-beta-thujone, and (-)-camphor (Said Mel et al. 2016). This approach provides a practical route to distinguish nominally identical GC-MS peaks that differ in absolute configuration. It is not evidence that *A. herba-alba* chemistry represents *A. absinthium*, or that any one enantiomer is a vermifuge; it shows why chiral identity should accompany abundance, voucher, and parasite-stage metadata.

Monoterpenes are the most method-accessible *Artemisia* terpenoids. Across oils, recurrent chemistries include pinene and limonene hydrocarbons, camphor, borneol, 1,8-cineole, thujones, and irregular-monoterpene products such as artemisia ketone. Volatile sesquiterpenes add germacrene D, β-caryophyllene, α-humulene, caryophyllene oxide, davanone, spathulenol, and related oxygenated products. These names should be treated as reported identities with confidence levels, not interchangeable biomarkers: retention-index agreement, authentic-standard co-injection, mass-spectrum library matching, and NMR confirmation provide different evidentiary strength.

The biological unit is often a glandular-trichome or aerial-tissue oil, not a purified constituent. In *A. annua*, three recombinant monoterpene synthases generated distinct product spectra, while wounding and hormone treatment changed transcript levels. This connects volatile chemistry to enzyme specificity and inducible regulation, but it does not show that the induced oil is antiparasitic. In *A. frigida* and *A. absinthium*, population, geography, and phenology alter the relative abundance of volatile compounds. A cross-species comparison should therefore model presence/absence, relative abundance, and analytical detectability separately.

Enzyme studies reveal why one-gene/one-compound annotation is often inadequate. Recombinant *A. annua* epi-cedrol synthase produced epi-cedrol as its major oxygenated C15 product but also generated several olefinic sesquiterpenes and showed lower activity with the C10 precursor GPP (Mercke et al. 1999). Two additional *A. annua* sesquiterpene-cyclase cDNAs and later transient-expression experiments expanded the functionally tested synthase repertoire (Van Geldre et al. 2000; Kanagarajan et al. 2012). Product multiplicity means that an expanded TPS clade can increase chemical opportunity without predicting a unique metabolite, while heterologous expression can reveal catalytic potential that may be constrained by native substrate supply or cell type.

The underrepresented irregular-monoterpene space is also phylogenetically informative. Structure-confirmed isolation from *A. tridentata* subsp. *vaseyana*, *A. cana* subsp. *viscidula*, and *A. tridentata* subsp. *spiciformis* yielded 26 regular and irregular monoterpenes, including artemisia triene, trans-chrysanthemal, lavandulol, and a lavandulyl-skeletal compound reported as new to the Anthemideae family (Gunawardena et al. 2002). The result shows that C10 diversity is not exhausted by routine pinene/camphor/thujone profiling, but isolated occurrence in selected western taxa does not establish abundance, pathway orthology, ecological function, or antiparasitic activity. It is a useful target for phylogeny-stratified metabolomics and enzyme assays.

Functional divergence can occur even among close homologs. An AaADS-like enzyme from *A. maritima* cyclized FPP to 4-amorphen-11-ol rather than amorpha-4,11-diene, and the product was recovered from engineered yeast (Muangphrom et al. 2018). This is a useful evolutionary experiment because it separates sequence homology from product identity: an ADS-like copy may preserve the substrate class while changing product outcome. The result is heterologous catalytic evidence, not proof of native abundance, artemisinin production, or parasite activity.

Downstream enzymes further diversify volatile products. A glandular-trichome-specific *A. annua* alcohol dehydrogenase oxidized monoterpene alcohol substrates in recombinant assays, demonstrating that volatile profiles reflect not only cyclization but subsequent redox chemistry (Polichuk et al. 2010). At the regulatory level, AabHLH112 perturbation changed β-caryophyllene, epi-cedrol, and β-farnesene-associated sesquiterpene metabolism, connecting a transcription factor to several non-artemisinin branches rather than a single endpoint (Xiang et al. 2022). Conversely, suppression of β-caryophyllene synthase increased artemisinin-pathway output in transgenic plants, providing direct evidence that FPP-consuming branches can compete for precursor under at least some engineered conditions (Chen et al. 2011).

Population studies show an orthogonal source of complexity. Oils of *A. campestris* and *A. herba-alba* varied among natural populations, and their non-parasitic bioassay profiles varied with composition (Younsi et al. 2017). In *A. herba-alba*, analysis of 80 individuals from eight Tunisian populations resolved four oil chemotypes and found substantial within-population marker diversity; global chemical divergence was not reducible to geographic or bioclimatic distance alone (Younsi et al. 2018). A controlled comparison of five Iranian species further illustrates the scale and limits of cross-species variation: 20-34 compounds were assigned by GC-MS per species, artemisinin ranged from 0.12 to 0.41 mg/g dry weight, and dominant volatiles differed markedly, including camphor-rich *A. fragrans*, beta-thujone-rich *A. absinthium*, and beta-farnesene-rich *A. biennis* (Jamshidi et al. 2024). The same study found no uniform correlation between artemisinin content and the relative expression of the surveyed pathway genes. Because the plants were grown under a common controlled regime and GC-MS assignments were not all confirmed by isolation, this is an accession- and method-bounded comparison rather than a species-wide chemotype or evolutionary gain/loss test. These observations argue for individual- or population-level sampling and against using a single oil chromatogram as a species diagnostic.

Spatial analysis adds an ecological-scale counterpart to accession-level comparisons. A province-level analysis of Chinese *A. annua* reported positive spatial autocorrelation of artemisinin content, with high-value clusters in southwestern regions and sunshine, temperature, and precipitation identified as major explanatory factors (Zhang et al. 2017). Because the record is an observational spatial analysis with region-level aggregation, it cannot separate genotype, seed source, agronomy, processing, and microclimate or establish a causal environmental response. It nonetheless supports explicitly modeling geography and climate when testing whether pathway-gene variation predicts malaria-relevant chemistry.

A fifteen-population study of *A. vulgaris* makes the safety implication concrete: davanone, germacrene D, 1,8-cineole, camphor, thujone, and cis-chrysanthenyl acetate each dominated different Lithuanian oils, and brine-shrimp toxicity varied across the profiles (Judzentiene and Garjonyte 2016). The result supports chemotype-stratified toxicology, but brine-shrimp LC0 values are not mammalian therapeutic windows and do not identify a single toxic constituent. Similarly, oral *A. argyi* oil produced rapid rat absorption and distribution of eight volatile constituents, including eucalyptol, alpha-thujone, camphor, borneol, bornyl acetate, beta-caryophyllene, and caryophyllene oxide, with prominent tissue exposure followed by prompt elimination (Hou et al. 2021). This is disposition evidence for a defined oil preparation, not proof of antiparasitic efficacy or a safe human dose.

Newer wormwood analyses reinforce the need to treat geography, phenology, and authentication as causal design variables rather than metadata. *A. absinthium* oils shifted among oxygenated monoterpene-, oxygenated sesquiterpene-, and monoterpene-hydrocarbon-dominated profiles across sites and flowering stages (Malaspina et al. 2025). A separate low-thujone oil was dominated by trans-sabinyl acetate, with trans-thujone only 1.2 +/- 0.5% in the reported sample (Judzentienė and Būdienė 2026). Conversely, authenticated and commercial oils formed distinct thujone/camphor/ocimene-epoxide clusters, and alpha- and beta-thujone produced tephritid behavioral or toxicity endpoints (Tabanca et al. 2025). These records support lot-specific chemical authentication and safety testing; they do not permit the species name "wormwood" to stand in for a defined thujone exposure or a vermifuge dose.

An additional Indian *A. absinthium* leaf oil shows how strongly the dominant class can change with provenance. The Western Ghats sample contained 64 identified constituents and was rich in oxygenated monoterpenes (39.7–41.1%), with borneol, methyl hinokiate, isobornyl acetate, beta-gurjunene, and caryophyllene oxide among the major compounds (Joshi 2013). Its tested phenotype was antibacterial and antifungal, not an anthelmintic assay. The record therefore strengthens the chemical case for regional chemotypes while weakening any attempt to infer vermifuge activity from the common name, traditional use, or a single dominant volatile.

An agricultural-invertebrate assay adds a useful but deliberately separate wormwood comparator. A Moroccan *A. absinthium* oil was rich in 3,3,5-trimethylcyclohexene and camphor and produced stage-dependent toxicity against the potato tuber moth *Phthorimaea operculella* (Mahboub et al. 2025). This supports a lot-specific volatile profile and an insect-control phenotype, but it is not evidence against helminths and does not identify a causal constituent; the result should not be used to strengthen a wormwood vermifuge claim.

Extraction technology is another source of apparent chemotype variation. In flowering *A. annua* aerial material, supercritical CO_2 extraction yielded 0.62-1.92% and produced artemisia ketone, camphor, 1,8-cineole, arteannuin B, and beta-selinene profiles that differed from hydrodistilled and steam-distilled oils (Mandić et al. 2025). This provides a controlled process comparison across mono- and sesquiterpene fractions, but it does not identify the best therapeutic preparation, establish antimalarial activity, or make the resulting profiles comparable to a species-wide oil mean.

Recent comparative assays extend the same logic from chemotype to arthropod control. Five *Artemisia* oils contained only 9–24 identified constituents per species and separated into distinct multivariate profiles; *A. ordosica* and *A. tangutica* were the strongest contact-toxic oils against the two tested stored-product insects (Zhang et al. 2022). In natural *A. scoparia* populations, altitude was associated with wide ranges of beta-myrcene, gamma-terpinene, and trans-beta-caryophyllene and with differences in mosquito repellency and larvicidal activity (Parveen et al. 2024). A camphor-rich *A. vulgaris* oil produced time- and concentration-dependent fumigant, contact, and repellent activity against the gall mite *Aceria pongamiae* (Kunnathattil et al. 2025). These records support environment- and species-stratified vector experiments, but the oils remain mixtures and the reported insect or mite endpoints cannot be used as proxies for helminth or malaria efficacy.

The chemical reference space is still expanding through structure-confirmed isolation. *A. tournefortiana* aerial parts yielded two novel sesquiterpenes and one new monoterpene, with NMR, HRESIMS, and calculated ECD used to assign structures and antibacterial testing showing only moderate activity for two compounds (Wu et al. 2024). A recent *A. argyi* leaf-oil study was dominated by 1,8-cineole, beta-caryophyllene, and longifolene and reported insecticidal activity against *Callosobruchus maculatus* at high oil concentration (Kačániová et al. 2026). These studies add authenticated C10/C15 occurrence and a contemporary *A. argyi* chemotype, but neither identifies a native pathway allele, a purified active, or a parasite target.

An earlier process benchmark similarly extracted artemisinin and artemisinic acid from six *A. annua* samples using supercritical CO_2 with 3% methanol at 50 degrees C and 15 MPa, with rapid recovery and no reported analyte decomposition under the tested conditions (Kohler et al. 1997). This is useful for reproducing analyte recovery and comparing extraction selectivity, but process yield remains distinct from native pathway flux, biological activity, and clinical suitability.

Older population and phenology studies remain valuable controls for the apparent stability of an oil profile. In the narrow endemic *A. molinieri*, 69 aerial-part volatiles were identified; ascaridole ranged from 19–76%, alpha-terpinene from trace levels to 36%, and germacrene D from 0.6–15% across season, phenological stage, and plant age (Masotti et al. 2003). A geographically matched *A. absinthium* comparison likewise separated French oils dominated by (Z)-epoxyocimene and chrysanthenyl acetate from Croatian oils dominated by (Z)-epoxyocimene and beta-thujone, with additional changes before and after anthesis (Juteau et al. 2003). The reported yeast assays in these studies are not parasite endpoints. Their main contribution is design logic: "wormwood oil" must be tied to geography, stage, organ, and a complete chromatographic fingerprint before a vermifuge experiment can be compared across lots.

2.3 Diterpenes and triterpenes

Diterpene evidence is comparatively sparse and often mixed into broad phytochemical surveys. The *A. annua* seed study provides isolation-level evidence for one diterpene among three monoterpenes and fourteen sesquiterpenes, while a dedicated *A. annua* study identified ten diterpene synthases in relation to glandular trichome formation, artemisinin production, and stress resilience. A multi-platform *A. absinthium* oil analysis additionally reported diterpene- and homoditerpene-associated constituents alongside GC and NMR confirmation (Judzentiene et al. 2009). These findings support a real C20 biosynthetic dimension but not a genus-wide diterpene chemotype. Diterpene records should preserve seed versus leaf or trichome tissue, extraction polarity, and whether the compound was isolated or only assigned by MS.

Triterpenes and sterols are especially important for avoiding a false "essential-oil-only" view of *Artemisia*. OSC2 and CYP716A14v2 were functionally connected to α-, β-, and δ-amyrin-related cuticular triterpenoids in *A. annua*, and triterpenoids have also been isolated from *A. igniaria*. Their cuticle or surface-associated localization suggests roles in barrier and stress biology, but current records do not support transferring those roles to antiparasitic action. Because triterpenes are less volatile, a GC-only survey can systematically under-report them.

The current diterpene and triterpene literature remains too sparse and methodologically heterogeneous for a genus-wide abundance synthesis. A defensible survey should distinguish a structurally isolated C20 or C30 metabolite from a tentative LC-MS feature, and it should separate pathway-gene discovery from measured products. The *A. annua* diterpene-synthase study is unusually informative because it combined gene discovery with developmental and physiological perturbation, whereas the OSC2/CYP716A14v2 work paired trichome comparisons with recombinant functional tests. These studies provide reference loci for comparative genomics, but their products must be re-measured in homologous tissues before copy-number differences are interpreted as ecological or antiparasitic adaptations.

Targeted primary isolations confirm that the sparse C20/C30 literature reflects under-sampling rather than an absence of chemistry. *A. sacrorum* yielded an ent-kauranoid diterpene diglycoside and two known ent-kauranoids; *A. caruifolia* yielded a new triterpene with co-isolated lignans; and whole-plant *A. sphaerocephala* yielded a new oleanane-type triterpenoid saponin (Li et al. 1990; Ma et al. 2001; Li et al. 2008). Petroleum-ether fractionation of *A. argyi* further identified nordammarane and cycloartane triterpenoids, while *A. vulgaris* yielded both new sesquiterpenoids and triterpenoids with structure confirmation (Zhang et al. 2020; Liu et al. 2022). These records establish chemical space and analytical identity; their cytotoxic, alpha-glucosidase, or anti-inflammatory endpoints do not establish antiparasitic activity. A genome-wide *A. annua* OSC survey identified 24 genes across CAS, LAS, LUS, and unknown subfamilies and reported differential expression under UV-B, but its bioinformatic and expression evidence still requires product-level and recombinant-enzyme validation (Guo et al. 2025).

An enzyme-resolved *A. argyi* example shows why C30 evolution should be studied at the active-site level. AaOSC-22030 converted 2,3-oxidosqualene into diverse triterpene skeletons; multiscale QM/MM analysis and mutagenesis implicated S728 in relaxed intermediate accommodation and catalysis, whereas S728Y made dammarendiol II the exclusive reported product and the N369/E371 dyad controlled terminal hydration of pentacyclic products (Liu et al. 2026). This is strong sequence-to-function evidence for catalytic plasticity in one recombinant enzyme, but engineered specificity and computationally resolved transition-state hypotheses do not establish native abundance, lineage-wide selection, or antiparasitic activity. It supplies a concrete comparative test: map active-site residues onto common-annotation OSC gene trees, then verify product spectra in matched tissues.

The class-balanced tranche adds further structure-confirmed evidence. Aerial parts of *A. suksdorfii* yielded five new polyol monoterpenes and seven new sesquiterpene lactones, with high-field NMR and X-ray support for one lactone (Ahmed et al. 2004). Across *A. giraldii* var. *giraldii*, *A. mongolica*, and *A. vestita*, isolation work recovered a new antifungal monoterpene, a new germacranolide, alpha-amyrin, and sterol-associated constituents (Tan et al. 1999). In *A. igniaria*, four triterpenoids were structurally assigned from an extract, including beta-glutinanol as a first natural occurrence report (Hu and Feng 2000). These studies materially widen the C10/C15/C30 inventory, but they remain specimen- and extraction-bounded: they do not supply genus-wide abundance estimates, shared pathway orthology, or antiparasitic efficacy. The antifungal results are retained as bioactivity context rather than recoded as malaria or helminth evidence.

An earlier cross-species phytochemical study makes the same class boundary explicit. *A. sieversiana* aerial parts yielded a new guaianolide, while antifungal fractions of *A. annua* contained arteannuin B, artemisinin, oleanolic acid, beta-sitosterol, stigmasterol, and several flavones; arteannuin B was active against the tested *Candida* strain at 100 microgram/mL, whereas most other isolates were inactive (Tang et al. 2000). This is structure- and activity-resolved evidence for C15 and C30 chemical space, but the fungal endpoint does not establish malaria, helminth efficacy, or a shared *Artemisia* pathway. It also illustrates why non-terpene co-isolates and sterols should remain visible in the evidence table rather than being forced into a terpene-only interpretation.

Historical records add depth to the C10/C15 literature while also showing where the evidence is thin. An indexed *A. filifolia* study explicitly addressed new monoterpene structures, synthesis, rearrangements, and biosynthesis, providing a foundational C10 reference for structure-function comparison, although detailed product extraction still requires full-text recovery (Torrance and Steelink 1974). Seasonal sampling is represented by a primary *A. feddei* study of sesquiterpene-lactone constitution in material collected in Kochi prefecture; because its abstract is unavailable in the current retrieval, it is retained as a phenology lead rather than a quantitative seasonal result (Matsueda et al. 1973). A separate *A. feddei* isolation study reported five sesquiterpene lactones, including two new guaianolides, from aerial parts (Tan et al. 1992). *A. sieversiana* aerial parts likewise yielded one known and three new guaianolides alongside lignans and sterols, with extensive spectral characterization; antifungal testing in that study is not an antiparasitic endpoint (Tan et al. 1998). Together these records support a broad structural and phenological search space, but the title-level and isolation-level evidence should not be converted into native pathway evolution, abundance, or malaria/vermifuge claims.

Structure-confirmed isolation from *A. sieberi* provides another class-balanced STL anchor. Two new sesquiterpene lactones and two known compounds were assigned by NMR, mass spectrometry, and X-ray diffraction; the tested activities were antimicrobial or antifungal rather than antiparasitic (Mohamed et al. 2017). The study therefore expands authenticated C15 lactone space and supplies a candidate set for future parasite assays, but it does not support a claim that these compounds are vermifuges or that their antimicrobial endpoints predict mammalian selectivity.

An earlier *A. annua* isolation study adds a distinct C30 anchor: friedelin and friedelan-3-beta-ol were isolated alongside artemisinin-related terpenoids, and friedelane-type triterpenoids were reported from the plant for the first time (Zheng 1994). The associated endpoint was cytotoxicity in human tumor cell lines, so this record expands nonvolatile triterpene coverage without supporting parasite activity or a safe therapeutic window. In a separate *A. herba-alba* fractionation study, a new monoterpene dimer, a known monoterpene, and three sesquiterpene lactones were characterized by multidimensional NMR and HRMS (Abou El-Hamd et al. 2013). The two records together illustrate why class-balanced sampling must include both structure-confirmed nonvolatile constituents and unusual terpene architectures, not only routine essential-oil percentages.

Sesquiterpene engineering supplies a complementary functional anchor. Coexpression of valencene synthase and valencene oxidase in *A. annua* produced nootkatone, and plastid-targeted constructs reached 12.11-47.80 microgram/g fresh weight without changing artemisinin production (Gou et al. 2021). The result demonstrates that substrate compartment and enzyme choice can redirect C15 flux in an engineered plant, but it is not evidence that nootkatone is a native *A. annua* chemotype or an antiparasitic constituent.

The first committed artemisinin-pathway step has also been demonstrated with unusually direct sequence-to-product evidence. An amorpha-4,11-diene synthase cDNA from *A. annua* produced amorpha-4,11-diene from FPP in recombinant *E. coli*, and transformed tobacco accumulated the product at 0.2–1.7 ng/g fresh weight (Wallaart et al. 2001). This anchors catalytic capacity and heterologous plant production, but it does not establish native flux, tissue localization, or conservation of the exact product across *Artemisia*. It is therefore a useful positive control for comparative ADS-like sequence-function analysis rather than a genus-wide artemisinin marker.

2.4 Sesquiterpene lactones

Structure-confirmed dimeric guaianolides extend the nonvolatile wormwood inventory beyond the monomeric lactones usually emphasized in pharmacognosy. Aerial parts of *A. absinthium* yielded five new absinthins (A–E) and seven known dimeric guaianolides; absinthin C and isoanabsinthin inhibited LPS-induced nitric-oxide production in BV-2 cells at 1.52 and 1.98 microM, respectively (Turak et al. 2014). The NMR, HRESIMS, and X-ray evidence supports authenticated C15-dimer chemical space, while the immune-cell endpoint remains distinct from helminth or malaria activity. These molecules are therefore priority standards for fractionated *A. absinthium* parasite assays, not evidence that “absinthin” is the active vermifuge principle.

Isolation from in-vitro shoot culture expands the non-artemisinin lactone reference set. *A. aucheri* yielded the eudesmanolide artemin (2,5-dihydroxy-12,6-eudesmanolide-4(15)-ene), reported from this plant for the first time; artemin showed micromolar cytotoxicity in prostate and breast cancer cell lines (Abbaspour et al. 2019). This is isolation-level sesquiterpene-lactone evidence, not antiparasitic efficacy, and the tissue-culture origin means native abundance and ecological function remain untested.

Structure-confirmed isolations widen the STL repertoire beyond the artemisinin branch. *A. anomala* aerial parts yielded a dimeric guaianolide, a glaucolide, a seco-guaiaretic acid, additional guaianolides, and known STLs (Wen et al. 2010). *A. dubia* yielded eight new artemdubolides and nineteen known STLs, while aerial parts of *A. myriantha* yielded ten guaianolides including artemyriantholide E, arglabin, dehydroleucodin, and related structures (Huang et al. 2010; Zan et al. 2018). In *A. argyi*, HPLC activity profiling followed by ECD/VCD and NMR resolved the absolute configuration of a diverse STL set, and a separate leaf study added new guaianolide and eudesmane structures (Reinhardt et al. 2019; Zhang et al. 2021). These records establish specimen-bounded chemical diversity and the need for stereochemical authentication; cytotoxic or immune-cell endpoints in several of them are not antiparasitic evidence. They also provide a tractable comparative resource: common-annotation gene trees can now be paired with authenticated STL profiles and recombinant oxidase/lactonization assays rather than inferred from compound names alone. The structure space continues into dimeric guaianolides and absinthin-related chemistry. *A. mongolica* aerial parts yielded seven undescribed and seventeen known sesquiterpenes, including artemongolide dimers, artemanomalides, absinthin-related compounds, and other guaianolides; NMR, HRESIMS, ECD, and first X-ray determinations supported the assignments (Zhu et al. 2024). A *A. gorgonum* study used santonin-derived photochemistry to prepare and revise seven seco-guaianolide/guaianolide structures, with X-ray-supported absolute configurations; the strongest reported endpoint was phytotoxicity in wheat coleoptiles rather than parasite activity (Macías et al. 2012). These studies are valuable for stereochemical and defense-chemistry coverage, but they should not be used to infer anthelmintic or antimalarial action. They also reinforce a practical design rule: dimeric and highly oxygenated STL features should be searched by LC-MS/MS and structure-guided isolation, not assumed from volatile-oil fingerprints.

A newer structure-confirmed tranche shows that this dimeric and oxygenated space is still expanding. *A. myriantha* yielded 19 new sesquiterpenolides spanning 13 germacranolides, four guaianolides, and two eudesmanolides, including uncommon cis-fused 10/5 systems and an unusual (7S) configuration (Tang et al. 2020). *A. dubia* yielded 20 previously undescribed guaiane-type dimers: A-Q were proposed as [4+2] Diels-Alder adducts of monomeric guaianolides, whereas R-T were ester-linked (Gao et al. 2022). *A. princeps* added 13 new dimeric sesquiterpenoids, also assigned a Diels-Alder origin, with X-ray/ECD-supported configurations (Su et al. 2023). In *A. porrecta*, 27 STLs covered eudesmanolide, germacranolide, melampolide, and heliangolide classes, while *A. myriantha* var. *pleiocephala* supplied eight new guaiane-type sesquiterpenoids plus 14 known compounds (Abdullajanov et al. 2025; Wang et al. 2025). The associated endpoints were cytotoxicity, anti-inflammatory activity, or structural elucidation—not malaria or helminth assays. These records therefore expand the candidate library and suggest experimentally testable assembly hypotheses, but dimer assignment, native biosynthetic provenance, parasite activity, and host selectivity still require LC-MS/MS, NMR/X-ray confirmation, pathway-enzyme assays, and matched parasite testing.

The next queue tranche further argues against treating named STL families as a closed inventory. *A. sacrorum* yielded 23 highly oxygenated guaiane-type lactones, including acetoxylated alpha-methylene-gamma-lactones and 1,10-rearranged structures; whole-plant *A. mongolica* yielded 12 new and seven known STLs with multiple X-ray assignments (He et al. 2024; Zhu et al. 2022). *A. leucophylla* added 12 new and 13 known eudesmane-type sesquiterpenoids, including unprecedented 1,2-seco-1-nor-eudesmane skeletons, and *A. heptapotamica* supplied both monomeric and dimeric lactones (Wang et al. 2023; Zhamilya et al. 2019). Dimeric chemistry was also reported from *A. selengensis* through proposed [2+2] and [4+2] couplings, from *A. myriantha* through additional Diels-Alder-like dimers, and from *A. myriantha* var. *pleiocephala* through a second variety-specific series (Chen et al. 2025; Wang et al. 2025; Wang et al. 2024). Finally, *A. ordosica* yielded cadinane-monoterpene heterodimers and unusual sesquiterpenoids, while *A. annua* yielded 14 cadinane-involved dimers spanning several coupling modes (Wang et al. 2025; He et al. 2024). These findings are valuable for comparative chemical evolution and target-library design, but their tumor-cell, inflammatory-cell, or computational endpoints do not establish malaria, helminth, or other parasite efficacy. A publishable synthesis should therefore report them as authenticated chemical-space expansions and explicitly preserve the gap between structural novelty, native pathway reconstruction, and antiparasitic mechanism.

Sesquiterpene lactones (STLs) include guaianolide, germacranolide, eudesmanolide, and related structural families. Their α-methylene-γ-lactone or other electrophilic motifs can react with nucleophiles, which provides a chemically plausible basis for bioactivity but not a demonstrated parasite target. In *Artemisia*, STL reports include artemisinin-related amorphane chemistry, santonin, and numerous guaianolide and germacranolide constituents. Analytical workflows differ from oil profiling: LC-MS, isolation, multidimensional NMR, and stereochemical assignment are often needed, while direct GC percentages are not a suitable universal denominator.

Wormwood provides an unusually useful analytical reference for this class. A validated HPLC-DAD method separated and quantified five sesquiterpene lactones, two lignans, and a polymethoxylated flavonoid in *A. absinthium* material and commercial preparations; HPLC-MS and HPLC-SPE-NMR also characterized absinthin degradation products and showed strong stability under several storage conditions (Aberham et al. 2010). This is important for the vermifuge question because measured lactone exposure depends on extraction and storage, but the analytical record does not by itself establish anthelmintic activity or a safe dose.

Santonin adds a historically important but safety-limited branch to the wormwood question. A validated HPLC-UV method measured santonin in *A. cina* extracts and leaves from eight Kazakhstan *Artemisia* taxa, including *A. absinthium*, providing a practical lot-specific control for a compound formerly used as an anthelmintic (Sakipova et al. 2017). In a separate *A. cina* fractionation study, high-speed countercurrent chromatography removed santonin from a CO2 subcritical extract while the santonin-free preparation retained antinociceptive and anti-inflammatory activity (Sakipova et al. 2020). Because the latter endpoints were not parasite assays, these studies do not prove that santonin or santonin-free extract is a vermifuge. They do show why historical anthelmintic reputation, measured santonin exposure, preparation-level activity, and human safety must be analyzed as separate variables.

A particularly relevant attribution lead comes from a parasite-oriented analytical study of *A. absinthium*. HPLC peak deconvolution and reconstructed spectra tentatively identified the sesquiterpene lactones alpha-santonin and ketopelenolid-A in methanolic wormwood extracts; the authors presented alpha-santonin as a possible active principle based on its established antiparasitic reputation (Perez-Souto et al. 1992). This links a named wormwood preparation to named lactone candidates, but the identification was tentative and the record does not provide a matched purified-compound assay, direct parasite-protein experiment, or standardized extract exposure. The result therefore motivates targeted LC-MS/MS confirmation and fractionation rather than proving that alpha-santonin explains any modern wormwood phenotype. Historical indexing also records treatment of ascariasis with santonin prepared from *A. transiliensis* (Guseinov 1954). Because that record is title- and metadata-level in the current retrieval, it documents historical vermifuge context only; it cannot support a dose, efficacy magnitude, product-quality, or current clinical claim.

The malaria literature has been repeatedly synthesized, but review recurrence should not be mistaken for new primary evidence. A foundational endoperoxide review documented the early chemical and antimalarial development of artemisinin, while a recent review revisited artemisinin chemistry, extraction, photochemistry, mechanisms, and translational applications across *Artemisia* sources (Ziffer et al. 1997; Das et al. 2024). These reviews justify the historical and mechanistic framing of the malaria section; the present synthesis retains primary plant, parasite-stage, pharmacokinetic, and safety records as the basis for quantitative or clinical statements.

Historical pharmacognosy also links santonin to an *A. maritima* source in India (Chopra and Chandler 1924). Because the indexed record is historical and not a modern standardized composition or controlled parasite study, it strengthens provenance for santonin's vermifuge reputation only. It cannot establish that contemporary *A. absinthium* preparations contain an equivalent dose, retain equivalent activity, or have an acceptable safety margin.

The best-defined biosynthetic model for a non-artemisinin STL branch begins with FPP cyclization to germacrene A, oxidation to germacrene A acid, and downstream P450-dependent oxidation and lactonization. The presence of a germacranolide-like metabolite is therefore compatible with several levels of evidence-from structural analogy to a fully demonstrated enzyme sequence-and those levels must not be collapsed. STL abundance should also be reported separately from volatile sesquiterpene abundance because extraction and storage can partition them differently. In *A. annua*, field measurements of glandular-trichome dynamics across leaf development and senescence, together with dead-versus-green leaf precursor profiles, show why tissue age and precursor conversion must be retained as explicit variables (Lommen et al. 2006; Lommen et al. 2007). Enzyme evolution adds a testable explanation for structural diversification: a germacrene-A oxidase accepted all seven alternate sesquiterpene substrates tested in yeast, whereas amorphadiene oxidase was comparatively rigid; engineered active-site substitutions increased artemisinic-acid activity per enzyme but reduced stability (Nguyen et al. 2019). This supports catalytic promiscuity as an evolutionary opportunity, not a claim that every GAO homolog makes a native lactone. A 2024 functional reanalysis adds an important correction to the AMO/GAO boundary. One *A. annua* CYP71AV1-like copy, termed AMOHAP, had catalytic-region features and a divergent transcript context more consistent with germacrene-A oxidase; recombinant protein assays demonstrated GAO activity rather than a canonical amorphadiene-oxidase interpretation (Chen et al. 2024). This result supports a pathway-evolution model in which a new artemisinin branch coexists with, or replaces, an ancestral costunolide-related branch through both catalytic specialization and expression-level divergence. It also supplies a concrete comparative test: common-annotation CYP71/GAO trees should be paired with cis-regulatory, tissue-expression, and matched germacrene-A/amorphadiene enzyme assays before assigning orthology or pathway completion in other *Artemisia* species.

Pathway organization is another experimentally testable layer between gene inventory and metabolite abundance. A Molecular Plant report explicitly investigated promoting artemisinin biosynthesis in *A. annua* by substrate channeling, but the current indexed retrieval lacks an abstract; it is therefore retained as a full-text-recovery lead rather than assigned a quantitative flux or compartment claim (Han et al. 2016). In an open-access transformation study, ADS plus the viral antisilencing factor P19 increased measured artemisinin to 0.18% in transformed leaves and 0.095% in transformed hairy roots, whereas control hairy roots had no detected artemisinin (Elfahmi et al. 2020). These engineered outcomes demonstrate production potential in *A. annua*, not native cross-species conservation or parasite efficacy. Environment can further separate transcript abundance from terminal product: dynamic cold-stress profiling found transient jasmonate-associated induction and coordinated PAL/C4H/ADS/DBR2 responses, while prolonged exposure increased dihydroartemisinic and artemisinic acids as terminal conversion was inhibited (He et al. 2024). This duration dependence is a useful design variable for comparative experiments and a reason not to encode "stress-induced artemisinin" as a single invariant trait.

A complementary combinatorial-engineering study showed that AaDBR2 and a *Tanacetum parthenium* CYP71CB1 could act on multiple sesquiterpene-lactone pathway substrates, enabling heterologous production of dihydrocostunolide, dihydroparthenolide, and hydroxylated derivatives (Beyraghdar Kashkooli et al. 2019). Such catalytic plasticity offers a plausible route by which duplicated or recruited downstream enzymes can enlarge lactone chemical space, but it also warns against inferring a unique product from a gene-family label: substrate access, stereochemistry, compartment, and downstream turnover must be experimentally matched.

Evolutionary sequence-function work provides a complementary explanation for terpene diversity outside the lactone branch. By recombining naturally occurring residues between cyclic ADS and linear beta-farnesene synthase, Salmon et al. identified a small epistatic network in which one dominant substitution could activate cyclization and additional residues could suppress or restore it; the corresponding network also affected a *Citrus* homolog (Salmon et al. 2015). This is direct evidence that a few interacting amino-acid changes can alter TPS product topology in a heterologous assay. It motivates gene-tree- and structure-aware tests across *Artemisia*, but does not by itself identify ancestral states, native selection, or a parasite-relevant product.

Recent structure-confirmed isolations show that the genus contains substantially more STL architecture than the artemisinin branch alone suggests. *A. vulgaris* yielded twelve new and ten known guaianolides, with NMR, electronic-circular-dichroism, and crystallographic support for stereochemistry; *A. verlotorum* yielded seven new and nineteen known sesquiterpenoids, including uncommon bicyclic, iphionane, and oxygen-bridged eudesmanolide structures (Chen et al. 2022; Wang et al. 2023). Basin big sagebrush leaf fractionation and *A. nitrosa* isolation add geographically distinct, structure-confirmed lactone records, including a new germacrane-type lactone and known sesquiterpene series (Anibogwu et al. 2024; Kazymbetova et al. 2022). The associated macrophage or cytotoxicity assays are useful for structure-activity context, but they are not antiparasitic evidence. These studies strengthen the evolutionary hypothesis that repeated germacrane, guaiane, and eudesmane diversification creates a large chemical search space; demonstrating lineage-specific pathway evolution still requires common annotations, gene trees, expression, and direct enzyme assays.

The artemisinin branch demonstrates how metabolite isolation, enzyme biochemistry, and nonenzymatic chemistry can be integrated. Dihydroartemisinic acid and its hydroperoxide were isolated as candidate late intermediates before genetic perturbation established a broader branch architecture (Wallaart et al. 1999a; Wallaart et al. 1999b). CYP71AV1 disruption subsequently redirected metabolites toward arteannuin X and supported a major role for spontaneous oxidative chemistry in terminal artemisinin formation. The evidence therefore supports a hybrid pathway model: enzyme-controlled scaffold formation and oxidation establish the precursor pool, while storage environment and post-biosynthetic oxidation help determine terminal products.

The chemical activation step has an independent physical-chemistry basis. EPR experiments detected carbon-centered radicals after iron-catalyzed decomposition of artemisinin and related endoperoxides, supporting a reactive-intermediate model for the peroxide bridge (Butler et al. 1998). In a parasite assay, extracellular hemin at pathologic concentrations enhanced artemisinin activity against *P. falciparum* and increased lipid-peroxidation products; vitamin E reduced the enhancement, whereas the same hemin manipulation did not potentiate quinoline drugs (Tangnitipong et al. 2012). These results connect iron/heme context to activation and phenotype, but neither identifies one necessary parasite receptor nor validates an unstandardized *Artemisia* preparation. The appropriate comparative-genomics question is therefore whether a plant accession changes the amount and exposure of an activatable peroxide-bearing metabolite, not whether an ADS-like gene alone predicts parasite killing.

Non-artemisinin lactones broaden both structural and pharmacological space. *A. gorgonum* yielded isolated guaianolide and germacranolide compounds with antiplasmodial testing, while *Artemisia*-derived or *Artemisia*-associated lactones such as dehydroleucodine have been tested against trypanosomatids. Dehydroleucodine inhibited cultured *T. cruzi* epimastigote growth and altered parasite physiology in a defined-compound study (Brengio et al. 2000). Ten additional germacrane-type sesquiterpenoids, artemyrianosins A-J, were isolated and structurally characterized from aerial parts of *A. myriantha*; selected compounds showed 43.7-89.3 microM cytotoxicity against human hepatoma cell lines (Zhang et al. 2022). That study expands confirmed STL structural space, but its endpoint is anticancer cytotoxicity and should not be recoded as antiparasitic activity. Such studies are stronger for constituent attribution than an oil assay, but species provenance, parasite stage, exposure, and host-cell selectivity still prevent direct translation to therapeutic efficacy.

3. Biosynthesis and compartmentalization

Production is also shaped by nutrient state and cultivation context. In cloned greenhouse plants, potassium deficiency increased artemisinin concentration from 0.93% to 1.62%, but total per-plant yield was not significantly different from complete nutrition, illustrating the difference between concentration and biomass-normalized output (Ferreira 2007). In a fed-batch hairy-root bioreactor, model-based sucrose feeding reached 1.0 mg/g artemisinin after 16 days, showing process control rather than native field flux (Patra and Srivastava 2015). AMF responses were similarly conditional: *Rhizophagus irregularis* increased artemisinin and essential-oil measures in semi-open soils, but open-field artemisinin differences were not detected and beta-farnesene/germacrene D shifted; a separate phosphate gradient found an inverse artemisinin response and lower leaf artemisinin with *Funneliformis mosseae* (Domokos et al. 2020; Todeschini et al. 2022). These studies make nutrient, symbiont, soil, and cultivation regime explicit covariates for comparative chemotype experiments.

Common-environment and plant–microbe experiments help separate genetic, ecological, and physiological components of this variation. In a Korean panel, 103 individuals grown under the same conditions formed six oil chemotype labels (artemisia-ketone, camphor, beta-cubebene, eucalyptol, alpha-pinene, and beta-selinene), demonstrating substantial intraspecific variation beyond immediate growth conditions (Hong et al. 2023). In contrast, AM colonization of *A. annua* did not change total terpene emission, but increased limonene and artemisia-ketone emission and altered selected leaf sesquiterpenes, emphasizing that volatile release and tissue accumulation are related but non-identical phenotypes (Rapparini et al. 2008). Geographic HPLC-DAD and LC-QTOF-MS/MS profiling likewise separated regional groups by artemisinin, arteannuin B, and artemisinic acid content (Fu et al. 2020). Together, these results justify common-environment designs followed by genotype, expression, and metabolite linkage rather than treating a location-level oil profile as a fixed species trait.

The plastidial MEP pathway and cytosolic mevalonate pathway supply IPP and DMAPP, which are condensed into prenyl diphosphates. GPPS and related enzymes provide C10 substrates for monoterpene synthases; FPPS provides the C15 substrate FPP for sesquiterpene synthases, including germacrene A synthase and ADS. Cytochrome P450s, reductases, dehydrogenases, oxidases, acyltransferases, glycosyltransferases, and spontaneous reactions expand the scaffold space.

Functionally characterized *A. annua* monoterpene synthases illustrate why copy number alone is insufficient: recombinant AaTPS2, AaTPS5, and AaTPS6 produced different product spectra, and transcript abundance changed after wounding and hormone treatments (Ruan et al. 2016). Product specificity, inducibility, and tissue localization therefore need to be modeled together in evolutionary comparisons.

The artemisinin pathway in *A. annua* is the strongest gene-to-metabolite reference. ADS converts FPP to amorpha-4,11-diene, a foundational enzyme result established in early biochemical work (Bouwmeester et al. 1999); CYP71AV1 oxidizes pathway intermediates; DBR2 and ALDH1 support the route toward dihydroartemisinic acid, after which terminal chemistry includes nonenzymatic steps. Glandular secretory trichomes provide a relevant biosynthetic and storage context. These statements are pathway anchors, not proof that homologous genes in another species produce artemisinin or confer antiparasitic activity.

Sequence-function experiments provide the bridge from comparative gene trees to testable catalytic hypotheses. A structure-based combinatorial protein-engineering workflow developed active-site and focused libraries for plant sesquiterpene synthases, using defined residue combinations to expose epistasis and alternative product spectra (Dokarry et al. 2012). Applied to *Artemisia* TPS and oxidase clades, this approach can distinguish a lineage-specific sequence change that alters product topology from a copy-number difference that merely expands catalytic opportunity. It should be paired with authentic-standard product detection, tissue expression, and substrate-matched assays; library activity alone cannot establish native evolution, in-planta flux, or parasite relevance.

Quantitative analytical validation helps connect pathway genes to measured chemistry. Rapid chloroform immersion followed by LC-ESI-Q-TOF-MS/MS recovered more than 97% of artemisinin, arteannuin B, artemisitene, and artemisinic acid from fresh *A. annua* material across a 0.1–3.0 microgram/mL calibration range (Van Nieuwerburgh et al. 2006). The authors used the recovery pattern to support localization in the subcuticular gland space, but analytical recovery is not itself proof of complete tissue localization or in-vivo pathway flux. In a historical vegetation-period study, artemisinin was highest immediately before flowering, one growth regulator increased content by approximately 30%, and most tested non-*annua* material lacked technically high concentrations (Liersch et al. 1986). These records support a stage- and method-aware sampling design rather than a universal preflowering rule.

Regulatory studies add a causal layer between stress signaling, trichome development, and pathway output. AaWRKY1 activated *ADS* and *CYP* promoter reporters, and overexpression increased pathway-gene expression and artemisinin in *A. annua* (Jiang et al. 2016). AaNAC1 was induced by dehydration, cold, salicylic acid, and methyl jasmonate; its overexpression increased artemisinin and dihydroartemisinic acid while improving drought and *Botrytis* tolerance (Lv et al. 2016). At the post-translational level, AaPP2C1 physically interacted with and dephosphorylated AaAPK1, reducing AabZIP1-mediated activation of artemisinin-biosynthesis genes (Zhang et al. 2019). Finally, exogenous beta-ocimene increased juvenile-plant artemisinin together with glandular-secretory-trichome size and density (Xiao et al. 2020). These are strong *A. annua* regulatory anchors, but overexpression, reporter, and elicitation results do not establish conserved function in other *Artemisia* lineages or increased parasite potency.

Developmental timing and culture conditions provide important negative and process controls. Constitutive *fpf1* expression advanced flowering in transgenic *A. annua* by about 20 days, yet artemisinin content did not differ significantly from the comparison plants, arguing against flowering time as a sufficient proxy for pathway output (Wang et al. 2004). In transformed hairy roots, autoclaving and the choice of filter-sterilized sugar changed biomass and artemisinin yield even at constant total carbon; glucose stimulated artemisinin production while sucrose supported better growth than glucose (Weathers et al. 2004). These studies strengthen the genotype-by-development-by-culture model and caution against interpreting a production increase as a conserved regulatory mechanism or as evidence of greater parasite activity.

Selection and engineering provide complementary tests of heritable versus inducible production. Four recurrent-selection cycles increased *A. annua* artemisinin from 0.15% to 1.16% and identified a molecular marker associated with high-yielding genotypes, linking standing genetic variation to a reproducible chemotype shift (Paul et al. 2010). In a separate transgenic experiment, co-overexpression of *CYP71AV1* and its reductase partner *CPR* increased artemisinin by up to 38% in one line (Shen et al. 2012). These results establish genetic and enzymatic leverage within *A. annua*, but neither identifies a conserved cross-species mechanism nor shows that a higher plant titer improves parasite selectivity or clinical efficacy.

Trichome regulators and plant–microbe interactions add another layer. The trichome-specific AP2/ERF factor AaORA positively regulated *ADS*, *CYP71AV1*, and *DBR2*; perturbation changed artemisinin and dihydroartemisinic acid, and disease-response phenotypes were also reported (Lu et al. 2013). Co-cultivation of *A. annua* shoots with *Piriformospora indica* increased biomass and artemisinin from 0.80% in controls to 1.30% in one treatment (Sharma and Agrawal 2013). These studies support a regulatory and symbiotic context for pathway output, while retaining the necessary boundary between production enhancement, native ecology, and antiparasitic phenotype.

The pathway was assembled through convergent evidence rather than a single experiment. ADS cloning and bacterial expression established the first committed cyclization; a trichome-enriched cDNA strategy identified CYP71AV1 and demonstrated sequential oxidation of amorpha-4,11-diene and related intermediates; and DBR2 cloning showed reduction at the branch that favors dihydroartemisinic-acid-derived chemistry (Chang et al. 2000; Teoh et al. 2006; Zhang et al. 2008). Stable-isotope feeding in intact growing plants supplied an independent carbon-flow perspective (Schramek et al. 2010). Together, these data are considerably stronger than pathway reconstruction from annotation alone, yet even this reference pathway remains sensitive to developmental state, precursor competition, and terminal nonenzymatic conversion.

The early pathway evidence also illustrates the value of preserving intermediate-level observations. Trichome and leaf analyses detected oxygenated amorpha-diene derivatives, while measured amorpha-4,11-diene hydroxylase, alcohol/dehydroaldehyde dehydrogenase, and related activities supported a proposed sequence toward dihydroartemisinic acid (Bertea et al. 2005). Recombinant

CYP71AV1 independently oxidized amorpha-4,11-diene, artemisinic alcohol, and artemisinic aldehyde (Teoh et al. 2006). The combined record is strong step-level evidence in *A. annua*, but a pathway diagram should still distinguish detected intermediates, measured activities, inferred order, and demonstrated in-vivo flux.

A 2025 enzyme study adds a distinct branch to the production model. AaDHAADH catalyzed bidirectional conversion between artemisinic acid and dihydroartemisinic acid; a P26L variant increased activity toward dihydroartemisinic acid production, and engineered yeast reached 3.97 g/L DHAA in a 5-L bioreactor (Guo et al. 2025). This is strong enzyme and metabolic-engineering evidence, but it does not demonstrate that AaDHAADH controls native *A. annua* flux in planta or that homologs in other *Artemisia* species have the same directionality or physiological importance. A production-perturbation study provides an additional tissue-level control: foliar application of the strigolactone analog GR24 increased growth, glandular-trichome attributes, and artemisinin content and yield in the tested *A. annua* plants (Wani et al. 2022). This is useful evidence that hormone signaling can alter the secretory capacity associated with a sesquiterpene-lactone phenotype, but it is not a conserved cross-species mechanism, a field recommendation, or evidence that increased artemisinin production necessarily increases parasite potency.

Chemotype regulation and precursor competition provide complementary constraints on pathway interpretation. In high- versus low-artemisinin *A. annua* chemotypes, DBR2L promoter variants, DBR2L expression, sesquiterpene profiles, and recombinant reduction activity were associated with different artemisinin yields (Wang et al. 2023). Separately, loss of an AaBPS-associated locus reduced camphor and other monoterpenes while increasing FPP and selected sesquiterpenes, but not artemisinin (Olofsson et al. 2022). Together these observations argue that chemotype evolution can act through both downstream branch control and precursor allocation; neither study alone establishes a universal natural tradeoff or a parasite mechanism.

Compartment evidence is equally important. Immunolocalization placed key artemisinin enzymes in apical cells of glandular secretory trichomes, and a trichome transcriptome enriched the candidate space for isoprenoid and downstream-modification genes (Olsson et al. 2009; Wang et al. 2009). Tissue-expression studies showed that terpene-metabolism genes are not uniformly expressed across roots, stems, leaves, and reproductive structures (Olofsson et al. 2011). A grafting study further implicated root-derived regulation of shoot trichomes and artemisinin-related metabolites, indicating that the producing cell is embedded in a whole-plant signaling system (Wang et al. 2016).

Transport and developmental regulation add two distinct layers to that compartment model. AaPDR3 was localized mainly to T-shaped trichomes of old leaves and roots; RNAi reduced beta-caryophyllene, overexpression increased it, and transformed yeast accumulated beta-caryophyllene preferentially over germacrene D and beta-farnesene (Fu et al. 2017). In parallel, AaMIXTA1 perturbation changed glandular-trichome initiation, cuticle-related transcription, and artemisinin content (Shi et al. 2018). These studies show that a metabolite phenotype can depend on compound-selective transport and secretory-cell development, not only on synthase copy number. They also provide experimentally tractable cross-species tests: orthologous transporter activity, trichome-cell expression, and matched leaf-surface metabolomics.

Promoter-reporter evidence sharpens this compartment model. A 1,151-bp *CYP71AV1* promoter drove GUS expression in glandular secretory trichomes of young leaves and flower buds, shifted toward T-shaped trichomes in older leaves, and differed among *A. annua* varieties; methyl jasmonate increased reporter activity (Wang et al. 2013). This supports age-, tissue-, and genotype-dependent regulation, but reporter localization is not a measurement of native protein abundance, pathway flux, or conserved promoter function in another species.

These tissue and redox relationships were synthesized early in the artemisinin literature, which emphasized the coupled roles of trichomes, roots, reactive oxygen species, and regulatory enzymes (Nguyen et al. 2011; Wen and Yu 2011). The reviews remain useful for organizing hypotheses, but pathway-step and quantitative claims are anchored here to primary enzyme, perturbation, and isotope-tracing studies.

Precursor supply and branch competition have been tested genetically. Bidirectional perturbation of AaHDR1 altered artemisinin and other terpenoids, supporting a role for MEP-pathway supply while also showing class-specific responses rather than a uniform increase (Ma et al. 2017). Suppression of β-caryophyllene synthase shifted carbon away from one sesquiterpene branch, and a T-DNA insertion line produced a broader change in terpenoid composition and crude-extract bioactivity (Chen et al. 2011; Karaket et al. 2014). These perturbations support network competition, but engineered responses should not be assumed to describe standing natural variation.

Regulation is multilayered. Jasmonate-responsive AaERF1/AaERF2, glandular-trichome-specific WRKY1, AaMYB108, and the AaGSW1-AaYABBY5/AaWRKY9 complex have each been connected to artemisinin-pathway expression or metabolite accumulation through transgenic, interaction, and expression experiments (Yu et al. 2012; Chen et al. 2017; Liu et al. 2023; Kayani et al. 2023). Cold response adds another regulatory layer: AabHLH112 is induced at low temperature, is enriched in glandular secretory trichomes, and promotes artemisinin biosynthesis indirectly by binding the AaERF1 promoter and activating the associated pathway program (Xiang et al. 2019). The emerging model is therefore not a simple linear pathway but a tissue-localized, environment-responsive network in which developmental programs and stress signals act on precursor supply, synthase branches, oxidation, chromatin state, and trichome capacity. The uncaptured production and stress literature makes the environmental term more quantitative. Hairy-root lines selected from hundreds of transformants differed in growth and artemisinin accumulation, showing that culture context and clonal background can generate large production differences even within one taxon (Liu et al. 1998). In field-grown *A. annua*, plant density changed the relationship between artemisinin and essential-oil yield under a defined North-Indian site and season (Ram et al. 1997). Soil copper produced a nonlinear, genotype-dependent response: a low treatment increased artemisinin content whereas higher doses reduced accumulation and impaired growth (Zehra et al. 2020). Hydrogen sulfide partially rescued copper-associated injury while increasing glandular-trichome traits and artemisinin in the tested material (Nomani et al. 2022). These studies support genotype-by-environment and canopy-by-tissue interaction terms, but they are not evidence that stress manipulation improves parasite control or that a production phenotype is conserved across *Artemisia*.

An earlier Ri-plasmid study independently showed that transformed *A. annua* roots produced artemisinin and that pH, sucrose, auxin, and gibberellin altered growth or product accumulation (Cai et al. 1995). In a complementary stress experiment, a nitric-oxide donor improved photosynthetic and redox traits, trichome attributes, and artemisinin under cadmium exposure (Wani et al. 2023). Both studies strengthen the case for culture- and environment-aware sampling, but neither establishes native aerial chemotype evolution, a field intervention, or antiparasitic efficacy. Defense and post-transcriptional regulation provide complementary causal anchors. AaWRKY17 bound W-box motifs in the *ADS* promoter, increased artemisinin in transgenic *A. annua*, and reduced susceptibility to *Pseudomonas syringae* (Chen et al. 2021). A small-RNA, degradome, and perturbation study found that miR160 suppressed glandular-trichome formation and artemisinin biosynthesis, whereas ARF1 activated *AaDBR2* (Guo et al. 2023). Together these results connect defense signaling, trichome development, and a pathway branch point, while leaving control of non-artemisinin terpenes, natural selection, cross-species orthology, and antiparasitic activity unresolved. An allele-aware *A. annua* perspective is therefore useful as a genome-interpretation frame, not as a substitute for primary abundance or enzyme evidence (Xu et al. 2022). An additional jasmonate perturbation adds a negative-regulatory node: AabHLH5 and two JAZ proteins were identified using transformation, yeast two-hybrid, and bimolecular-fluorescence assays, and AabHLH5 was verified as a negative regulator of jasmonate-linked artemisinin biosynthesis (Xu et al. 2025). In parallel, an *A. argyi* hairy-root system and recombinant assay showed that AYFOMT2 methylates jaceosidin to eupatilin and that transformation increases eupatilin accumulation (Peng et al. 2025). The AYFOMT2 result is a non-terpene branch control, but it demonstrates the kind of tissue transformation and step-level enzyme validation that comparative terpene studies need. Neither study supports transfer of regulator or enzyme function across species by homology alone.

The regulatory network has continued to expand in a direction that is relevant to comparative experiments. AaMYC2-type directly bound E-boxes in the *AaADS* and *AaCYP71AV1* promoters, while co-expression of AaMYC2 with AaMYC2-LIKE increased trichome density and artemisinin more than either factor alone (Khan et al. 2024; Hurrah et al. 2025). AaSHI1 activated the same core pathway promoters in transient assays, AaWRKY6 bound the *AaDBR2* promoter during growth-stage experiments, and AaABI5 connected light and ABA signaling to multiple pathway genes (Yang et al. 2024; Wang et al. 2023; Li et al. 2023). AabZIP1 adds ABA-JA cross-talk through AaMYC2 and also affects cuticular-wax genes under drought (Shu et al. 2022). These studies establish a dense, experimentally tractable *A. annua* network, but their engineered increases in artemisinin are not evidence that the same regulators control volatile terpenes, are conserved in other *Artemisia*, or improve parasite outcomes.

Post-translational and catabolic controls further complicate the genotype-to-metabolite map. AaAPK1, an ABA-responsive SnRK2-family kinase, interacted with and phosphorylated AabZIP1; the phosphorylated regulator increased transactivation of artemisinin-pathway genes, and AaAPK1 overexpression increased artemisinin in the reported experiments (Zhang et al. 2018). A class III peroxidase, Aa547, showed modeled affinity for artemisinin and deoxyartemisinin, localized to peroxisomes, and tracked inversely with artemisinin but directly with lignin in silencing and overexpression experiments (Nair et al. 2019). The kinase result supports a regulatory node and the peroxidase result a candidate turnover branch, but neither permits direct transfer to a second species without phosphorylation, enzyme-product, compartment, and flux measurements.

UV-B provides a related example of coordinated but not necessarily synergistic regulation. A leaf transcriptome identified 10,705 differentially expressed genes and 533 transcription factors after UV-B; reporter assays implicated AaMYB4 in activating *AaADS*, *AaDBR2*, and a flavonoid-pathway gene (Li et al. 2021). Parallel transcriptional regulation can explain correlated accumulation without proving that flavonoids potentiate artemisinin at the parasite. Comparative studies should therefore measure both transcript and chemical interaction terms, including matched parasite exposure, rather than treating co-induction as pharmacological synergy. Production studies add two important causal boundaries. Co-transformation of *HMGR*, *FPS*, and *DBR2* with the trichome regulators *AaHD1* and *AaORA* increased artemisinin up to 3.2-fold in T0 plants and up to 2.2-fold in selected T1 progeny, showing that pathway and compartment engineering can be combined while progeny stability remains variable (Hassani et al. 2023). Separately, endophytes isolated from environmentally distinct plants differed in their microbial conversion of artemisinin: *Colletotrichum* converted the peroxide bridge to an ether, whereas an environment-dependent isolate did not (Maehara et al. 2023). These results support a plant-microbe and engineering layer in the model, but neither proves that endophytes control native plant flux or that engineered/formulated material has improved antiparasitic efficacy.

Protein-level validation remains uneven. A CYP71AV1 expression and purification study established a recombinant heme-protein preparation and preliminary crystallization conditions for the enzyme that oxidizes amorpha-4,11-diene toward artemisinic acid (Shi et al. 2018). This is useful structural-method evidence, but a preliminary crystallization condition is not a solved structure, and protein production alone does not establish native activity in another *Artemisia* species. The comparison should retain protein purification, product formation, and pathway flux as separate evidence fields.

Chromatin accessibility provides a complementary tissue-level mechanism. ATAC-seq comparison of glandular trichomes and leaves identified thousands of differentially accessible regions, with greater accessibility near DBR2 and CYP71AV1 in glandular trichomes and enrichment of MYB, bZIP, C2H2, and AP2-like regulatory motifs (Zhou et al. 2021). These data support an epigenomic hypothesis for pathway compartmentalization, but accessibility is associative: causal chromatin perturbation, protein abundance, flux, and cross-species conservation remain to be established.

Regulatory perturbations also connect artemisinin chemistry to broader plant signaling. Salicylic-acid treatment produced an early reactive-oxygen burst and increased artemisinin, artemisinic acid, and dihydroartemisinic acid while changing HMGR and ADS responses more than FDS or CYP71AV1 expression (Pu et al. 2009). RNAi reduction of cinnamate-4-hydroxylase similarly accumulated trans-cinnamic acid and salicylic acid and increased artemisinin while reducing phenylpropanoid outputs (Kumar et al. 2016). These experiments support pathway cross-talk and redox-sensitive terminal chemistry, not a direct parasite mechanism or a field-production prescription.

Germacranolide biosynthesis offers a second comparative framework: FPP can be cyclized by germacrene A synthase, oxidized by germacrene A oxidase, and converted through P450-dependent chemistry and lactonization. The broad presence of a scaffold family does not imply identical product profiles because enzyme specificity, tissue localization, substrate competition, and downstream modification differ.

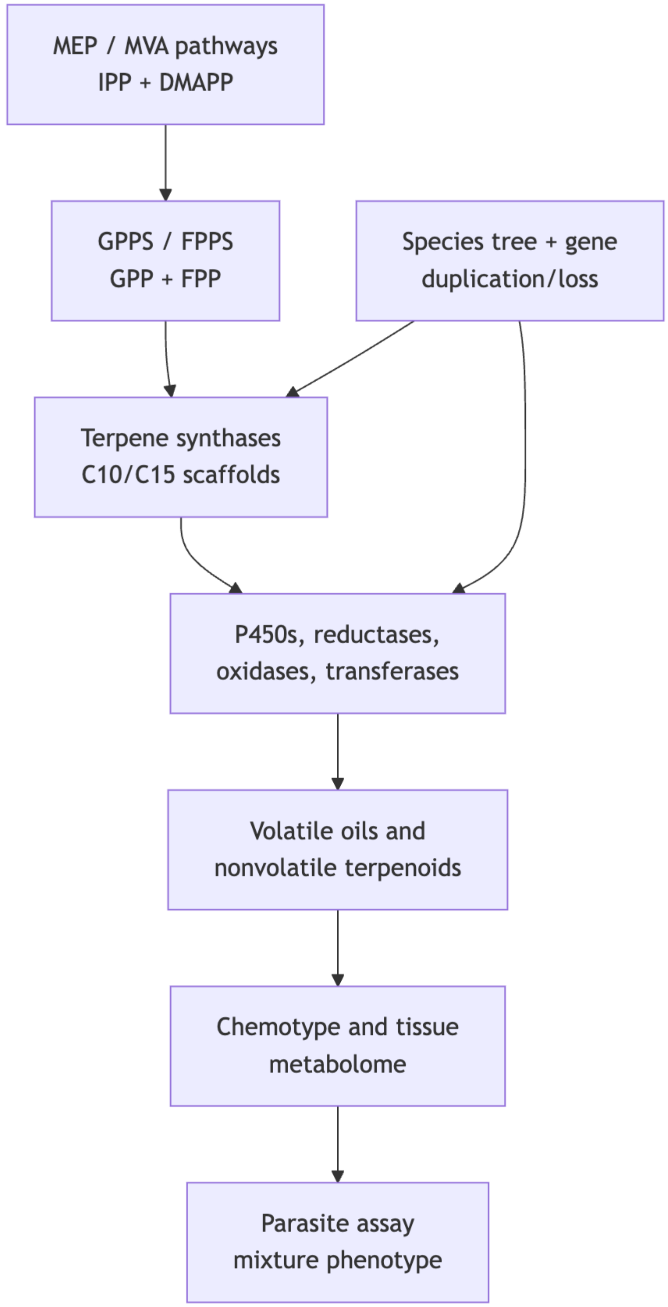
In *A. annua*, a glandular-trichome cDNA library yielded a germacrene A synthase that converted FPP to germacrene A in heterologous assays (Bertea et al. 2006). This creates an experimentally anchored comparator for germacranolide-like pathways and illustrates a general strategy for other species: first establish synthase product, then test downstream oxidases, tissue expression, and matching metabolites. Sequence similarity to germacrene synthase, by itself, is insufficient because closely related TPS proteins can differ in product spectra and because downstream enzymes may redirect a shared scaffold.

A historical biochemical anchor sits downstream of this framework: a purified *A. annua* leaf protein converted arteannuin B to artemisinin with 58% conversion under the reported assay conditions, with a Km of 0.5 mM and a pH optimum near 7.0–7.2 (Dhingra and Narasu 2001). This supports a transformation step at the protein-assay level, but the paper does not resolve a modern gene identity, native in-planta flux, subcellular transport, or conservation of the activity in other *Artemisia* species.

Chemical structure work provides an earlier precursor anchor: artemisinic acid was isolated from *A. annua* hexane extract and characterized by X-ray crystallography as a possible biogenetic precursor of artemisinin (Misra et al. 1993). Isolation and structural plausibility are important evidence for pathway substrate space, but they are not equivalent to isotope-traced conversion, enzyme assignment, or malaria activity.

Pathway evidence boundary. The ordered pathway model is a set of evidence-qualified hypotheses. Direct enzyme characterization supports individual steps; a homologous sequence or transcript supports candidate function; a detected metabolite supports presence in the sampled material. None of these alone establishes flux, compartment, or in-vivo production in a second species. Multi-substrate requirements, physiological direction, stereochemistry, and missing transport or cofactor information must remain explicit in the Terpedia pathway records.

Figure 1. Conceptual route from precursor supply to phenotype.



4. Species and chemotype diversity

A. annua combines artemisinin-pathway chemistry with variable volatile oils that may include artemisia ketone, camphor, germacrene D, and 1,8-cineole. *A. absinthium* can show thujone-, camphor-, cineole-, davanone-, or chamazulene-associated profiles depending on provenance and stage. *A. herba-alba*, *A. vulgaris*, and *A. argyi* likewise show regional and tissue-level variation. Multi-species Canadian oils reinforce that composition can differ substantially among taxa and regions (Lopes-Lutz et al. 2008). *A. dracunculus* is a useful boundary case because estragole may dominate some oils; estragole is not a terpene and should remain in a broader volatile-metabolite record.

Preparation can redistribute the same plant's volatile evidence. In Serbian sweet wormwood, the essential oil was dominated by artemisia ketone and trans-caryophyllene, whereas the hydrosol concentrated the more water-soluble camphor, 1,8-cineole, and artemisia ketone; a synthesis of 61 accessions placed *A. annua* oils into four broad chemotype groups (Aćimović et al. 2022). This is a practical warning for malaria and wormwood comparisons: oil, hydrosol, tea, alcoholic extract, and dried leaf are different exposure matrices, not interchangeable labels for the same terpene dose.

Geography and climate can alter both the relative abundance and the dominant chemical class. Phenological shifts may change monoterpene and sesquiterpene proportions between vegetative and flowering stages. Drying and distillation can remove, transform, or concentrate constituents. These effects explain why percentage tables cannot be pooled without specimen-level metadata and analytical harmonization.

Developmental chemistry is not restricted to wormwood. In davana (*A. pallens*), a two-season, three-stage study found that oil content was higher at full emergence of the flower heads; davanone and linalool declined toward seed set, whereas methyl/ethyl cinnamates, bicyclogermacrene, davana ether, 2-hydroxyisodavanone, and farnesol increased (Mallavarapu et al. 1999). These trajectories make harvest stage and season necessary covariates in cross-species volatile comparisons, but they do not establish antiparasitic activity for any changing constituent.

Genetic and environmental components can be partially separated but rarely ignored. A linkage map identified loci affecting artemisinin yield in *A. annua*, providing direct evidence that segregating genetic variation contributes to the phenotype (Graham et al. 2010). In contrast, salt exposure, phytohormone treatment, reproductive development, and methyl jasmonate each changed pathway or oil phenotypes within experimental genotypes (Yadav et al. 2017; Maes et al. 2011; Arsenault et al. 2010; Wu et al. 2011). The DBR2-promoter study further linked promoter activity to high- and low-artemisinin chemotypes, showing how cis-regulatory variation at a branch point can contribute to pathway allocation (Yang et al. 2015).

Quantitative genetics adds an older but useful within-species benchmark: artemisinin content in *A. annua* showed predominantly additive inheritance and high narrow-sense heritability, and selected hybrid lines reached up to 1.4% of dry-leaf mass (Delabays et al. 2001). This supports selection as a source of reproducible chemotype variation, while leaving causal loci, linkage drag, genotype-by-environment stability, and terpene-network consequences unresolved.

Abiotic-stress studies extend this interaction beyond elicitor shorthand. Salt, cold, and waterlogging increased reported artemisinin content in one *A. annua* experiment, whereas drought did not, and each treatment produced a distinct transcriptome (Vashisth et al. 2018). In a separate two-variety experiment under arsenic stress, carrageenan oligomers and salicylic acid further changed antioxidant physiology and artemisinin yield (Naeem et al. 2021). These studies support environment-by-genotype regulation hypotheses, not universal stress responses or production recommendations; contamination, elicitor dose, harvest timing, and non-target metabolites must be measured alongside artemisinin.

Direct trichome phenotyping places a useful limit on stress-based explanations. Across field and greenhouse experiments on full-grown *A. annua*, sandblasting, leaf cutting, jasmonate, salicylate, chitosan, hydrogen peroxide, and salt produced mostly minor effects on glandular-trichome density; responses depended on clone and leaf position, and the hypothesis that stress generally induces larger or more numerous trichomes was rejected (Kjær et al. 2012). Thus, a stress-associated increase in artemisinin cannot be assumed to arise from more trichomes. Comparative experiments should measure trichome initiation, maturation, secretory capacity, pathway transcripts, and metabolites separately.

The same principle applies to apparently stable markers. Germacrene D is dominant in many *A. frigida* samples, yet the distribution of that dominance and its relationship to climate must be evaluated across populations and years. Recent *A. absinthium* work similarly shows that geography and phenological stage can change the dominant oxygenated mono- versus sesquiterpene class. These studies support stratified comparisons rather than a single canonical “wormwood oil.”

Post-harvest handling is part of the phenotype that reaches an assay. Controlled drying changed concentrations of artemisinin and its acid precursors in *A. annua* leaves (Ferreira and Luthria 2010). Hydrodistillation, solvent extraction, headspace analysis, and direct tissue extraction sample different chemical spaces and may also create or remove products through heat, oxidation, hydrolysis, or volatility. A reproducible comparison must therefore treat extraction method, storage duration, temperature, moisture endpoint, and abundance denominator as model variables rather than descriptive footnotes.

The oil fraction itself is sensitive to drying conditions. In wild Iranian *A. annua* harvested at full bloom, increasing drying temperature from room temperature to 65 °C reduced hydrodistilled oil yield from 1.12% to 0.37%; monoterpenes progressively declined as sesquiterpenes increased, and the dominant profile shifted from artemisia ketone/1,8-cineole toward beta-caryophyllene/germacrene D (Khangholi and Rezaeinodehi 2008). This is a processing effect within one sampled population, not evidence for a universally superior drying regime or a change in antimalarial potency. It does, however, make drying temperature a necessary covariate in any comparison of volatile chemistry or wormwood preparations.

Developmental series in *A. annua* make the confounding concrete: artemisinin and related sesquiterpenes were highest at particular vegetative stages, while the essential-oil mono:sesquiterpene ratio shifted from approximately 55:45 near the preflowering maximum to approximately 30:70 at other stages (Woerdenbag et al. 1994). A separate two-genotype time series identified genotype- and stage-dependent marker sets that included artemisinic acid, arteannuin B, borneol, beta-farnesene, camphor, and lanceol (Wang et al. 2009). In a broader Pakistan comparison of eight *Artemisia* species and two varieties, flowering-stage leaves generally had higher artemisinin, with the highest measured value in *A. dracunculus* var. *dracunculus* (Mannan et al. 2011). This is useful for a cross-species harvest hypothesis, but the HPLC endpoint and field/geographic design do not establish a universal optimum for volatile terpenes or all *Artemisia* taxa. Thus, “chemotype” should be modeled jointly with development and genotype rather than treated as a static label.

A five-species comparison provides a useful cross-species test of this principle: variation in glandular-trichome density, artemisinin content, and expression of pathway genes was observed among *A. annua*, *A. deserti*, *A. absinthium*, *A. sieberi*, and *A. khorassanica* (Salehi et al. 2018). This is stronger comparative evidence than extrapolating from a single *A. annua* genotype, but expression and metabolite association still require common-annotation orthology, enzyme assays, and matched specimen chemistry before they can be interpreted as evolutionary gain or loss.

Contrasting high- and low-artemisinin *A. annua* chemotypes show why pathway intermediates must be measured alongside the end product. NMR-based profiling characterized more than 80 natural products and found that the low-artemisinin chemotype, with low *DBR2* expression, accumulated artemisinic acid, arteannuin B, and other unsaturated amorpha-4,11-diene derivatives, whereas the high-artemisinin chemotype accumulated more dihydroartemisinic acid, artemisinin, and related saturated products (Brown et al. 2018). The contrast supports branch-point and terminal-conversion hypotheses, but it is not an antiparasitic assay and does not prove that DBR2 alone determines field yield or parasite potency.

An independent Japanese accession study reinforces this seasonal boundary: wild-derived *A. annua* grown in planters showed seasonal and positional variation in artemisinin, artemisinic acid, arteannuin B, and artemisitene, with leaf artemisinic acid reaching 1.0% of dry weight (Kawamoto et al. 1999). This is a useful accession-level comparator for precursor-rich chemotypes, but it does not define a universal harvest optimum or separate photoperiod, temperature, developmental, and genotype effects.

Regional production studies add a useful origin-by-chemistry layer. In Ugandan material, highland plants contained approximately 0.8% artemisinin and 2.6% total flavonoids versus 0.4% and 1.5% in lowland material, while camphor dominated relatively similar aromatic-oil profiles across regions (Engeu et al. 2015). The study supports source-specific standardization, but its component associations do not prove flavonoid or volatile synergy. Developmental regulation can be equally large: three-month leaves showed higher artemisinin than two-week leaves alongside miR156/SPL changes and increased *ADS*, *CYP71AV1*, *ADH1*, *DBR2*, and *ALDH1* transcripts (Li et al. 2021). These data make age, leaf position, and trichome state necessary covariates rather than interchangeable labels for "high-artemisinin" material.

Breeding and propagation can stabilize a selected chemotype without demonstrating a natural evolutionary mechanism. Selected *A. annua* genotypes exceeded 2% artemisinin and clonally propagated field material improved uniformity, with an estimated productivity of up to 70 kg/ha in the reported system (Wetzstein et al. 2018). A cross-species SCAR-marker study likewise proposed a roughly 1-kb marker associated with higher HPLC artemisinin in selected *A. annua* and *A. absinthium* samples (Asghari et al. 2015). Both are useful selection resources, but marker transfer, pathway integrity, genotype-by-environment stability, and parasite potency require independent validation.

4.1 *Artemisia argyi* as a tissue- and time-resolved comparator

The expanding *A. argyi* literature provides a useful counterweight to *A. annua*. De novo and full-length transcriptomes identified candidate genes across the MVA/MEP backbones, TPS families, P450s, and regulatory families, but these resources differ in tissue composition, assembly strategy, and annotation granularity (Liu et al. 2018; Cui et al. 2021). Targeted terpenoid measurements paired with bHLH-family analysis nominated regulatory candidates without claiming direct enzyme function (Yi et al. 2021). A second reference genome, assembled at 4.15 Gb with 147,248 predicted genes and 5,121 reported species-specific gene families, paired genome analysis with MeJA-responsive transcriptomics of terpenoid, flavonoid, and phenylpropanoid pathways (Gao et al. 2024). Its unusually large gene-model count and repeat-rich assembly are valuable resources but also illustrate why gene-family counts must be normalized to assembly, ploidy, and annotation quality before evolutionary interpretation. The cytogenomic context is now more explicit. Genome-map-based karyotyping supports an allotetraploid interpretation for *A. argyi* and identifies a new octoploid germplasm class, with chromosome-doubling experiments affecting moxa-related traits (Li et al. 2025). This result does not prove terpene-gene dosage, but it changes the comparative design: homeolog retention, allele dosage, chromosome pairing, and cytotype must be modeled before a larger *A. argyi* TPS count is interpreted as lineage-specific innovation. In practical terms, diploid, tetraploid, and octoploid material should be analyzed with ploidy-aware assemblies and read-depth-normalized copy-number estimates, then linked to secretory-tissue expression and quantitative chemistry.

Tissue-resolved metabolomics and transcriptomics showed that roots, stems, and leaves differ in both terpene-associated metabolites and candidate pathway expression; leaf-enriched MEP-pathway genes and co-expressed P450s formed hypotheses for subsequent functional testing (Zhang et al. 2022). Temporal profiling across four harvest months reported changing volatile-oil levels and identified full-length transcripts correlated with monoterpene and sesquiterpene profiles (Xu et al. 2022). These studies demonstrate that tissue and harvest date can produce differences as large as those attributed to genotype.

Analytical harmonization can make those tissue differences testable. A targeted UPLC-QQQ-MS/MS method quantified amorpha-4,11-diene, artemisinic aldehyde, dihydroartemisinic acid, artemisinic acid, artemisinin B, artemisitene, and artemisinin in roots, stems, leaves, and lateral branches of four-month-old *A. annua* (Wang et al. 2024). The value of this record is methodological as much as chemical: simultaneous precursor/intermediate/product measurement can distinguish branch-point allocation from a terminal-product change. It does not establish in-vivo flux or antimalarial potency without isotopic, enzyme, and parasite-exposure data, but it provides a practical assay template for cross-species tissue comparisons.

An environmental perturbation study gives this tissue-and-genotype framework a mechanistic test. Low-phosphorus treatment increased caryophyllene in *A. argyi* volatile oil and fresh leaves, increased *AaTPS3* expression, and implicated *AaWRKY1* and *AaWRKY2* as negative regulators of *AaTPS3* (Ma et al. 2026). The result supports nutrient-responsive regulation of a defined terpene branch in the tested material, but it should not be generalized to all *A. argyi* accessions or interpreted as an antiparasitic mechanism without matched parasite and host-selectivity assays.

The East Asian essential-oil literature adds a useful review-level comparison. A synthesis of *A. argyi*, *A. princeps*, and *A. montana* describes oils containing monoterpenes, sesquiterpenes, and derivatives, while reporting that species-specific *A. argyi* constituents were often monoterpenoid, *A. princeps* had many species-specific aliphatic hydrocarbons, and *A. montana* was distinguished by sesquiterpenes (Yu et al. 2023). This supports geographic and taxonomic stratification of volatile sampling, but the review itself identifies unresolved mechanisms, toxicity, and quality control; it is not a substitute for voucher-linked primary chromatograms or a terpene-specific parasite assay.

Not every trichome study is a terpene study, and that distinction is experimentally useful. Integrated multi-omics in *A. argyi* nominated AaMYB1 and AaYABBY1 for flavonoid biosynthesis; overexpression increased flavonoid or rutin readouts, and AaYABBY1 increased trichome density in tobacco and *Arabidopsis* (Cui et al. 2024). These results provide a non-terpene developmental control for interpreting trichome-associated chemistry: a regulatory perturbation can change both cell number and a specialized-metabolite branch without proving a corresponding change in terpene flux or parasite activity.

Genome size is another source of apparent chemical variation. Flow cytometry across 42 Chinese *A. argyi* accessions measured 8.428–11.717 pg and separated the accessions into three clusters without a significant geographic relationship; a genome survey estimated approximately 7.852 Gb and provided heterozygosity, repeat, GC, and ploidy context (Luo et al. 2023). Because genome size can reflect chromosome dosage, repetitive DNA, and assembly representation, this result argues for cytotype-aware normalization before relating TPS abundance to chemistry. It does not itself demonstrate TPS expansion, pathway dosage, or adaptive antiparasitic function.

Organ and time are equally important environmental covariates. Dynamic cold-exposure and recovery experiments in *A. annua* found different transcription-factor and calcium-signaling responses in leaves and roots, disrupted then restored carbon transport from source leaves to sink roots, accelerated root-to-leaf nitrogen transport, and earlier activation of cold-response programs in leaves (He et al. 2024). The experiment supports organ-resolved sampling of terpene-associated stress responses; it does not establish that a cold-induced transcript predicts a particular volatile or sesquiterpene-lactone product, nor that the response is conserved across *Artemisia*.

An *A. argyi* soil-cadmium study provides a complementary stress boundary: plants maintained normal growth and intact cellular structure up to the reported 10 mg/kg treatment, while proline, cadmium translocation, and transcriptomic pathways involving terpenoid, flavonoid, amino-acid, and sugar metabolism changed with exposure (Yang et al. 2024). This is valuable for separating environmental transcriptome effects from genetic differences and for flagging contaminated-material risks, but pathway transcription is not native flux, adaptive fitness, product safety, or parasite efficacy.

The same design logic extends to non-medicinal ecological comparators. Drought and rewatering in *A. sphaerocephala* changed water status, oxidative-stress measures, soluble metabolites, and hormone-, carbohydrate-, glyoxylate-, and nitrogen-related transcript programs in a full-length transcriptome study (Xiao et al. 2025). Although the reported endpoints do not resolve a terpene product, they establish why drought history and recovery must be recorded when comparing arid-zone *Artemisia* accessions. A stress transcript is a sampling covariate and candidate regulator, not evidence of an antiparasitic molecule.

Single-cell transcriptomics and matched metabolomics now resolve this problem at higher spatial resolution. Glandular trichomes and non-glandular tissues of *A. argyi* showed strongly different metabolite profiles, with sesquiterpenoids prominent among trichome-enriched terpenoid differences, and the single-cell atlas nominated developmental and TPS-expression programs (Dong et al. 2026). This is powerful candidate evidence, but co-expression edges and cell-type enrichment remain hypotheses until recombinant enzymes, gene perturbations, and isotope or product-feeding experiments establish catalytic sequence and flux.

An *A. annua* single-nucleus and spatial-transcriptomic atlas provides a complementary cell-resolved reference for the cross-species design. The study defined three glandular-secretory-trichome developmental states and identified six secretory cells within a ten-cell trichome as the primary artemisinin-production site, together with hundreds of hub-gene candidates (Zhang et al. 2025). This sharpens the sampling requirement: bulk-leaf orthology should be paired with matched secretory-cell expression and metabolite localization. The atlas remains a candidate map, however; hub-gene function, protein abundance, isotope-traced flux, and conservation outside *A. annua* still require direct tests.

5. Comparative evolutionary genomics

The initial controlled comparison is *A. annua* versus *A. argyi*. The former supplies a pathway-rich reference genome and functional artemisinin literature. The latter has a chromosome-scale genome with reported whole-genome duplication and terpene-synthase expansion, plus tissue transcriptomes. The apparent ADS contradiction is now interpretable as an assembly-, ploidy-, annotation-, and function-level discrepancy rather than a simple presence/absence result. The earlier chromosome-scale study selected a 3.89-Gb, 17-pseudochromosome haplotype-A projection, annotated 122 TPS genes, and described partial ADS deletion and loss of function. A later 7.88-Gb haplotype-resolved assembly predicted 451 TPS models, of which 261 were manually classified as intact, and identified a six-copy tandem ADS homolog cluster; recombinant AarADS proteins produced α-bisabolol rather than the amorpha-4,11-diene made by AanADS (Chen et al. 2023; Qi et al. 2025). Thus, the later result supports ADS-family retention with functional divergence, while neither paper alone establishes an intact, expressed, artemisinin-producing pathway in *A. argyi*.

The genus-scale phylogenomic scaffold is now substantially better than the earlier marker-limited picture. Nuclear SNP data from 228 *Artemisia* species and 258 samples recovered eight supported clades, nested *Kaschgaria* within *Artemisia*, and showed that several traditional subgenera are not monophyletic (Jiao et al. 2023). This is the appropriate backbone for testing whether terpene traits are phylogenetically clustered, but it does not itself measure terpene genes or chemistry. Cytogenomic sampling adds a second constraint: flow cytometry across 67 species and 84 populations showed that genome-size increase is not proportional to ploidy and that 1Cx values tend to decline at high ploidy (Pellicer et al. 2010). Copy-number comparisons should therefore use cytotype-aware, read-depth-normalized assemblies rather than treating genome size as a direct proxy for TPS expansion.

Within-species plastome structure can improve accession sampling while remaining distinct from nuclear pathway evolution. Complete chloroplast genomes from 38 *A. annua* strains resolved two lineages and 14 haplotypes, with 60 polymorphic sites but no chloroplast gene-content presence/absence difference (Ding et al. 2024). This supports accession discrimination and a possible geographic sampling covariate for artemisinin studies; chloroplast haplotype structure is not evidence for nuclear ADS, CYP71AV1, DBR2, or volatile-TPS dosage or function.

Resource-level plastome data in *A. selengensis* show how this scaffold can be extended without overclaiming. Ten complete chloroplast genomes were highly conserved in gene content and order, spanning 151,148–151,257 bp and 133 genes, while variable regions and eight mutation hotspots improved phylogenetic discrimination among resources (Wang et al. 2024). A paired comparison of *A. maritima* and *A. absinthium* likewise reported approximately 151-kb plastomes, high structural similarity, and 20 candidate polymorphic marker regions, with a maximum-likelihood tree placing *Artemisia* as monophyletic and sister to *Chrysanthemum* (Shahzadi et al. 2020). These data can distribute accessions for common-garden chemistry and nuclear sequencing, but plastome markers cannot resolve nuclear TPS/OSC orthology, WGD-derived copy number, or enzyme function. The appropriate model treats plastid topology as a sampling covariate and tests terpene evolution with phased nuclear assemblies, reconciled gene trees, expression, and product assays.

Broader *Seriphidium* sampling reinforces the distinction between taxonomic scaffold and pathway evolution. Eighteen newly sequenced plastomes from 16 *Seriphidium* species, combined with a published taxon, showed conserved structure and eight variable loci but recovered the subgenus as polyphyletic (Jin et al. 2023). This result is useful for choosing representatives and testing whether wormwood-associated chemistry tracks a plastid-defined clade; it cannot establish nuclear TPS duplication/loss, sesquiterpene-lactone biosynthesis, or shared vermifuge chemistry. In particular, plastome-based non-monophyly should trigger taxonomic and sampling review, not a claim that parasite-active terpene pathways evolved convergently.

The comparative analysis should infer orthogroups and gene trees for GPPS, FPPS, monoterpene synthases, sesquiterpene synthases, ADS, CYP71AV1-like P450s, DBR2, ALDH1, OSCs, and selected sesquiterpene-lactone enzymes. Duplications and losses should be mapped to a broad *Artemisia* species tree. Expression should be interpreted by tissue and developmental context, with glandular trichome data prioritized where available. Missing annotation is unresolved data, not evidence of biological absence.

An older cross-species PCR screen illustrates the cost of inferring pathway completion from a single marker. Among ten *Artemisia* species, selected ADS active-site regions amplified only in *A. annua*, *A. aucheri*, and *A. chamaemelifolia*, while the complete gene was not amplified in any tested species (Hosseini et al. 2011). Negative amplification can reflect primer mismatch, indels, gene fragmentation, or missing template; positive amplification of a short motif cannot establish an intact enzyme. The result should therefore be reinterpreted as a historical candidate screen and replaced by long-read assemblies, common annotation, reconciled gene trees, expression, and recombinant product assays.

Ploidy provides a second, experimentally anchored confounder. Colchicine-induced tetraploid *A. annua* showed 39-56% higher artemisinin than diploid plants across the vegetation period, with increased expression of selected pathway genes including *FPS*, *HMGR*, and *Aldh1* (Lin et al. 2011). This supports a ploidy-associated production phenotype, but it does not distinguish dosage from altered development, trichome density, or regulation. The observation strengthens the case for cytotype-aware comparative sampling across *A. annua*, *A. argyi*, *A. tridentata*, and wormwood rather than interpreting raw gene counts as evolutionary innovation.

Plastid phylogenomics can provide a sampling scaffold, but it cannot replace nuclear pathway phylogenies. A 2025 comparative analysis of *Artemisia* plastomes addressed limited chloroplast sampling and inferred infra-generic relationships that can help distribute accessions across the genus (Guo et al. 2025). Those relationships are useful for choosing representatives and testing whether chemical traits are phylogenetically clustered; they do not establish nuclear TPS/OSC orthology, WGD-derived copy number, or functional evolution. The article therefore treats plastome topology as a taxonomic covariate and reserves gene-family conclusions for common nuclear annotations and reconciled gene trees.

Published comparative results already provide testable anchors: the later *A. argyi* analysis reports 451 predicted TPS models split between two haplotypes (221 and 230), with 261 intact models after manual correction, while the chromosome-scale study links WGD and duplication modes to TPS expansion and identifies a germacrene D synthase cluster. The latter study also reports pathway elucidation for germacrenes, (+)-borneol, and (+)-camphor. These observations are catalogued in the KB's gene-family-evidence.csv and ADS/TPS reconciliation table; they are not treated as a substitute for a common-annotation, orthology-aware rerun.

The transcript and metabolite literature provides a validation ladder for genomic predictions. A candidate duplicated TPS is more compelling when it is intact under a common annotation, placed within a supported gene tree, retained in synteny or a tandem cluster, expressed in the relevant secretory cell, and correlated with a product that the recombinant protein can generate. The *A. argyi* tissue, temporal, and single-cell studies supply several of these intermediate layers, but none independently establishes orthology or ancestral function. In *A. annua*, by contrast, the mature ADS/CYP71AV1/DBR2 evidence chain shows what eventual convergence should look like. A functional *A. argyi* example is beta-caryophyllene synthase: four candidate synthases were identified, and engineered yeast expressing AarTPS88 reached 15.6 g/L beta-caryophyllene in fed-batch fermentation (Li et al. 2025). This validates a production-capable enzyme/cell-factory route, not native leaf abundance, ecological adaptation, orthology, or antiparasitic action.

The genome-scale anchors also illustrate why "expansion" must be treated as a layered observation. The *A. annua* draft genome combined a 1.74-Gb highly heterozygous assembly, 63,226 predicted protein-coding genes, terpene-enzyme diversification, and transcriptomic evidence for a regulatory network associated with artemisinin biosynthesis. The chromosome-scale *A. argyi* study instead placed a recent lineage-specific WGD, duplicated TPS families, germacrene-D/(+)-borneol/(+)-camphor pathway evidence, and partial ADS deletion or homolog loss-of-function in a 3.89-Gb, 17-pseudochromosome context (Luo et al. 2018; Chen et al. 2023). The contrast is biologically informative, but the reported gene counts and ADS states are not a common-denominator orthology result: assembly scope, haplotype representation, annotation, ploidy, and product validation differ. These studies therefore motivate, rather than replace, the KB's common-reannotation and reconciled-gene-tree gates.

Regulatory evolution should be evaluated with the same caution as enzyme evolution. The abundance of MYB, bHLH, WRKY, AP2/ERF, and light- or jasmonate-responsive candidates in transcriptomic analyses may reflect both genuine network expansion and permissive co-expression thresholds. Functional studies of AaERF1/AaERF2, WRKY1, AaMYB108, AabHLH112, and the AaGSW1 complex define experimentally testable motifs and interactions in *A. annua*, but homologous regulators in *A. argyi* should not inherit those functions by name alone. Conserved target motifs, protein interactions, cell-type expression, perturbation phenotypes, and metabolite responses are required to claim network conservation.

The Terpedia KB now carries an execution specification for this analysis (comparative-genomics/). Its required outputs include frozen input checksums, software versions, orthology-aware gene trees, synteny/tandem-cluster calls, and a tiered gene-family evidence table. A completed two-proteome OrthoFinder run is now archived, but the broader genus-wide comparison remains a design-supported hypothesis framework because the input contains only *A. annua* and one *A. argyi* annotation release. As a bounded extension, six pathway-associated candidate groups were aligned with MAFFT and analyzed with IQ-TREE3 under LG+G4 with 1,000 ultrafast bootstrap and 1,000 SH-aLRT replicates. The groups were annotated in the *A. annua* reference as germacrene D synthase, germacrene A synthase, sesquiterpene synthases 1 and 6, CYP71AV1, and DBR2; they contained 4–9 *A. tridentata*, 1–4 *A. argyi*, and 1–6 *A. annua* leaves per group. These trees are exploratory candidate-family diagnostics: they do not establish orthology, duplication/loss, ancestral state, enzyme function, pathway flux, or antiparasitic mechanism. The exact alignments, trees, manifests, software versions, and input hashes are archived in comparative-genomics/results/three-species-pathway-gene-trees-2026-09-02/. AabHLH2 and AabHLH3 provide a regulatory anchor for this pathway-centered comparison: in *A. annua* they acted as nuclear repressors that competed with AaMYC2-like cis-regulatory recognition and negatively regulated artemisinin biosynthesis (Shen et al. 2022). This result makes bHLH copy, sequence, binding-site, and cell-type comparisons testable, but a homologous gene name is not evidence of conserved regulation in *A. argyi*, *A. tridentata*, or any other species. A separate engineering study shows that the genus can serve as a platform for non-native C15 chemistry: farnesyl diphosphate synthase and patchoulol synthase expression in *A. annua* produced patchoulol, and signal-peptide targeting increased the reported accumulation to 273 microgram/g dry weight (Fu et al. 2020). This is evidence for engineered production and compartment targeting, not native patchoulol ecology, pathway evolution, or antiparasitic activity.

An NCBI Datasets metadata audit was run without downloading sequence archives. It returned one *A. annua* assembly report and three separate *A. argyi* reports. The literature resource PRJCA010808 was resolved to its external NGDC/CNCB project record, not to an NCBI assembly accession; the published 3.89-Gb, 17-pseudochromosome haplotype must therefore be treated as a distinct input until sequence and annotation files are obtained and checksummed. This is itself a result for reproducibility: assembly choice must be documented before copy-number or presence/absence comparisons are interpreted.

The KB's assembly comparison table makes the confounders explicit: the *A. annua* NCBI reference is scaffold-level, the literature *A. argyi* input is a 3.89-Gb haplotype-A projection, and NCBI also exposes larger or differently assembled *A. argyi* resources. Genome length and chromosome count should therefore not be interpreted as evidence for terpene-family expansion without a common annotation and orthology workflow.

The execution gate is also explicit: NCBI reports an annotation with 63,226 protein-coding genes for the *A. annua* reference, and the earlier NGDC/CNCB *A. argyi* haplotype-A annotation exposes 62,844 protein-coding genes. These counts are not directly comparable to the later 7.88-Gb haplotype-resolved annotation. A seed-centered DIAMOND screen of 31 annotated *A. annua* pathway/TPS queries returned 105 *A. argyi* candidate hits; the exact inputs and hashes are archived in the KB. These are homology candidates only. A cross-species copy-number or orthology result is not reported until a common annotation, reciprocal/phylogenetic validation, and gene-tree reconciliation are completed.

Reciprocal checking re-searched 67 unique *A. argyi* candidates against the *A. annua* proteome and produced 26 reciprocal-best-hit candidate rows. Because a duplicated *A. argyi* locus can share the same best *A. annua* seed, this count is not a one-to-one ortholog count and is not interpreted as gene-family expansion.

As a diagnostic next step, the KB built MAFFT alignments and IQ-TREE3 exploratory trees for the three reciprocal groups that retained at least four unique protein sequences (PWA60940.1, PWA68009.1, and PWA85324.1). The run used the LG+G4 model, 1,000 ultrafast-bootstrap replicates, 1,000 SH-aLRT replicates, and one thread; the exact alignments, tree files, reports, software versions, and input checksums are archived under `comparative-genomics/results/seed-gene-trees-2026-09-01/`. Eight of eleven reciprocal groups were not tree-built because they had fewer than four leaves or fewer than four unique sequences. These small, seed-biased trees are quality-control diagnostics only: they do not establish orthology, duplication/loss, or pathway function.

A restricted OrthoFinder diagnostic then clustered the 31 *A. annua* seed proteins and 67 unique *A. argyi* forward-hit candidates. It produced 16 candidate orthogroups, 11 shared clusters, and five *A. argyi*-only clusters. The full tables, input FASTAs, software context, and checksums are archived in `comparative-genomics/results/candidate-orthofinder-2026-09-01/`. This result is useful for prioritizing follow-up gene trees and synteny, but the seed-biased, two-taxon candidate set cannot support genome-wide orthology, copy-number, duplication/loss, or gene-function claims.

To broaden pathway prioritization without overstating the unfinished full-proteome run, a separate annotation-keyword and homology-filtered analysis selected 890 *A. annua* pathway-associated proteins and 1,273 unique *A. argyi* candidates. OrthoFinder `-M msa -og` grouped these filtered inputs into 388 candidate orthogroups: 271 shared, 17 *A. annua*-only, and 100 *A. argyi*-only; 83 groups were single-copy in both filtered inputs. Because the query set is annotation-biased, the *A. argyi* set is homology-selected, and `-og` stops before species-tree and gene-tree inference, these counts are a reproducible prioritization scaffold rather than evidence of genome-wide orthology, copy-number change, duplication/loss, ancestral state, enzyme function, pathway flux, or antiparasitic mechanism. The complete tables, group sequences, command, software versions, input hashes, and checksums are archived in `comparative-genomics/results/pathway-focused-orthofinder-2026-09-02/`.

The full two-proteome run was then completed by resuming the finished DIAMOND searches with native ARM64 MCL and one OrthoFinder analysis thread (Emms and Kelly 2019). It assigned 111,235 of 129,761 proteins (85.7%) to 28,696 orthogroups, including 25,039 groups containing both species, 7,086 single-copy groups, 3,657 species-specific groups, and 18,526 unassigned proteins. The inferred tree, `(A. annua, A. argyi)`, is a valid two-tip species tree but contains no information about relationships among the wider genus. The complete compressed result, log, input checksums, and execution note are archived in `comparative-genomics/results/artemisia-of-results-run3-full-2026-09-02.tar.gz` and `comparative-genomics/results/full-orthofinder-2026-09-02.md`.

Two published genome-scale studies provide important context for interpreting that restricted two-proteome comparison. An integrated *A. annua* multi-tissue dataset identified 88 metabolites across 14 tissues—26 monoterpenes, 22 sesquiterpenes, two triterpenes, and one diterpene—and connected developmental expression patterns with pathway-associated genes (Ma et al. 2015). In *A. argyi*, an earlier 8.03-Gb allotetraploid assembly reported 135 TPS genes, tandem clusters, and tissue-biased expression alongside broad secondary-metabolism expansion (Miao et al. 2022). These studies support a class-balanced gene-to-metabolite design, but their assembly sizes, ploidy interpretations, annotation releases, and gene models are not directly interchangeable with the later 3.89-Gb or 7.88-Gb *A. argyi* resources. TPS counts and gene-expansion claims must therefore be re-evaluated after common annotation, homeolog-aware gene-tree placement, synteny, expression, and product confirmation.

An allele-aware *A. annua* assembly provides a complementary within-species evolutionary anchor. Two chromosome-level haplotype maps, paired with resequencing of 36 samples spanning artemisinin content, found a positive association between *amorpha-4,11-diene synthase* (ADS) copy number and artemisinin yield and enabled paralog-aware transcript detection (Liao et al. 2022). This supports ADS dosage as a testable contributor to chemotype variation in *A. annua*, but it does not establish that every ADS-like copy is intact, expressed, or catalytically equivalent. In comparative work, copy number should therefore be modeled jointly with haplotype, gene-tree placement, transcript abundance, trichome localization, and measured artemisinin or precursor levels.

An annotation-anchored extraction mapped 35 *A. annua* proteins carrying curated pathway/TPS header labels to 21 pathway-label/group combinations, representing 17 assigned orthogroups plus one *A. annua* protein that remained unassigned. Several groups contain many copies in both species, but this is not evidence of a lineage-specific expansion: the labels are *A. annua* annotation anchors, *A. argyi* members have not inherited function by name, and OrthoFinder duplication counts are not equivalent to intact, expressed, biochemically active terpene loci. One particularly important caution is that the CYP71AV1 and germacrene-A-oxidase anchors fall in the same broad P450 orthogroup, illustrating why family membership must be followed by gene-tree placement and biochemical validation. The full per-protein mapping and grouped summary are archived in `comparative-genomics/results/full-pathway-family-mapping-2026-09-02.tsv` and `comparative-genomics/results/full-pathway-group-summary-2026-09-02.tsv`.

Annotation quality is itself an evolutionary-comparison variable. Manual correction of the *A. annua* genome annotation using second- and third-generation transcript evidence changed gene structures, alternative-splicing calls, expression estimates, and functional predictions (Liu et al. 2023). A common annotation pipeline should therefore be treated as a prerequisite for comparing TPS, P450, DBR2, ALDH, and OSC copy number; missing or mis-modeled loci are unresolved data rather than biological absences. The plant-to-yeast literature provides a complementary production context: engineered *E. coli* and *S. cerevisiae* systems, plant pathway engineering, and fermentation optimization have moved artemisinin precursor production from plant tissue toward microbial chassis, with reported artemisinic-acid titers reaching 25 g/L in yeast (Zhao et al. 2022). These outputs demonstrate engineering feasibility and supply-chain relevance, not native *Artemisia* evolution, plant abundance, or parasite efficacy.

An important third genome-scale context is *A. tridentata*, a North American sagebrush with a haploid pseudo-chromosome assembly of approximately 4.2 Gbp, 43,377 predicted genes, 91.37% complete BUSCOs, and 77.99% repetitive sequence (Melton et al. 2022). A comparative resequencing study found extensive chromosomal rearrangements relative to *A. annua*, conserved large-scale synteny in selected regions, and population-level transposable-element differentiation among three diploid *A. tridentata* genomes (Melton et al. 2024). More recent cytological and reduced-representation sequencing data support both one-step and two-step whole-genome-duplication pathways, including a possible triploid bridge and a reported 0.22% polyploidization rate in the study material (Grossfurthner et al. 2026). These records materially strengthen the evolutionary design by showing that genome architecture and cytotype can vary within the genus, but they do not establish *A. tridentata* terpene-locus function or parasite activity. The NCBI assembly has no NCBI protein annotation, so the accession-matched Maker GFF3 released with the genome study was translated locally and retained with its source genome URL, assembly-name map, derivation script, and checksums in the Terpedia KB. This makes *A. tridentata* usable in a bounded candidate-family diagnostic while preserving the common-reannotation limitation.

The resulting three-species diagnostic searched the same 890 *A. annua* pathway-associated queries against the frozen *A. argyi* proteome and the 41,700-protein *A. tridentata* translation, recovering 1,273 and 768 candidate homologs, respectively. OrthoFinder `-S diamond -M msa -og` grouped the filtered inputs into 478 candidate orthogroups, of which 205 contained all three species and 31 were single-copy in all three. Pairwise shared-group counts were 291 for *A. annua*–*A. argyi*, 236 for *A. annua*–*A. tridentata*, and 265 for *A. argyi*–*A. tridentata*. These values are useful for selecting loci for gene trees, synteny, expression, and enzyme assays, but the query and target sets are homology-selected, annotations are not common-reannotated, and `-og` stops before gene-tree/species-tree reconciliation. They therefore do not support genome-wide copy-number, duplication/loss, ancestral-state, pathway-flux, or antiparasitic-mechanism claims. The full candidate tables, filtered FASTAs, OrthoFinder log, provenance manifest, and checksums are archived in `comparative-genomics/results/three-species-pathway-orthofinder-2026-09-02/`.

At broader genus scale, flow cytometry across 21 *Artemisia* species and 24 populations measured 2C DNA contents from 3.5 to 25.65 pg, with strong associations to karyotype length and ploidy and additional systematic/ecological structure (Torrell and Vallès 2001). Genome size is therefore a relevant covariate for sampling and orthology interpretation, especially where homeolog retention and chromosome-scale duplication can mimic lineage-specific terpene-gene expansion. It is not, however, a proxy for TPS copy number, pathway completeness, metabolite abundance, or antiparasitic function.

Within *A. annua*, induced autotetraploids provide a more direct ploidy perturbation. A transcriptomic comparison reported higher mean artemisinin in tetraploids, altered glandular-secretory-trichome morphology, and 8,763 differentially expressed genes involving metabolism, defense, hormone response, and methylation (Xia et al. 2018). This supports a ploidy-associated chemotype hypothesis, but the induction background, biomass tradeoff, and causal gene set require independent validation and should not be generalized to natural polyploids or other species.

Table 4. Selected annotation-anchored groups from the completed two-proteome run. Counts are OrthoFinder group membership and duplication diagnostics, not functional copy-number calls.

A. annua annotation anchor	Orthogroup	*A. annua* / *A. argyi* genes	Duplications	Gene-tree leaves	Boundary
CYP71AV1; germacrene-A oxidase	OG0000295	19 / 8	21	27	Broad P450 group; shared group does not distinguish CYP71AV1 from GAO function

A. annua annotation anchor	Orthogroup	*A. annua* / *A. argyi* genes	Duplications	Gene-tree leaves	Boundary
DBR2; artemisinic-aldehyde reductase labels	OG0006177	3 / 1	2	4	Compact shared group; exact enzyme assignment requires sequence/function checks
Germacrene-A synthase	OG0011988	2 / 1	not reported	not built	Three total members fell below the gene-tree output threshold
Germacrene-A synthase 2	OG0001344	5 / 6	7	11	Candidate TPS family; *A. argyi* function unverified
Germacrene-D synthase; sesquiterpene synthase 4	OG0000309	17 / 10	19	27	Mixed TPS labels and duplication-rich group; no product transfer inferred
Farnesyl diphosphate synthase 2	OG0000439	16 / 5	16	21	Prenyl-transferase group; not a direct flux or abundance comparison
Oxidosqualene cyclase 3	OG0001412	9 / 1	8	10	Candidate triterpene branch group; *A. argyi* member remains uncharacterized
Geranylgeranyl pyrophosphate synthase	OG0009299	1 / 3	2	4	Shared prenyl-transferase group; copy number is annotation- and assembly-dependent

An independent *A. argyi* leaf, root, and stem transcriptome supplies candidate terpenoid-biosynthesis transcripts and tissue contrasts (Liu et al. 2018). It extends the comparison beyond *A. annua*, but de novo transcript annotation cannot distinguish paralogs, homeologs, and allelic copies reliably enough for gene-family evolution without genome-guided placement. Experimental work on an *A. annua* (E)-β-farnesene synthase further shows that a sparse set of epistatic sequence changes can alter terpene-enzyme behavior (Ballal et al. 2020); this motivates testing sequence-function effects within orthologous clades rather than treating all TPS copies as functionally equivalent.

Environmental and perturbation studies provide useful mechanistic tests within *A. annua*. Short-term UV-B exposure increased artemisinin production and induced ADS, CYP71AV1, and stress/hormone-associated transcripts (Pan et al. 2014). A CYP71AV1 mutant redirected flux toward the alternate sesquiterpene epoxide arteannuin X and supported nonenzymatic terminal conversion in glandular trichomes (Czechowski et al. 2016). These results demonstrate inducibility and metabolic plasticity, but they do not establish that a homologous response or alternate product occurs in another *Artemisia* species.

Recent within-species studies add actionable controls for this comparative design. In a 17-bioecotype panel, gene-specific STS markers and HPLC identified marker-yield associations and bioecotypes exceeding 2.7 mg/g artemisinin, but the correlations remain selection signals rather than causal variant assignments (Hallajian et al. 2026). AaSPATULA perturbation connected glandular-trichome initiation with biomass and whole-plant artemisinin yield, while AaMYC3 linked jasmonate response, trichome density, and pathway-gene activation (Yuan and Zhou 2026; Yuan et al. 2025). TLR3 and differentially expressed lncRNAs provide additional plant-side regulatory candidates (Dong et al. 2024; Ma et al. 2024). The common inference boundary is important: improved trichome capacity or artemisinin yield does not, by itself, predict antiparasitic potency unless the resulting chemical fingerprint and parasite exposure are measured.

The older regulatory literature provides a useful perturbation ladder for interpreting these newer candidates. TAR1 altered trichome morphology, cuticular wax, ADS/CYP71AV1 expression, and artemisinin; trichome-specific AaWRKY1 increased CYP71AV1 and artemisinin without changing several upstream transcripts; and AabHLH1, AaSPL2, and AabZIP9 supplied promoter-binding or transactivation evidence at ADS, CYP71AV1, or DBR2 (Tan et al. 2015; Han et al. 2014; Ji et al. 2014; Lv et al. 2019; Shen et al. 2019). DBR2 overexpression increased both artemisinin and competing-branch metabolites, showing why pathway engineering should be evaluated by a complete metabolite panel rather than target-product abundance alone (Yuan et al. 2015).

Tissue and environment can uncouple a genomic prediction from a harvested phenotype. Bulk-tissue measurements found the highest artemisinin in leaves but distinct patterns for upstream and pathway genes across leaves, flowers, roots, and stems (Xiang et al. 2012); imaging and LC-MS later detected pathway products in non-glandular cells and glandless material (Judd et al. 2019). Mycorrhizal colonization, ABA rescue under copper stress, and combined UV-B/GA treatment each increased artemisinin-associated traits through partially overlapping stress and hormone networks (Mandal et al. 2015; Zehra et al. 2020; Ma et al. 2020). In field germplasm, ART and dihydroartemisinic acid often peaked before flowering and differed among Brazilian, Chinese, and Swiss chemotypes, whereas forced early flowering did not itself increase artemisinin (Ferreira et al. 2018; Wang et al. 2007). These findings support a model in which copy number, cell type, developmental state, environment, and precursor competition jointly determine chemistry.

Negative comparative controls are especially valuable. Eighteen *A. khorassanica* populations included diploids and tetraploids, yet neither class contained detectable artemisinin; DBR2 was low in diploids and silenced in autotetraploid material (Hamidi et al. 2018). This result constrains the inference that ploidy or expression of an ADS-like gene is sufficient for an artemisinin phenotype. In the proposed cross-species analysis, absence of artemisinin should therefore be modeled alongside sequence integrity, expression, tissue localization, precursor abundance, and assay detection limits.

6. Antiparasitic evidence

Two parasite-facing stories deserve priority in this review. Malaria is the strongest mechanistic case because artemisinin has a chemically defined scaffold, a reconstructed *A. annua* pathway, extensive parasite pharmacology, and a clinically established artemisinin-based combination-therapy context. Wormwood (*A. absinthium*) is a separate, important veterinary anthelmintic case: traditional vermifuge use is supported by several extract, oil, feed, and sheep studies, but the active chemistry, dose standardization, and human safety remain unresolved. The evidence should be developed in parallel rather than merged under the generic label "antiparasitic *Artemisia*."

6.1 Terpene-protein interaction map

Vector-control evidence provides a preparation-level bridge between terpene chemistry and parasite transmission. Nanoliposomes carrying essential oils from *A. annua*, *A. dracunculus*, and *A. sieberi* showed species-dependent larvicidal activity against *Aedes aegypti* and the malaria vector *Anopheles stephensi*, with the *A. dracunculus* formulation most active in the reported comparison (Sanei-Dehkordi et al. 2023). This supports formulation- and oil-specific vector control, but not malaria treatment, a single-terpene mechanism, or non-target safety. A newer defined-constituent study similarly found fumigation toxicity of *A. annua* oil, 1,8-cineole, and camphor against adult *Tetranychus urticae*, accompanied by inhibition of esterase, GST, and P450 detoxification enzymes (Nasr Azadani et al. 2025). The result strengthens the case for matched oil/monoterpenoid assays while remaining an invertebrate acaricide phenotype rather than parasite-protein engagement. An older *A. annua* larvicide comparison extends this vector-control evidence to *Culex*. Chloroform, methanol, ethanol, and acetone leaf extracts, together with purified artemisinin, killed late-stage larvae of *Culex quinquefasciatus* and *Culex tritaeniorhynchus*; the chloroform extract was more potent than artemisinin in the reported LC50/LC90 comparisons (Sharma et al. 2012). The result supports an extract-versus-compound vector phenotype, but it does not identify a causal terpene, establish malaria treatment, or provide non-target safety. Process chemistry is similarly preparation-specific: pressurized cyclic solid-liquid extraction recovered artemisinin and scopoletin from *A. annua* more efficiently and rapidly than traditional maceration in the tested material (Zarrelli et al. 2019). Higher extraction yield should not be equated with higher biological efficacy or clinical suitability without stability, composition, exposure, and assay validation.

The new Terpedia interaction map separates target engagement from target hypotheses (`parasite-protein-interactions.csv`; extraction rules in `protein-interaction-protocol.md`). The most experimentally anchored case is artemisinin against *Plasmodium falciparum*: haem activation supports covalent labeling of a broad parasite protein set, with 124 reported targets and orthogonal validation of selected metabolic and structural proteins, including OAT, PyrK, LDH, SpdSyn, SAMS, plasmepsins, MSP1, and actin (Wang et al. 2015). This is a promiscuous, chemically activated interaction model, not evidence for one universal artemisinin receptor.

Resistance should also be modeled as a dynamic parasite state rather than as an intrinsic label on a plant preparation. Intraerythrocytic oxidative-stress conditions significantly reduced artemisinin and dihydroartemisinin efficacy in matched *P. falciparum* assays, while heat shock altered the response to lumefantrine rather than artemisinin (Pires et al. 2024). A complementary chemical-genetic study showed that early unfolded-protein-response signaling and proteasome-mediated protein degradation determine DHA survival, with proteasome perturbations increasing susceptibility independently of *Kelch13* genotype (Rosenthal et al. 2024). These findings strengthen the parasite-side resistance framework and define a critical experimental interaction for the plant comparison: a chemotype or fraction should be tested under matched parasite redox/proteostasis states before an apparent difference is attributed to terpene composition. Neither study establishes that an *Artemisia* constituent causes or reverses resistance.

Additional malaria studies provide mechanistic anchors at different evidence levels. DHA inhibited immunopurified PfPI3K at nanomolar concentration, and PfK13 resistance biology was linked to altered PfPI3K/PI3P regulation; a mammalian-kinase counter-screen supported parasite selectivity in that experiment (Mbengue et al. 2015). Artemisinin binding to recombinant PfTCTP was mapped by surface plasmon resonance, mass spectrometry, and structural analysis, although the contribution of this interaction to parasite killing remains unresolved (Eichhorn et al. 2013). Functional oocyte and parasite-imaging experiments supported iron-dependent PfATP6 inhibition, but later multi-target evidence argues against treating PfATP6 as the sole receptor (Eckstein-Ludwig et al. 2003). Separately, DHA caused protein damage and compromised parasite proteasome function, producing proteostasis stress (Bridgford et al. 2018). A stage-resolved photoaffinity study reported covalent and noncovalent target classes associated with protein synthesis, glycolysis, and oxidative homeostasis (Gao et al. 2024); cryo-EM and MS-CETSA subsequently showed that artemisinin treatment increases PfRACK1 occupancy on Pf80S ribosomes (Go et al. 2025). Together these records favor a multi-target, stage- and activation-dependent model; they do not make PfPI3K, the proteasome, PfTCTP, PfATP6, or PfRACK1 a sole mechanism.

Computational chemistry adds a hypothesis layer to this malaria mechanism rather than a new level of target proof. Docking and three-dimensional QSAR studies modeled artemisinin-analogue interactions with hemin and related antimalarial activity, while a later DFT/QSAR analysis related electronic descriptors and peroxide reactivity to activity against chloroquine-resistant *P. falciparum* (Cheng et al. 2002; Rajkhowa et al. 2013). These studies can prioritize analogues and testable activation hypotheses, but docking poses, calculated descriptors, and model fit do not establish parasite target engagement, causal heme binding in vivo, or a therapeutic window. They are therefore stored in the KB as E2 computational evidence, separate from direct binding, recombinant-enzyme, and activity-based proteomics records.

The target-class signal is not restricted to *Plasmodium*. In *Toxoplasma gondii*, artemisinin inhibited TgSERCA-associated Ca2+-ATPase activity in a heterologous yeast system and triggered calcium-dependent microneme secretion and calcium perturbation in parasites (Nagamune et al. 2007). This supports a conserved parasite calcium-homeostasis vulnerability, but it is functional target-class evidence rather than a purified artemisinin-affinity measurement. On the plant-chemistry side, bioassay-guided fractionation of *A. afra* identified 1alpha,4alpha-dihydroxybishopsolicepolide and yomogiartemin as principal gametocytocidal sesquiterpene-lactone leads, including activity against asexual parasites for the former (Moyo et al. 2019). The result provides a stronger constituent-to-stage phenotype bridge than a crude infusion, while leaving parasite target engagement, host selectivity, and in-vivo transmission effects unresolved.

A plant-side perturbation illustrates why trichome chemistry and parasite phenotype must be measured together. In *A. annua*, calcium/magnesium silicate applied at 400 kg ha-1 increased glandular-trichome size and leaf/infusion artemisinin concentration, but the resulting infusion did not impair *T. gondii* growth more effectively than infusion from untreated plants; higher silicate doses instead reduced artemisinin concentration (Rostkowska et al. 2016). The result is a controlled dissociation between a plant chemical endpoint and a host-cell parasite assay. It supports environment-by-trichome-by-matrix experiments, not a claim that more artemisinin or larger trichomes necessarily produce greater activity in a non-*Plasmodium* system.

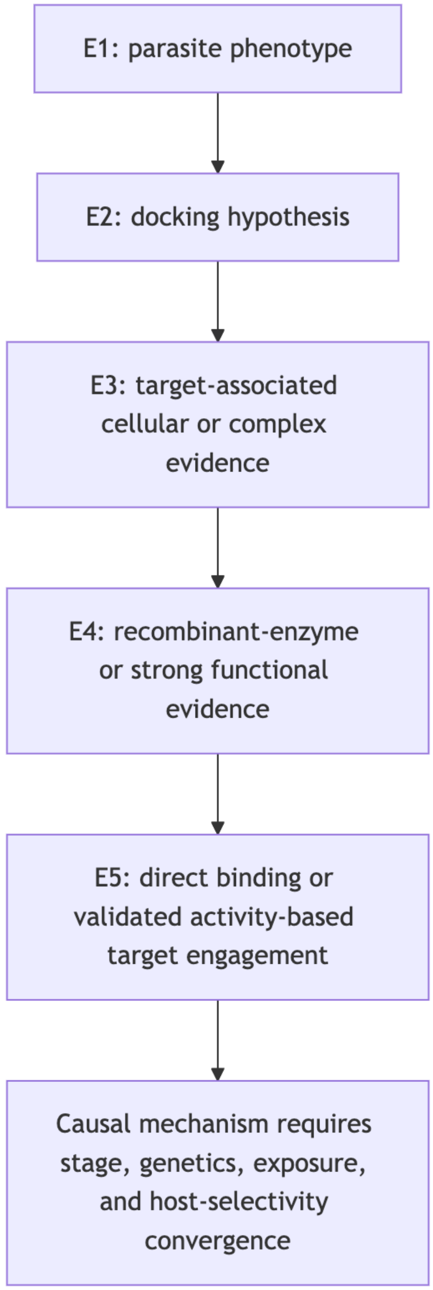
Evidence is weaker outside malaria. A 123-compound sesquiterpene docking study proposed *Leishmania* PTR1, carbonic anhydrase, NMT, and trypanothione synthetase as binding hypotheses; xanthanolides 75-76 had the strongest reported predicted PTR1 scores (-10.6 kcal/mol), but this is E2 docking evidence and the compounds are not established as *Artemisia* constituents in that record (Mishra et al. 2014). In *T. brucei*, recombinant-enzyme assays strengthened the case for target-class activity: cnicin inhibited TbPTR1, while cynaropicrin inhibited both TbPTR1 and TbDHFR, with IC50 values of 12.4 and 7.1 μM, respectively; both compounds came from non-*Artemisia* Asteraceae sources (Possart et al. 2021). A *Leishmania major* study provides a more direct *Artemisia*-provenance STL bridge: four purified eudesmane/elemane lactones from *Seriphidium khorassanicum* (syn. *A. khorassanica*) inhibited amastigote-infected macrophages at 4.9-25.3 μM, while normal-macrophage CC50 values were 432.5-620.7 μM; the strongest compound had IC50 4.9 +/- 0.6 μM and selectivity index 88.2 (Fattahian et al. 2022/2024). This is defined-compound evidence in a related taxon, not evidence that the same lactones occur in *A. absinthium* or engage a known parasite protein. A cruzain docking paper is retained as *T. cruzi* context only, not as a demonstrated *Artemisia* interaction. A 2026 poultry study proposed ATP synthase and glutathione transferase as camphor docking targets in *Ascaridia galli*, but the same experiment tested crude *A. herba-alba* and *A. judaica* ethanolic extracts rather than purified camphor; the result is therefore E2 target prioritization, not direct target engagement or terpene causality (Abdelqader et al. 2026). For *H. contortus*, the current *Artemisia* records establish anthelmintic phenotypes but no parasite protein target, making target discovery a high-priority gap rather than an inferred mechanism.

The practical conclusion is that artemisinin supplies a validated parasite-protein interaction framework, while most non-artemisinin *Artemisia* terpene claims still need purified-compound assays, parasite permeability measurements, target-rescue or genetic tests, and host-protein counterscreens. The KB therefore stores interaction mode, evidence tier, compound origin, parasite stage, and inference boundary for every record.

Table 2. Cross-parasite terpene-protein evidence landscape.

Parasite	Compound context	Protein or complex	Strongest current evidence	Boundary
P. falciparum	Artemisinin/DHA	Broad covalent target set	E5 activity-based proteomics with selected orthogonal validation	Probe- and activation-dependent; not all targets are equally causal
P. falciparum	Artemisinin/DHA	PfPI3K, PfTCTP, PfATP6, proteasome, PfRACK1-Pf80S	E3-E5 biochemical, functional, genetic, structural, and complex-level evidence	Supports a multi-target model, not a single universal receptor
Leishmania spp.	Sesquiterpene-related panel	PTR1, carbonic anhydrase, NMT, TryS	E2 docking	Requires direct binding, enzyme inhibition, target engagement, and *Artemisia* provenance
T. brucei	Cnicin and cynaropicrin	TbPTR1 and TbDHFR	E4 recombinant-enzyme inhibition plus docking	Compounds were sourced from non-*Artemisia* Asteraceae; parasite target engagement remains open
T. cruzi	Terpenoid panel	Cruzain	E2 docking with prior inhibition context	Not an *Artemisia*-matched direct interaction experiment
A. galli	Crude *A. herba-alba/A. judaica* extracts; camphor docking hypothesis	ATP synthase; glutathione transferase	E2 docking paired with in-vitro/in-vivo extract phenotype	Camphor abundance, direct binding, and causal contribution remain untested
Leishmania spp.	*A. absinthium* phenolic profile; apigenin	N-myristoyltransferase	E2 docking plus HPLC context	Apigenin is non-terpene; no direct binding, enzyme inhibition, or parasite efficacy was shown
T. cruzi	Dehydroleucodine from *A. douglassiana*	Unresolved	E3 defined-compound stage-resolved phenotype	Programmed-cell-death readouts do not establish a protein target or host selectivity
H. contortus	*Artemisia* oils, extracts, and one isolated sesquiterpene	Unresolved	E1 phenotype	Target discovery and host-selectivity experiments are required

Figure 2. Evidence ladder used for terpene-parasite target claims.



The arrows indicate increasing evidentiary requirements, not an automatic promotion path. A docking result advances only when the same compound-protein pair is tested experimentally, and even direct binding does not establish that the interaction is necessary or sufficient for parasite killing.

Published essential-oil studies provide useful phenotype anchors. *A. annua* leaf oil has been tested against *Leishmania donovani* in in-vitro and in-vivo contexts. Oils from *A. campestris* and *A. herba-alba* have been tested against *L. infantum* promastigotes, with apoptosis-like and cell-cycle-associated effects reported. These results support mixture-level activity under the reported experimental conditions. They do not establish a single active terpene, whole-life-cycle efficacy, host safety, or clinical benefit.

Whole-leaf and aqueous-extract studies occupy a different evidentiary category from essential oils. *A. annua* leaf powder was evaluated against *Leishmania* in cell and animal models, but the complex material prevents attribution to artemisinin or another terpene (Mesa et al. 2017). *A. nilagirica* extracts were screened against *P. falciparum*, again providing a species- and preparation-specific phenotype rather than a defined terpene mechanism (Panda et al. 2018). Most recently, aqueous *A. afra* and *A. annua* extracts were compared against artemisinin-resistant *P. falciparum* with selectivity measurements (Roesch et al. 2025). This resistant-parasite design is valuable, but activity of a chemically complex infusion cannot be projected onto an isolated constituent or clinical regimen.

The quantitative assay records are separated in `antiparasitic-evidence.csv` so that parasite stage, preparation, endpoint, dose, host control, and translation boundary remain visible. For example, the camphor-rich *A. annua* oil reported IC50 values of 14.63 +/-

1.49 microgram/mL against promastigotes and 7.3 +/- 1.85 microgram/mL against intracellular amastigotes, plus an approximately 90% liver/spleen burden reduction at 200 mg/kg in infected mice. Those observations belong to one hydrodistilled oil and its experimental model; they are not evidence that camphor alone, or *A. annua* generally, produces the effect.

An Ethiopian study of *A. abyssinica* further illustrates why selectivity must be reported with activity: its oxygenated-monoterpene-rich oil showed activity against *Leishmania* promastigote and amastigote systems but also variable toxicity in mammalian-cell and erythrocyte assays. This is a valuable comparative record because it carries parasite stage and host-toxicity information together, although it still does not resolve which constituents or interactions drive the result.

The broader antiparasitic record is heterogeneous but not limited to Leishmania. *A. campestris* oil has been tested against *Haemonchus contortus* eggs and adults and against *Heligmosomoides polygyrus* in a nematode model (Abidi et al. 2018); *A. vestita* and *A. maritima* extracts have been tested against *H. contortus* in vitro and in infected sheep (PubMed). A hydrodistilled *A. gilvescens* oil and selected constituents have also shown larvicidal activity against *Anopheles anthropophagus* (PubMed). Seven geographic *A. argyi* oils provide a population-level vector comparison against *Anopheles sinensis*, but the shared and variable constituents remain mixture-level predictors rather than identified actives (Luo et al. 2022). These studies widen the phenotype space to helminths and vectors, but they also emphasize that extract polarity, dose, life stage, host model, and formulation prevent direct pooling. An Ethiopian extract comparison against bloodstream *Trypanosoma brucei* adds another protozoan anchor while demonstrating why cytotoxicity and extract composition must accompany activity (PubMed).

Tick and insect studies also show why formulation and host counterscreens must remain attached to the evidence record. An *A. herba-alba* oil nanoemulsion showed the lowest reported LC50 for tick eggs and nymphs among the tested *Hyalomma dromedarii* stages, but oral exposure in mice was accompanied by liver and kidney histopathological changes despite largely unremarkable blood chemistry (Abdel-Ghany et al. 2021). The formulation therefore supplies a useful stage-specific acaricide phenotype and a cautionary host signal, not proof that a native oil or one terpene is safe. Together with altitude-associated *A. scoparia* mosquito activity and the *A. vulgaris* gall-mite assay described above, it supports a comparative design in which oil composition, formulation, exposure route, stage, non-target toxicity, and parasite relevance are modeled separately.

Within helminth research, defined-compound and whole-preparation studies can be compared without treating them as equivalent. Artemisinin and *Artemisia* extracts were tested against *H. contortus* in a gerbil model, directly exposing the distinction between purified compound and botanical matrix (Squires et al. 2011). *A. lancea* essential oil was tested against egg and larval stages in vitro (Zhu et al. 2013), whereas bio-guided *A. cina* isolation later produced a defined sesquiterpene lead for L3 larvae. These assays suggest tractable phenotypes for fractionation and target discovery, but host pharmacokinetics, adult-worm activity, and safety remain separate questions.

An additional *A. herba-alba* tick study tested Libyan leaf oil against *Ixodes ricinus* nymphs using open-filter-paper exposure. At 1 microliter/cm2, complete mortality was reported within two hours, whereas lower exposure produced slower or incomplete effects (Elmhalli et al. 2021). This is a useful vector-borne-disease comparator for the wormwood evidence stream, but it remains a hydrodistilled mixture assay without purified-constituent attribution, mammalian safety data, or evidence against gastrointestinal helminths.

Historical structure-activity work on artemisinin and related sesquiterpenes in rodent malaria already showed that close chemical relatives can differ substantially in activity (Peters et al. 1986). That observation remains relevant to comparative genomics: detecting a pathway-related scaffold or ADS-like sequence does not establish an artemisinin-equivalent phenotype. Structure, peroxide status, stereochemistry, exposure, and activation environment all matter.

Two additional high-priority records sharpen the chemistry-phenotype boundary. In *A. indica*, a hydrodistilled oil contained 92 detected compounds, with camphor (13.0%) and caryophyllene oxide (10.87%) as major constituents; the oil and solvent extracts showed activity across *Plasmodium falciparum*, *Trypanosoma brucei*, and *T. cruzi* systems, with several preparations showing low mammalian cytotoxicity (Tasdemir et al. 2015). The study is valuable because chemistry, parasite panels, and cytotoxicity were measured together, but it remains a mixture/extract study. A separate activity-guided *A. indica* study isolated exiguaflavones A and B, maackiain, and a benzofuran; the reported antimalarial principles were non-terpene compounds, so the record is retained as a boundary control rather than evidence for terpene activity (Chanphen et al. 1998). In *A. scoparia*, a BA plus 2,4-D callus extract inhibited *Leishmania tropica* promastigotes with an IC50 of 19.13 mg/mL and produced apoptotic morphology; GC-MS reported nerolidol alongside non-terpene metabolites (Yousaf et al. 2019). This is evidence for a tissue-culture-derived crude extract, not proof that nerolidol or any single terpene accounts for the phenotype.

The strongest causal bridge would require matched specimen chemistry, parasite-stage assays, host-cell cytotoxicity, fractionation or defined mixtures, and ideally purified-compound plus combination testing. Gene expression or a predicted pathway can prioritize candidates but cannot replace those experiments.

Additional species and preparation types reinforce this boundary. *A. pedemontana* subsp. *assoana* essential oils and hydrolate were evaluated across insect, nematode, protozoan, and antiplasmodial assays, making it a useful multi-phenotype specimen record while retaining oil and hydrolate as distinct preparations (Sainz et al. 2019). Isolated *A. sieversiana* sesquiterpenoids provide a compound-level trypanocidal lead set (Nurbek et al. 2020), whereas *A. absinthium* fractionation studies link selected oil fractions to *T. cruzi* and *T. vaginalis* activity with cytotoxicity measurements (Martínez-Díaz et al. 2015). These records increase taxonomic and mechanistic coverage, but they do not make mixture, purified-compound, or host-selectivity endpoints interchangeable. A bio-guided *A. cina* study adds a stronger constituent-to-phenotype bridge by isolating a sesquiterpene active against *Haemonchus contortus* L3 larvae (Arango-De la Pava et al. 2024); an *A. brevifolia* study extends the nematode comparison across life stages (Hussain et al. 2025). Both still require host-selectivity and in-vivo confirmation.

Purified-compound evidence provides an important contrast to oil studies: helenalin, mexicanin, and dehydroleucodine inhibited cultured *Leishmania mexicana* promastigotes at approximately low-micromolar concentrations, but that result cannot be assumed for an *Artemisia* oil or a different parasite stage (Barrera et al. 2008). Likewise, methanolic aerial-part extracts of *A. sieberi* and *A. khorassanica* have been evaluated in *Plasmodium berghei* mouse models (PubMed; PubMed). These records broaden the evidence base while reinforcing the need to retain extraction polarity, dose, parasite model, and compound identity.

Isolation and fractionation can provide a stronger causal bridge than crude-oil screening. Leaves and flowers of Cape Verdean *A. gorgonum* yielded guaianolides and germacranolides; ridentin had a *P. falciparum* FcB1 IC50 of 3.8 +/- 0.7 microgram/mL against a Vero-cell IC50 of 71.0 +/- 3.9 microgram/mL (Ortet et al. 2008). Conversely, a Cuban *A. absinthium* oil dominated by trans-sabinyl acetate inhibited *L. amazonensis* promastigotes and amastigotes and reduced lesion and parasite burden after intralesional dosing in mice (Monzote et al. 2014). The contrast is informative: purified-lactone selectivity and oil-level in-vivo activity are both stronger than an uncharacterized extract claim, but neither supports extrapolation to all *Artemisia* preparations. A separate *A. absinthium* oil study tested the oil and three major constituents against six mosquito vectors while measuring acute toxicity in non-target aquatic organisms (Govindarajan and Benelli 2016); the oil contained (E)-beta-farnesene, (Z)-en-yn-dicycloether, and (Z)-beta-ocimene as major constituents, and their larvicidal potency varied by mosquito species. This is valuable paired efficacy-ecotoxicity and constituent-resolved evidence, but still preparation- and dose-specific.

Phylogeny can improve discovery prioritization without replacing chemical validation. Sampling of 117 ethnobotanically reported species detected artemisinin in nine species, including eight reported for the first time, but detection does not establish abundance, pathway completeness, or therapeutic equivalence (Pellicer et al. 2018). This finding supports a comparative design in which phylogenetic position selects accessions for targeted LC-MS and pathway-genomics follow-up rather than serving as a proxy for efficacy.

6.2 Malaria: artemisinin as the primary validated case

The modern review literature frames artemisinin as a defined endoperoxide drug rather than as a generic synonym for *Artemisia* material. A 2021 review places the 1,2,4-endoperoxide bridge, artesunate development, and Kelch13-associated resistance in the clinical-drug context (Khanal 2021). We use this review for scope and mechanism framing; quantitative claims about plant chemistry, parasite stages, pharmacokinetics, or safety remain anchored to primary records.

A 2025 plant-matrix study adds a useful preclinical malaria comparator: normalized *A. annua* extracts produced approximately ten-fold greater antimalarial effect than equivalent-dose artemisinin in two murine models, while rat pharmacokinetics showed enhanced oral exposure and Caco-2 experiments implicated reduced P-glycoprotein efflux; deoxyartemisinin, artemisinic acid, and dihydroartemisinic acid affected transporter behavior in the reported assays (Xu et al. 2025). This is matrix and host-transporter evidence, not human efficacy, clinical equivalence, a single active terpene, or direct parasite-protein engagement. A matched origin comparison reinforces the same boundary. Water infusions from Cameroon and Luxembourg *A. annua* plants differed in HPLC profiles and beta-haematin inhibition; the Cameroon material showed stronger inhibition at the reported dilution and more major chromatographic features (Akkawi et al. 2014). Because the study used a hemozoin surrogate and preliminary chromatographic assignments rather than a purified terpene or a clinical endpoint, it supports origin- and matrix-dependent malaria chemistry, not a universal chemotype or treatment equivalence.

Malaria should be the anchor phenotype for the evolutionary-genomics component because it links a defined sesquiterpene lactone to a validated plant pathway and a clinically important parasite. The artemisinin scaffold contains a peroxide bridge whose activation is central to parasite damage, and the *A. annua* ADS-CYP71AV1-DBR2-ALDH1 network provides the best-resolved plant-side genotype-to-metabolite framework in the genus. Historical structure-activity work showed that related sesquiterpenes are not functionally interchangeable, while modern target studies support activation-dependent, multi-target parasite injury rather than one universal receptor (Peters et al. 1986; Wang et al. 2015).

The non-malaria protozoan evidence includes a host-response-mediated veterinary coccidiosis study. In *Eimeria tenella*-infected chickens, artemisinin and dried *A. annua* leaves reversed infection-associated apoptosis markers and reduced inflammatory-response signals; leaf treatment produced stronger clinical and pathological improvement than artemisinin in the reported comparison (Jiao et al. 2018). Because the readouts were cecal host responses and the leaf preparation was chemically complex, this record supports a veterinary comparator and a testable host-mediated mechanism, not a direct *Eimeria* protein interaction or evidence that the whole leaf is interchangeable with an artemisinin drug.

Mechanistic uncertainty is not a minor footnote. A dedicated review of carbon-radical and heme hypotheses emphasized that artemisinin activation, iron/heme chemistry, parasite stage, metabolism, and combination partners all influence interpretation, while competing hypotheses cannot be collapsed into a single receptor model (Haynes et al. 2013). In parallel, microbial transformation of artemisinin and related terpenoids has been used to generate derivative candidates for medicinal chemistry (Parshikov et al. 2012). These transformed products are valuable experimental probes, but neither their activity nor their pharmacokinetics can be assigned to native *Artemisia* chemistry without direct measurement.

The clinical boundary must remain explicit. WHO guidance supports quality-assured artemisinin-based combination therapy for malaria; it does not validate arbitrary teas, capsules, essential oils, or unstandardized *Artemisia* materials. Whole-plant experiments remain scientifically valuable for understanding matrix effects and resistance evolution. In rodent models, dried whole-plant *A. annua* overcame selected artemisinin resistance and delayed resistance acquisition relative to pure artemisinin (Elfawal et al. 2015). A historical PubMed-indexed report of an *A. annua* gelatin-capsule preparation likewise described parasitemia clearance in *P. berghei* and reported clinical *P. vivax* cure with high recrudescence, but its crude preparation, limited modern metadata, and absence of constituent attribution make it historical context rather than current efficacy evidence (Wan et al. 1992). These records motivate fractionation, exposure, and resistance-mechanism studies; they do not establish human efficacy, dosing, quality control, or safety of a nonpharmaceutical whole-plant product.

The resistance result is especially informative because it was tested against both an existing resistant parasite and experimentally evolving populations. Dried whole plant overcame selected *P. yoelii* resistance to pure artemisinin, and stable resistance arose three times more slowly under whole-plant than pure-artemisinin selection in *P. chabaudi* (Elfawal et al. 2015). The authors' redundant-defense interpretation is biologically plausible, but the experiment does not identify whether the relevant co-components are sesquiterpenes, flavonoids, volatile constituents, altered absorption, or their combination. Older mouse isobolograms make the same interaction principle explicit: artemisinin was potentiated by some partners, including mefloquine, tetracycline, spiramycin, and selected primaquine or clindamycin contexts, but antagonism occurred with several sulfonamide/pyrimethamine combinations and the result changed with parasite resistance strain (Chawira et al. 1987). Thus, plant-matrix and drug-combination effects should be modeled as partner-, strain-, dose-, and exposure-specific rather than summarized as generic synergy.

The matrix hypothesis has a useful non-terpene comparator. A critical review proposed that *A. annua* leaf flavonoids might influence artemisinin activity, absorption, CYP-mediated metabolism, and host immunomodulation, but emphasized that only a limited number of matched synergy studies existed (Ferreira et al. 2010). Endophyte-associated chemistry provides a separate discovery route: metabolomics-guided screening of fungi from *A. annua* identified emodin and physcion as validated antimalarial candidates with IC50 values of 0.9 and 1.9 micromolar, respectively (Alhadrami et al. 2021). Both records are scientifically relevant but non-terpene: neither demonstrates that a plant volatile or sesquiterpene lactone drives whole-leaf activity. They instead motivate matched fractionation, host-exposure, and parasite-stage experiments that can distinguish plant terpenes from co-extracted or microbially derived chemistry.

Historical reviews and production analyses explain why the quality boundary is central to malaria translation. Early syntheses described plant production and biotechnology as ways to increase artemisinin supply (Van Geldre et al. 1997; Segura 2010), while a later development review emphasized crop quality, formulation, bioequivalence, and combination validation as recurrent failure points (Jambou et al. 2011). A historical Myanmar report is retained as a title-level record of an extract antimalarial trial, but its full text must be recovered before any preparation, dose, or efficacy value is extracted (Lwin et al. 1991). These sources support a reproducibility and stewardship framework, not a clinical endorsement of unstandardized extracts.

That stewardship concern has also been raised specifically for unapproved *A. annua* use in malaria prevention. A risk-focused review identified uncontrolled exposure as a potential contributor to artemisinin drug pressure and future resistance, but it did not measure resistance evolution in a defined plant product (Ataba et al. 2022). The inference is therefore precautionary and complementary to the experimental whole-plant selection result: it strengthens the case for quality-assured ACTs, surveillance, and controlled comparative studies rather than treating a tea or capsule as a resistance-resilient therapy.

Non-*annua* malaria studies are useful as bounded comparators rather than substitutes for the *A. annua* pathway. Iranian *A. sieberi* aerial-part extracts reduced *P. berghei* parasitemia in mice in a study that reported no toxicity in the tested NMRI-mouse assessment, while a flowering-stage *A. khorassanica* extract showed a more limited response and contained chrysanthenone and cis-thujone among GC-MS features (Nahrevanian et al. 2012; Nahrevanian et al. 2010). These records strengthen the comparative malaria evidence base, but both are crude, species-specific preparations: detection of a volatile, absence of short-term toxicity, or reduction of murine parasitemia does not establish a shared artemisinin pathway, a single active terpene, or human antimalarial efficacy. Earlier screening studies add two useful non-*annua* boundaries. Dichloromethane extracts of *A. ciniformis*, *A. biennis*, and *A. turanica* inhibited beta-hematin formation, with *A. ciniformis* most active at an IC50 of 0.92 +/- 0.01 mg/mL; this is a hemozoin-surrogate screen rather than a parasite-stage result (Mojarrab et al. 2015). In a separate fractionation study, *A. armeniaca* and *A. aucheri* dichloromethane extracts were active in the same surrogate assay, and a fraction of *A. armeniaca* reached an IC50 of 0.47 +/- 0.006 mg/mL (Mojarrab et al. 2014). These records prioritize medium-polar extracts for chemical follow-up but do not identify a terpene, establish blood-stage efficacy, or support clinical translation.

An extract-versus-compound comparison in *A. sieberi* is more biologically direct but still bounded. Both an aqueous plant extract and purified artemisinin reduced *Leishmania major* promastigote and intracellular-amastigote burdens in vitro, with the plant extract reported as more active in that experiment and low effects on uninfected macrophages (Heydari et al. 2013). The result supports a preparation-specific protozoan phenotype and a testable extract/compound comparison; it does not identify the active constituent, establish a parasite protein target, or demonstrate in-vivo efficacy. Together with the *P. berghei* records, it shows why non-*annua* activity should be retained as comparative evidence without being collapsed into artemisinin-pathway equivalence. Formulation evidence provides a further boundary between plant chemistry and parasite phenotype. Green silver nanoparticles synthesized with *A. abrotanum* or *A. arborescens* extracts were characterized as sub-50-nm particles and tested against *P. falciparum* cultures; the *A. abrotanum*-derived particles showed lower parasitemia and lower hemolysis than the comparison preparations in the reported dose range (Avitabile et al. 2020). This is a formulation-level antiplasmodial result, not evidence that a terpene is the active principle or that the particles are clinically safe. Conversely, dry leaves, artemisinin, and several *A. annua* extracts that were active against *Histomonas meleagridis* clones in vitro failed to improve experimental histomonosis in turkeys or chickens (Thøfner et al. 2012). The paired records make in-vitro/in-vivo discordance an explicit translation variable rather than an unexplained exception.

For the comparative study, malaria-related claims should be separated into three levels: (1) *A. annua* artemisinin production and parasite mechanism, which have the strongest evidence; (2) detection of artemisinin or related lactones in another *Artemisia* species, which supports chemical occurrence only; and (3) an *Artemisia* extract phenotype against *Plasmodium*, which supports a preparation-level assay result. A species can satisfy level 2 or 3 without possessing the same pathway flux, peroxide abundance, pharmacokinetics, or resistance profile as *A. annua*. A useful negative control is the matched *A. annua/artemisinin* study in which artemisinin, artesunate, and dried-leaf slurry failed to reduce *Babesia microti* parasitemia in mice at the tested 100 microgram artemisinin/kg regimen, and activity against six *Candida* species was not sustained or was essentially absent up to 180 micromolar (Elfawal et al. 2021). This does not weaken the *Plasmodium* evidence; it demonstrates that parasite identity, stage, dose, and host/model context are causal design variables. This tiering is central to the proposed orthology and metabolomics workflow.

Recent cross-species measurements illustrate why the second level must remain separate from therapeutic equivalence. In *A. tournefortiana*, RP-HPLC-PDA measured artemisinin most strongly in flowers, reaching 1.43%, with flower values ranging from 0.05% to 1.43% across geographic sites; roots and seeds contained lower amounts. Comparative sequencing of pathway-associated genes supported biosynthetic potential (Sahoo et al. 2026). These data make *A. tournefortiana* a useful accession- and tissue-stratified comparator, not a demonstrated substitute for *A. annua*. Two earlier regional surveys widen the occurrence map but reinforce its limitations. A Tajikistan collection contained HPLC-quantified artemisinin at 0.07-0.45% dry mass in seven sampled species, including *A. vachanica* as a proposed new source (Numonov et al. 2019). A northern Pakistan survey compared flowers, leaves, roots, and stems from *A. annua* and fourteen other species; *A. annua* leaves and flowers were highest in that dataset, while several other species showed lower but detectable levels (Mannan et al. 2010). These data support phylogenetically distributed, tissue-resolved follow-up, but occurrence in a hexane extract or HPLC peak does not prove an intact ADS-CYP71AV1-DBR2-ALDH1 pathway, stable field chemotype, or antimalarial equivalence.

Whole-leaf mutant experiments refine the frequently proposed “artemisinin plus flavonoid synergy” explanation. A *CHI1* mutant that lacked major flavonoids but retained wild-type artemisinin levels showed antiplasmodial activity comparable to wild type, whereas *CYP71AV1*- and ADS-impaired lines with severely reduced artemisinin biosynthesis showed little or no activity against intraerythrocytic *P. falciparum* Dd2 (Czechowski et al. 2019). In that matched genetic system, flavonoids were not a major independent driver of rapid in-vitro activity. This does not eliminate matrix effects in other cultivars, extraction methods, parasite strains, or exposure windows; it establishes a stronger causal comparator than correlation of unrelated metabolites.

Environment can also move both chemistry and apparent antimalarial potency. Under controlled LED treatments, white and blue light increased artemisinin and artemisinic acid relative to red light and shifted mono-, sesqui-, di-, and triterpene profiles. Crude-leaf extract IC50 values against *P. falciparum* NF54 were 0.51 and 0.52 microgram/mL under white and blue light, 1.11 microgram/mL under red light, and 2.23 microgram/mL for greenhouse-grown plants (Sankhuan et al. 2022). Because the experiment changed multiple metabolites together, the result supports an environment-chemotype-phenotype relationship rather than a causal assignment to germacrene D, trans-beta-farnesene, artemisinin, or any other correlated compound.

Recent matrix studies sharpen, rather than settle, the synergy question. A dedicated review of isolated *A. annua* and *A. afra* metabolites explicitly separates direct antiplasmodial compounds from metabolites affecting PK/PD, inflammation, immunomodulation, and toxicity (Shinyuy et al. 2023). An *A. annua* extract tested with equivalent artemisinin showed no synergy against *P. falciparum* Pf3D7 at 150 nM, but produced a biphasic effect in *P. yoelii* mice, including a lower ED90 and higher artemisinin exposure at an early infection burden; the authors identified chrysosplenetin as a candidate matrix component through LC-HRMS and fractionation (Xu et al. 2026). Chrysosplenetin is a polymethoxy flavonol, not a terpene, and the result is therefore a matrix/pharmacokinetic comparator rather than evidence for a terpene-protein interaction. It also reinforces the need to model infection stage, host metabolism, and extract standardization when interpreting whole-plant malaria data. A separate defined-component mouse study combined artemisinin with arteannuin B, arteannuic acid, and scopoletin; reported parasitemia reduction was about 93% for the combination versus about 31% for pure artemisinin, and repeated-dose pharmacokinetics showed higher artemisinin exposure (Li et al. 2018). This is a useful preclinical combination anchor, but scopoletin is non-terpene and the result cannot be used to assign synergy to any single constituent or to justify unstandardized botanical treatment.

Independent metabolomics supports a species-specific malaria boundary. *A. afra* extracts contained only trace artemisinin and produced a parasite metabolic response distinct from *A. annua*; the *A. annua* response more closely resembled purified artemisinin and included glutathione-associated changes, whereas *A. afra* affected lipid precursors (Mamede et al. 2025). This result is valuable because it separates a non-artemisinin *Artemisia* phenotype from the artemisinin-centered *A. annua* comparator, but it was based on aqueous extracts and metabolomics rather than a terpene-specific perturbation, direct target engagement, or a clinical endpoint.

An in-vivo *A. afra* test supplies an important negative control for claims based on artemisinin-independent infusions. In *P. berghei*-infected mice, aqueous powder suspensions of *A. annua* showed activity but *A. afra* did not support the proposed mammalian-prodrug hypothesis (Walz et al. 2024). This does not invalidate in-vitro *A. afra* findings, which may depend on parasite stage, exposure, metabolism, or assay matrix; it does mean that in-vitro activity should not be promoted to in-vivo efficacy without a chemically defined and host-relevant exposure experiment.

Extraction chemistry can also create an apparent antimalarial advantage. Hydrotrope-assisted aqueous extraction with sodium salicylate or cholinium salicylate produced artemisinin-rich extracts with greater *P. falciparum* activity than pure artemisinin in the tested system, while artemitin, chrysosplenol D, arteannuin B, and arteannuin J co-occurred in the extracts (Ferreira et al. 2024). The result is useful for sustainable extraction and matrix testing, but the solvent, extraction yield, co-metabolite composition, and assay exposure are inseparable confounders; it is not evidence for a clinical botanical product or for any one terpene-protein interaction.

A complementary top-down study compared ethyl-acetate and petroleum-ether *A. annua* extracts at equivalent artemisinin exposure. Reported *P. yoelii* ED50 values were 2.9 and 1.6 mg/kg versus 5.6 mg/kg for pure artemisinin, while only the petroleum-ether extract increased rat artemisinin AUC0-t by 1.7-fold. Arteannuin B and artemisinic acid were isolated as differential sesquiterpenes and tested in the matched petroleum-ether proportion (Cai et al. 2024). The result is a useful sesquiterpene/matrix-PK hypothesis, but the lack of in-vitro enhancement, rat-only PK verification, and preclinical design prevent clinical or unstandardized-botanical inference.

A related component-resolved study points to host metabolism as one route by which an *A. annua* matrix can alter apparent artemisinin potency. Fractionation identified nine major components, including five sesquiterpenes; arteannuin B inhibited CYP3A4 with an IC50 of 1.2 microM, and selected matrix fractions approximately doubled artemisinin exposure in healthy rats. The study also reported support for antiplasmodial enhancement in a preclinical combination experiment (Cai et al. 2017). This is a valuable bridge between terpene co-occurrence, host pharmacokinetics, and malaria phenotype, but it does not establish a parasite-protein interaction, a clinical benefit, or a safe standardized botanical dose.

In a separate resistant-mouse study, artemisinin plus chrysosplenetin improved survival and reduced parasitemia-related measures in *Plasmodium berghei* K173, with effects associated with ABC-transporter and heme-ROS/GSH responses and dependent on host genotype and parasite context (Ji et al. 2025). This is valuable resistance-biology evidence, but it is not a direct parasite-protein target demonstration, not a terpene result, and not a human regimen. Together with the matched mutant study above, it argues for testing terpenes against malaria as chemically defined comparators within a standardized *A. annua* matrix, rather than treating every matrix effect as proof of terpene synergy.

The parallel regulation review is useful context for these matrix experiments because it distinguishes shared transcriptional control of artemisinin and flavonoid pathways from demonstrated chemical synergy (Hassani et al. 2020). That distinction is retained in the present evidence model: co-regulation and co-occurrence are hypotheses until matched compounds, exposures, and parasite endpoints are tested.

Preparation and host context can change the apparent malaria phenotype before any new parasite target is invoked. In 14 healthy men, one liter of tea prepared from 9 g of dried *A. annua* leaves produced a mean artemisinin Cmax of 240 +/- 75 ng/mL and AUC of 336 +/- 71 ng/mL*h; the authors explicitly concluded that the tea should not be treated as equivalent to modern artemisinin drugs (Räth et al. 2004). In mice, dried-leaf delivery likewise produced higher exposure than pure artemisinin, and infection altered absorption and deoxyartemisinin metabolism: AUC was 299.5 m*gmin/L in healthy mice versus 435.6 mg*min/L in* P. chabaudi*-infected mice (Weathers et al. 2014). These studies make host pharmacokinetics a required covariate in any terpene/protein interaction experiment using plant material.

The literature contains a real, informative contradiction about whether artemisinin explains most activity in an *A. annua* preparation. Ancient fresh-herb soaking or pounding produced extracts with higher artemisinin concentration than dried-herb tea, but only the pounded juice contained sufficient artemisinin to suppress *P. berghei* parasitemia in mice (Wright et al. 2010). A matched in-vitro comparison across varieties found tea and nonpolar extract activity closely tracked artemisinin concentration and did not differ significantly from pure artemisinin (Mouton et al. 2013). By contrast, *A. annua* and *A. afra* infusions inhibited blood and liver stages, including *P. vivax* and *P. cynomolgi* hypnozoites, in a study that reported essentially artemisinin-independent activity (Ashraf et al. 2022). These findings should not be averaged into a single "synergy" estimate: they differ in plant species, preparation, parasite stage, exposure, and likely non-terpene chemistry. They instead define a testable interaction between chemotype, matrix, parasite stage, and host metabolism.

Gametocyte and strain-resolved work further narrows the hypothesis. At 5 g dry weight/L, *A. annua* and *A. afra* teas reduced *P. falciparum* asexual parasites and early/late gametocytes, with effects mainly attributable to artemisinin content and only a weak signal from artemisinin-deficient *A. afra* (Snider and Weathers 2021). A separate panel of sensitive, resistant, and reversion strains found exposure-time and strain-dependent dried-leaf effects and suggested that malaria-patient sera can contribute to inhibition in some assays (Gruessner and Weathers 2021). Taken together, "artemisinin-dependent" and "artemisinin-independent" are best treated as conditional labels. The comparative experiment should therefore test purified artemisinin, structurally related sesquiterpenes, quantified volatile fractions, and recombined matrices at matched parasite-stage exposures, while measuring serum protein binding and host CYP modulation.

Cross-species chemical detection adds a complementary but narrower boundary. HPLC analysis of Algerian wild *A. herba-alba*, *A. campestris* subsp. *glutinosa*, and *A. judaica* subsp. *sahariensis* reported artemisinin yields of 0.34%, 0.64%, and 0.04%, respectively (Ahmed-Laloui et al. 2022). These values support species- and lot-specific occurrence, not equivalent pathway completion, bioavailability, or antimalarial efficacy. The same study also measured antioxidant activity, which is a separate chemical phenotype and cannot be used as a proxy for parasite killing or host selectivity. A validated HPTLC method for artemisinin and artemisinic acid in *A. annua* callus and leaves provides a practical quality-control bridge for such comparisons (Khan et al. 2015). The appropriate comparative design is therefore voucher-linked quantitation followed by matched parasite-stage assays, rather than ranking species by a single chemical measurement. Repeated-stress experiments likewise changed artemisinin-related precursor ratios differently in upper and lower leaves and linked metabolite levels to glandular-trichome traits, while volatile fingerprints separated artemisinin-rich cultivars from wild type (Kjær et al. 2013; Reale et al. 2011). These are useful chemotype and sampling controls, but neither volatile correlation nor stress-induced precursor conversion establishes a causal parasite mechanism.

The malaria evidence also has a clear historical and analytical continuity. Artemisinin was isolated from *A. annua* growing in the United States in an early primary report (Klayman et al. 1984). Subsequent reviews describe the rapid antimalarial action and clinical development of artemisinin derivatives, while emphasizing chemistry and derivative pharmacology rather than equivalence of crude plant material to defined drugs (Balint 2001; Jung et al. 2004). A later review synthesizes the restricted distribution of artemisinin within *Artemisia* and its dependence on ontogeny and geography (Nabi et al. 2023). These records support malaria context and a distribution hypothesis, but only voucher-linked primary measurements can establish occurrence in a particular taxon or lot. Preparation chemistry is equally important: artemisinin recovery from *A. annua* tea infusions was strongly temperature-dependent and exceeded 90% under tested conditions, whereas an accuracy-profile-validated HPLC-UV method established a transferable workflow for hydroalcoholic extracts (van der Kooy and Verpoorte 2011; Diawara et al. 2011). Neither result makes tea clinically equivalent to artemisinin combination therapy; both support recording preparation, extraction, and analytical uncertainty as covariates.

Quality-control methods are not interchangeable, but their convergence supports a practical cross-study measurement ladder. HPLC-ELSD with microwave-assisted extraction, TLC, GC-ECD, and HPLC-UV have each been validated for artemisinin in plant material or crude extracts, while rapid HPLC-MS/MS can resolve artemisinin together with 9-epi-artemisinin, artemisitene, dihydroartemisinic acid, artemisinic acid, and arteannuin B (Liu et al. 2007; Marchand et al. 2008; Liu et al. 2008; Ferreira and Gonzalez 2009; Suberu et al. 2013). HPLC-MS/MS has also been applied to tea, biscuit, and porridge products made from *A. annua* leaves (Carrà et al. 2014). This convergence enables low-resource screening plus confirmatory multi-metabolite analysis, but an analytical method cannot by itself establish tissue localization, pathway direction, bioavailability, or therapeutic efficacy.

The ladder now includes assays suitable for living material and for structurally related analogs. A microscale GC-ECD workflow quantified artemisinin from a single living leaf or flower with greater than 95% recovery, enabling non-destructive or low-input screening during biosynthesis experiments (Liu et al. 2009). Ultrasound-assisted extraction paired with rapid-resolution LC-triple-quadrupole MS achieved 0.20 ng/mL detection and 92.45–103.8% recovery in *A. annua* samples (Wang et al. 2011). Capillary electrophoresis–MS further separated artemisinin from analogs including ascaridole, artemisia ketone, deoxyartemisinin, artemisinic acid, artemetin, and dihydroartemisinic acid, while revealing diastereomeric features in ethanolic extracts (Nagy et al. 2021). These methods strengthen lot- and tissue-specific exposure measurement, but detection of an analog or diastereomer is not evidence of a defined pathway or parasite mechanism.

Quality control also has a distinct downstream drug-product layer. Two color-reaction TLC field assays detected as little as 10% artemisinin derivatives in several ACT formulations and identified counterfeit products in a small survey (Ioset and Kaur 2009). This is important for malaria treatment stewardship, but it applies to defined combination medicines rather than crude *Artemisia* material; it should not be used as evidence that a plant preparation is clinically interchangeable with an ACT.

6.3 Wormwood (*A. absinthium*): vermifuge evidence and limits

A focused wormwood review summarizes reported constituents, pharmacological and anthelmintic activity, pharmacokinetics, and toxicity, including important cautions around essential-oil exposure (Batiha et al. 2020). A veterinary network meta-analysis synthesized 256 animals across six studies and identified *A. absinthium* among more effective botanical candidates, while emphasizing heterogeneity and the need for adequately powered randomized trials (Calzetta et al. 2020). These reviews support the vermifuge hypothesis but do not provide a standardized wormwood terpene effect or human-use evidence.

An independent review of more than 300 studies on edible *Artemisia* species and selected sesquiterpene lactones likewise integrates nutritional, pharmacological, and clinical-context evidence while identifying raw-extract standardization and adverse-effect trials as major gaps (Trendafilova et al. 2020). Its breadth is useful for locating wormwood and STL evidence, but review-level health claims remain distinct from chemically standardized vermifuge assays and cannot supply a safe dose.

An Iranian full-bloom collection illustrates why the wormwood label cannot substitute for a chemical fingerprint. Shade-dried aerial material yielded 1.3% hydrodistilled oil; 28 identified constituents represented 93.3% of the oil, with beta-pinene at 23.8%, beta-thujone at 18.6%, and hydrocarbon monoterpenes comprising 47.8% (Rezaeinodehi and Khangholi 2008). This is a well-defined regional beta-pinene/beta-thujone chemotype, but it supplies no anthelmintic endpoint and cannot be generalized to thujone-poor, camphor-rich, or oxygenated-sesquiterpene-rich wormwood oils. Any vermifuge experiment should therefore quantify the actual lot rather than infer exposure from the species name or traditional preparation.

The broader "wormwood" record illustrates why common names and parasite endpoints must be separated. A review of *A. sieberi* describes a Middle Eastern wormicide reputation and commonly thujone-dominant oils, but its synthesized efficacy evidence is primarily antifungal, including topical clinical studies; it does not establish a nematode assay or transfer to *A. absinthium* (Mahboubi 2017). Conversely, a primary study of crude ethanolic *A. inculta* extract in schistosomiasis-infected mice reported reduced worm load and tegumental damage at 500 or 800 mg/kg, with preserved bone-marrow morphology, cellularity, and maturation under the tested regimen (Lotfy et al. 2006). The latter is a useful helminth and safety comparator, but its species, route, crude chemistry, and incomplete organ toxicology prevent attribution to a wormwood terpene or extrapolation to human treatment. Two newer reviews help situate, but not replace, the primary malaria and vermifuge evidence. A medicinal-chemistry review surveys natural and nature-inspired antimalarial compounds from *Artemisia* across in-vitro and in-vivo systems and highlights unresolved questions in mechanism, combinations, administration, and translation (Rauf et al. 2025). A production review synthesizes *A. annua* breeding, genetic engineering, tissue culture, elicitation, environmental regulation, and whole-plant strategies, while a second review covers supercritical-CO2 extraction and biotechnological production of artemisinin precursors (Vashisth and Mishra 2025; Oladipo and Ojumu 2025). We use these as discovery and process-context records: neither establishes that a particular extraction method maximizes native flux, that a whole-plant product is clinically equivalent to an artemisinin drug, or that a terpene is responsible for wormwood anthelmintic activity. An additional sheep study strengthens the broader *Artemisia* vermifuge comparison without being evidence for *A. absinthium* specifically. Oral extracts of *A. vestita* and *A. maritima* reduced fecal egg counts in *H. contortus*-infected sheep, with maximum reported reductions of 87.2% and 84.5% at 100 mg/kg; the extracts also affected infective larvae and adult worms in vitro (Irum et al. 2015). This is useful in-vivo veterinary evidence, but the abstract does not provide a standardized terpene composition, purified active constituent, parasite-protein target, or human safety bridge. It should be used as a cross-species anthelmintic comparator, not as proof that wormwood oils share the same effect.

Wormwood has a long-standing traditional reputation as a vermifuge, but the scientifically appropriate term for the current evidence is preparation-specific anthelmintic activity. The strongest early veterinary study compared crude aqueous and ethanolic aerial-part extracts with albendazole against ovine gastrointestinal nematodes. Both extracts impaired adult *Haemonchus contortus* motility in vitro and reduced fecal egg output in sheep; the ethanolic extract reached 90.46% fecal egg-count reduction at 2.0 g/kg body weight on day 15, whereas the aqueous extract was less active (Tariq et al. 2009). This is meaningful in-vivo veterinary evidence, but it does not identify a terpene, define a standardized extract, or establish use in humans. A separate caprine coccidiosis comparison is a useful negative or limited translation control: a single 2 g/kg dose of ethanolic *A. absinthium* extract was less effective than toltrazuril or amprolium, and oocysts persisted through day 28 (Iqbal et al. 2013). The result does not disprove wormwood activity against other parasites or preparations, but it shows why a traditional vermifuge label cannot substitute for dose-, host-, and parasite-matched trials.

A closely related negative control is provided by *A. annua*, whose malaria-active sesquiterpene lactone should not be assumed to be an ovine anthelmintic. In vitro, sodium-bicarbonate leaf extract had the lowest reported *H. contortus* LC99 among the preparations tested, but in naturally infected sheep a single 2 g/kg extract dose and 100 mg/kg artemisinin dose produced only nonsignificant day-15 fecal-egg-count reductions of 19% and 28%, respectively; levamisole produced 97% reduction (Cala et al. 2014). The extract used in vivo contained no artemisinin. This discordance between in-vitro preparation activity and limited single-dose efficacy is a critical translation boundary for both artemisinin and wormwood claims.

Essential-oil evidence adds a different chemistry and exposure model. Oils from two cultivated, thujone-free *A. absinthium* populations were dominated by (Z)-epoxyocimene and chrysanthenol, varied quantitatively and qualitatively between populations, and reduced *Trichinella spiralis* infectivity ex vivo at 0.5-1 mg/mL without mammalian-cell cytotoxicity in that assay. The same study reported a 66% reduction of intestinal adults in vivo, while the oils were ineffective against the plant-parasitic nematode *Meloidogyne javanica* (García-Rodríguez et al. 2015). The population dependence and parasite selectivity are important: they argue for chemotype-by-parasite testing, not for a universal "wormwood oil" mechanism.

An ectoparasite study adds a thujone-rich *Artemisia* comparator. *A. sieberi* oil contained α-thujone (31.5%), β-thujone (11.92%), camphor (12.3%), and 1,8-cineole (10.09%) and produced contact LC50, LC90, and LC99 values of 15.85, 26.63, and 35.42 microgram/cm3 against adult poultry red mites, with repellency and residual effects in the reported assays (Tabari et al. 2017). This is a mixture-level ectoparasite-control phenotype, not direct evidence for *A. absinthium* vermifuge activity; thujone, camphor, cineole, exposure route, poultry safety, and single-constituent causality remain unresolved. A separate extraction-method comparison in *A. absinthium* showed that direct-steam oil had a lower *Tetranychus urticae* LC50 (0.04 mg/cm2) than microwave-assisted or water-distilled oils (0.13 mg/cm2); a C15H24 sesquiterpene was reported only in the more potent direct-steam profile (Chiasson et al. 2001). The candidate association is useful for fractionation, but it does not identify the active molecule or establish a helminth or human vermifuge effect.

Artemisinin occurrence and veterinary antiparasitic activity can also diverge by species and parasite. *A. sieberi* contained 0.20% artemisinin in summer and 0.14% in autumn dry weight, while the tested plant extract reduced severity or oocyst output for *Eimeria tenella* and *E. acervulina* but not *E. maxima* (Arab et al. 2006). This is a useful artemisinin/host-parasite comparator, but the result is season-, extraction-, species-, and dose-specific and cannot be transferred to *A. absinthium*, human malaria, or a single-constituent mechanism.

Feed and field studies are more heterogeneous. Dried wormwood supplementation and aqueous extracts produced strong in-vitro egg-hatch effects against *H. contortus*, with ED50 and ED99 values of 1.40 and 3.76 mg/mL for the wormwood extract, but fecal egg counts did not differ significantly at the group level in experimentally infected lambs (Mravčáková et al. 2020). A low-input swine-farm study reported antiparasitic effects after *A. absinthium* powder administration, but the co-occurring plant intervention and complex powder chemistry prevent attribution to terpenes (Băieş et al. 2024). A recent Awassi-sheep study of crude ethanolic extract reported 76.77% fecal egg-count reduction on day 15 compared with 100% for albendazole and no observed adverse effects during the small trial (Irehan et al. 2026). The latter is useful contemporary evidence, but it remains a single regional study with 18 animals and requires independent chemical, residue, dose, and safety confirmation.

Wormwood also has a defined-compound trematode signal that should not be conflated with the ovine nematode results. A dichloromethane *A. absinthium* extract caused 100% mortality of adult *Schistosoma mansoni* at 200 microgram/mL in vitro, and fractionation yielded the flavone artemetin and the sesquiterpene lactone hydroxypelenolide (Almeida et al. 2016). The study supports a preparation-level schistosomicidal phenotype and identifies chemically tractable leads, but it does not establish the isolated compounds' potency, selectivity, pharmacokinetics, or efficacy in an infected host. The constituent controls sharpen that boundary: artemetin and hydroxypelenolide from *A. absinthium* were inactive at 100 micromolar, whereas parthenolide isolated from the related *Tanacetum parthenium* killed adult worms at 12.5-100 micromolar and altered the tegument without affecting mammalian cells in the reported assay (Almeida et al. 2016). Thus, "sesquiterpene lactone" is not a sufficient attribution; structure, species, preparation, concentration, and host counterassay must remain linked.

Formulation can change the exposure boundary independently of plant chemistry. A Cuban *A. absinthium* essential oil formulated in nanocochleates showed an intracellular-*L. amazonensis* IC50 of 21.5 +/- 2.5 microgram/mL against a macrophage IC50 of 27.7 +/- 5.6 microgram/mL and suppressed murine cutaneous infection by approximately 50% after intralesional dosing at 30 mg/kg (Tamargo et al. 2017). This is a useful formulation and host-counterassay record, not evidence that unformulated wormwood oil is a human vermifuge or that a particular terpene is responsible.

The preparation-specific wormwood evidence is also visible in a recent comparative screen. Ethanolic tinctures from 46 traditional European medicinal plants were tested against *Strongyloides papillosus* and *Haemonchus contortus* larval stages; *A. absinthium* was among only five plants classified as nematicidal (Boyko and Brygadyrenko 2025). The abstract does not provide the exact *A. absinthium* concentration, and the screen does not assign activity to a terpene or establish in-vivo efficacy. It is best used as an independent qualitative support for wormwood's vermifuge hypothesis and as a rationale for dose-resolved, chemically fingerprinted follow-up.

Wormwood volatiles also show a vector-specific, compound-resolved phenotype that is relevant to malaria control but distinct from treatment of malaria infection. Leaf oil from *A. absinthium* contained (E)-beta-farnesene, (Z)-en-yn-dicycloether, and (Z)-beta-ocimene as major components; the oil and each compound were tested against six mosquito vectors, including *Anopheles stephensi*, with species-specific larvicidal potency, while non-target aquatic organisms were less sensitive in the reported comparison (Govindarajan and Benelli 2016). The result supports vector-control and selectivity experiments, not gastrointestinal anthelmintic efficacy, systemic antimalarial action, or safe human exposure. Similarly, *A. argyi* oil and isolated eucalyptol, beta-pinene, beta-caryophyllene, and camphor showed contact, fumigant, and repellency effects against adult *Lasioderma serricorne*, with the whole-oil phenotype attributed to combined major and minor constituents (Zhang et al. 2014). This is a useful constituent-resolved arthropod comparator, but it should not be converted into a helminth or human parasite mechanism.

A 2026 poultry study adds a comparative helminth phenotype outside *A. absinthium*: ethanolic extracts of *A. herba-alba* and *A. judaica* reduced *A. galli* adult-worm motility in vitro and reduced female fecundity, fecal egg counts, and worm burden in infected chicks; camphor docking to ATP synthase and glutathione transferase was reported as a target hypothesis (Abdelqader et al. 2026). Because the preparation was crude, the docking was computational, and the species/model differs from wormwood and human infection, this record strengthens target prioritization but does not establish camphor causality or human vermifuge efficacy. It belongs in comparative helminth evidence, not as a direct extension of the *A. absinthium* vermifuge literature.

An older *A. brevifolia* sheep study provides a second comparative dose-response anchor: crude aqueous and methanol whole-plant extracts caused paralysis or mortality of live *H. contortus* in vitro, while crude aqueous extract at 3 g/kg produced a maximum reported 67.2% day-14 fecal-egg-count reduction in naturally infected sheep versus 99.2% for levamisole (Iqbal et al. 2004). This supports a graded veterinary phenotype, but the plant material, mixed infection, small study scale, and lower comparator performance do not identify an active terpene or support human use.

A separate *A. herba-alba* study found concentration- and time-dependent activity of crude methanolic flower and leaf extracts against *H. contortus* adult motility and egg hatch; the flower extract reached 98.67% egg-hatch inhibition at 1 mg/mL (Ahmed et al. 2020). This strengthens the comparative helminth rationale but does not establish that the signal is caused by camphor, a sesquiterpene lactone, or another terpene, and it cannot be transferred directly to *A. absinthium* or human vermifuge use.

Two additional wormwood records strengthen the claim that *A. absinthium* has a reproducible vermifuge hypothesis while also exposing its limits. An aqueous extract caused dose-dependent paralysis, death, tegumental damage, and significant reductions in egg counts and worm burden in *Hymenolepis nana*-infected mice, although praziquantel performed best (Beshay 2018). In naturally infected cats, seven days of 300 or 600 mg/kg extract dosing was accompanied by declining *Toxocara cati* fecal egg counts and reported clinical-chemistry values within physiological ranges, but the same extract did not inhibit egg embryonation in vitro (Yıldız et al. 2011). These are stronger than traditional-use assertions because they include infected hosts, but both remain crude-preparation studies without a terpene-resolved exposure, a standardized thujone measurement, or human efficacy evidence.

Two independent extract screens extend the wormwood life-cycle evidence beyond *H. contortus*. In a 13-plant comparison, methanolic *A. absinthium* was among the most effective preparations against *H. contortus* eggs and larvae, while UPLC-MS/MS quantified selected plant secondary metabolites; the study therefore supports preparation-level activity but not a terpene-specific mechanism (Váradyová et al. 2018). In a separate concentration-series assay against *Ascaris suum*, alcoholic *A. absinthium* extract showed strong, concentration-dependent anti-embryogenic and larval effects; sesquiterpene lactones were among the mixed metabolite classes detected, but the authors explicitly called for deeper phytochemical attribution (Băieș et al. 2022). These results broaden the vermifuge hypothesis across nematode taxa while reinforcing that co-extracted phenolics, sterols, flavonoids, and lactones cannot be disentangled without fractionation and matched chemical exposure.

Comparative species studies help identify which helminth signals are likely to be terpene-independent. *A. campestris* aqueous and ethanolic extracts completely inhibited *H. contortus* egg hatching near 2 mg/mL and the ethanolic extract produced 100% adult mortality at the highest 24-hour dose, yet the chemical profile was dominated by flavonoids (Akkari et al. 2014). Methanolic *A. indica* and *A. roxburghiana* extracts showed more modest activity against mixed gastrointestinal nematodes and were explicitly proposed for purification and in-vivo follow-up (Khan et al. 2015). In *A. cina*, the n-hexane extract produced 75% and 82.6% activity against *H. contortus* L3 at 1 and 2 mg/mL, altered *H. contortus* Hc29 expression, and produced 100% reduction in an infected gerbil assay; isoguaiacine was also active in transitional larvae (Higuera-Piedrahita et al. 2021). Hc29 expression is a response biomarker rather than proof of direct binding, and none of these comparative results identifies the active constituent in *A. absinthium*. The appropriate next step is chemotype-matched fractionation with direct parasite-protein assays and host-cell counterscreens.

The newer comparative evidence clarifies both positive and negative translation. Powdered *A. herba-alba* shoots produced marked clinical, fecal, and necropsy responses in most of six experimentally infected goats, although two goats retained some egg production or adult worms (Idris et al. 1982). Aqueous white-wormwood extracts also reduced mature *H. contortus* motility in vitro, with complete killing at one treatment, but the abstract does not provide an in-vivo endpoint or terpene attribution (Hoshiki et al. 2025). In contrast, an ethanolic *A. absinthium* extract caused 81.79% larval motility inhibition and 82.2% larvicidal activity at 200 mg/mL in vitro but no significant fecal-egg-count reduction at 600 mg/kg in free-grazing goats (Flores-Prado et al. 2025). This positive-in-vitro/negative-in-vivo pattern is a direct warning against equating a vermifuge phenotype with a field-ready therapy.

Across the sheep evidence, the direction is therefore suggestive but not uniform: an ethanolic aerial-part extract produced 90.46% fecal-egg-count reduction in one ovine study, dry-wormwood supplementation failed to separate treated lambs from controls despite strong in-vitro egg-hatch activity, and a small Awassi-sheep trial reported 76.77% reduction versus 100% for albendazole (Tariq et al. 2009; Mravčáková et al. 2020; Irehan et al. 2026). This pattern makes extraction polarity, delivery matrix, host population, parasite burden, and endpoint timing essential moderators; it does not support a pooled wormwood effect or a terpene-specific conclusion.

Wormwood oil evidence also extends into protozoan assays without resolving the vermifuge chemistry. An *A. absinthium* oil containing 27.40% camphor inhibited promastigote and axenic-amastigote forms of *L. aethiopica* and *L. donovani*, but weak hemolysis and study-specific selectivity indices were reported (Tariku et al. 2011). This is a regional, mixture-level signal: it supports testing camphor-rich and thujone-rich chemotypes side by side, while the active constituent, host-relevant exposure, and relationship to *H. contortus* or other helminths remain unknown. At the purified-compound level, helenalin, mexicanin, and dehydroleucodine inhibited *L. mexicana* promastigotes at estimated low-micromolar IC50 values (Mesquita et al. 2008). These lactone results support a chemically defined parasite assay ladder, not an inference that the compounds are co-abundant in *A. absinthium*.

Standardization and formulation add two additional boundaries. A standardized *A. annua* extract showed activity against *Leishmania infantum* promastigotes and amastigotes without reported RAW264.7 macrophage cytotoxicity in the tested range, while also changing inflammatory cytokine readouts (Morua et al. 2025). This is stronger than an uncharacterized extract screen because the preparation and host counterassay were specified, but it remains mixture-level evidence and does not identify a terpene/protein interaction. Likewise, *A. cina* aqueous-extract silver nanoparticles caused high *H. contortus* L3 mortality and oxidative-enzyme changes, but the active entity was a synthesized nanoparticle rather than native *A. cina* material (Hernández Guerrero et al. 2025). Nanoparticle surface chemistry and silver toxicity therefore prevent direct comparison with wormwood oil, tea, or isolated terpenes.

The strongest new compound-level helminth signal is not from *A. absinthium* but from *A. cina*. Chromatographic isolation identified peruvin, 1-nonacosanol, hentriacontane, and cinic acid; binary peruvin mixtures with the hydrocarbon/alcohol pair or cinic acid increased *H. contortus* L3 lethality 1.42-fold and 4.87-fold, respectively, at matched LC25 proportions (Arango-De-la Pava et al. 2024). This provides an experimentally tractable mixture-synergy model, but the interaction is pair- and concentration-specific, and the compounds are not evidence for *A. absinthium* chemistry without direct isolation and quantitation in wormwood.

Fasciolid assays extend the comparative parasite space while preserving the same boundary. *A. vulgaris* extract reduced *Fasciola gigantica* egg hatch and produced time-dependent adult-fluke mortality with tegumental and internal morphology changes (Nurlaelasari et al. 2023); an ethyl-acetate extract of *A. ludoviciana* subsp. *mexicana* likewise showed ovicidal and fasciolicidal activity against *F. hepatica* (Ezeta-Miranda et al. 2023). These records support a comparative Artemisia helminth phenotype, not a shared wormwood terpene or a defined protein mechanism. Bioguided *A. kermanensis* fractionation adds low-macrophage-toxicity antileishmanial flavonoids, demonstrating that some Artemisia parasite leads are non-terpene constituents and should not be forced into the terpene hypothesis (Soleimanifard et al. 2023).

The follow-up bio-guided *A. cina* study provides a cleaner constituent attribution. The ethyl-acetate extract had an LC90 of 3.30 mg/mL against *H. contortus* L3 larvae, whereas the isolated sesquiterpene cinic acid had LC50 and LC90 values of 0.13 and 0.40 mg/mL, respectively; the authors reported approximately 256-fold and 15.71-fold greater activity at the corresponding endpoints (Arango-De-la-Pava et al. 2024). This is still an in-vitro *A. cina* result, not evidence that cinic acid occurs in *A. absinthium*, reaches a host-relevant exposure, or engages a defined parasite protein. It does, however, provide a concrete assay template for testing wormwood fractions against purified or recombinant parasite targets.

Defined sesquiterpene-lactone evidence also reaches beyond malaria. Dehydroleucodine isolated from *A. douglassiana* aerial parts induced phosphatidylserine exposure and DNA fragmentation in *Trypanosoma cruzi* epimastigotes and trypomastigotes, and combinations with benznidazole or nifurtimox increased the reported effects (Jimenez et al. 2014). This strengthens the claim that selected *Artemisia* lactones can produce parasite-stage phenotypes as purified molecules, while leaving protein target, mammalian selectivity, pharmacokinetics, and equivalence to *A. absinthium* unresolved.

An important negative attribution control comes from a 2024 *A. absinthium* computational study. HPLC profiling found nine prominent phenolics, and apigenin was the top predicted ligand for *Leishmania* N-myristoyltransferase at -9.6 kcal/mol after docking and molecular-dynamics analysis (Boudou et al. 2024). Apigenin is not a terpene, and no direct enzyme-binding or parasite target-engagement experiment was reported. The record is therefore useful for prioritizing a *Leishmania* experiment and for preventing non-terpene docking results from being misreported as wormwood terpene mechanisms.

Species-level oil comparisons further support chemotype-specific testing. Hydrodistilled *A. absinthium* oil was dominated by myrtenyl acetate (47.10%) and beta-thujone (11.41%) and produced high mortality against *Sitophilus granarius* but limited root-knot nematode activity, whereas *A. dracunculus* oil, with methyleugenol, beta-asarone, and elemicin among major constituents, reached 100% mortality against *Meloidogyne incognita* and 94.9% against *M. chitwoodi* (Evlice et al. 2026). The contrast is consistent with species- and mixture-dependent activity, but docking and partial-least-squares prioritization do not establish which constituent is causal, nor do these assays establish livestock efficacy or human safety.

The wormwood chemistry needed to interpret these assays is itself phenology-dependent. Across four harvest stages, alcoholic extracts were measured for alpha- and beta-thujone by SPME-GC-MS and for absinthin, artemisetin, and dihydro-epi-deoxyarteannuin B by UHPLC-HR-MS (Bach et al. 2016). This analytical design is directly relevant to vermifuge studies: a plant name and extraction label do not specify the actual exposure to thujones, bitter triterpene lactones, or other co-metabolites. It supports harvest- and lot-specific chemical fingerprints, not a conclusion that higher thujone means greater anthelmintic activity or safety. Independent developmental sampling reinforces the same point: in two wormwood chemotypes, flowers contained more oil than leaves, and the trajectories of alpha/beta-thujone and trans-sabinyl acetate differed across vegetative, budding, flowering, and post-flowering stages (Nguyen et al. 2018). An ex-situ comparison of 11 Polish populations identified 41 volatile compounds and separated sabinyl-acetate, mixed, and thujone chemotypes while also finding population differences in phenolic acids, tannins, and polyphenols (Kosakowska et al. 2025). Together these studies make tissue, stage, population, and nonvolatile co-metabolites mandatory covariates in any wormwood parasite experiment.

Table 3. Focused malaria and wormwood evidence anchors. Values are retained at the preparation, parasite-stage, and host-model level; they are not pooled effect estimates.

Case	Preparation and observed signal	Evidence boundary
A. annua / *P. falciparum* Dd2	Matched whole-leaf mutant extracts: a flavonoid-deficient CHI1 mutant retained wild-type artemisinin and antiplasmodial activity; artemisinin-impaired lines showed little or no activity	Supports artemisinin dominance in this genetic system; matrix effects elsewhere remain unresolved
A. annua / *P. falciparum* NF54	LED-conditioned leaf extracts: IC50 0.51, 0.52, 1.11, and 2.23 microgram/mL for white, blue, red, and greenhouse plants	Environment-chemotype-phenotype association; no single terpene assigned as causal
A. annua / rodent malaria	Dried whole plant versus artemisinin: whole plant overcame selected *P. yoelii* resistance and slowed stable *P. chabaudi* resistance	Preclinical matrix result; does not replace quality-assured ACT or establish human efficacy

Case	Preparation and observed signal	Evidence boundary
A. annua / *P. yoelii*	Equivalent-artemisinin ethyl-acetate and petroleum-ether extracts: ED50 2.9 and 1.6 mg/kg versus 5.6 mg/kg for pure artemisinin; petroleum-ether extract increased rat artemisinin AUC0-t 1.7-fold	Defined sesquiterpene/matrix-PK hypothesis; no in-vitro enhancement, clinical evidence, or unstandardized-botanical inference
A. annua / *H. contortus*	Sodium-bicarbonate extract LC99 1.27 microgram/mL in egg-hatch and 23.8 microgram/mL in larval development; single-dose extract and artemisinin produced nonsignificant 19% and 28% day-15 EPG reductions in sheep	Important negative/limited translation control; malaria efficacy does not imply anthelmintic efficacy
A. absinthium / ovine gastrointestinal nematodes	Crude aqueous and ethanolic aerial-part extracts reduced fecal egg output; ethanolic extract reached 90.46% FECR at 2.0 g/kg on day 15	Direct veterinary vermifuge evidence; active terpene, residue, and human translation unresolved
A. absinthium / *H. contortus*	Aqueous extract ED50/ED99 were 1.40/3.76 mg/mL in vitro, but dry-wormwood dietary supplementation did not significantly reduce group-level lamb EPG	Positive in-vitro/limited in-vivo translation boundary
A. absinthium / Awassi-sheep gastrointestinal nematodes	Crude ethanolic extract produced 76.77% day-15 FECRT versus 100% for albendazole in a six-sheep treatment group	Contemporary regional veterinary result; small, crude-preparation, short-follow-up study
A. absinthium / *T. spiralis*	Thujone-free population oils reduced larval infectivity ex vivo by 72–100% and adult burden in vivo by 66%, with no effect on *M. javanica*	Chemotype- and parasite-specific oil phenotype; not a universal wormwood effect
A. absinthium / *H. nana*	Crude aqueous extract caused dose-dependent adult-worm paralysis/death and reduced egg counts and worm burden in infected mice at 400 and 800 mg/kg	Rodent anchor; praziquantel performed best and active chemistry is unresolved
A. absinthium / *T. cati*	Crude extract at 300 or 600 mg/kg for 7 days reduced fecal egg counts in naturally infected cats; egg embryonation was unaffected in vitro	Host-level signal with a negative life-cycle endpoint; no terpene attribution or human evidence
A. absinthium / *S. mansoni*	Dichloromethane extract/fractions caused 100% adult-worm mortality at 200 microgram/mL; artemetin and hydroxypelenolide were isolated	Extract/fraction signal; isolated-compound potency, selectivity, and host efficacy unresolved
A. absinthium / *L. amazonensis*	Essential oil in nanocochleates: intracellular-amastigote IC50 21.5 +/- 2.5 versus macrophage IC50 27.7 +/- 5.6 microgram/mL; approximately 50% murine suppression at 30 mg/kg intralesional	Formulation- and route-specific result; not evidence for a single terpene or human vermifuge
A. absinthium / *L. aethiopica* and *L. donovani*	Camphor-rich oil inhibited promastigote and axenic-amastigote forms, with weak hemolysis reported	Regional mixture signal; does not identify camphor as causal or establish helminth efficacy
Purified natural sesquiterpene lactones / *L. mexicana*	Helenalin, mexicanin, and dehydroleucodine showed estimated 2–4 micromolar promastigote IC50 values	Defined-compound evidence; source co-occurrence, amastigote activity, and target identity remain unresolved
A. herba-alba/*A. judaica* / *A. galli*	Crude ethanolic extracts reduced fecundity, fecal egg counts, and worm burden in chicks; camphor docking to ATP synthase/GST was reported	Comparative poultry extract phenotype and computational target hypothesis; not purified-camphor, direct-binding, or human evidence
A. brevifolia / ovine gastrointestinal nematodes	Whole-plant crude aqueous extract reached 67.2% day-14 EPG reduction at 3 g/kg, below levamisole	Comparative veterinary phenotype; no terpene attribution or human evidence
A. campestris / *H. contortus*	Aqueous and ethanolic extracts completely inhibited egg hatching near 2 mg/mL; ethanolic extract caused 100% adult mortality at the highest 24-hour dose	Strong comparative extract phenotype, but chemistry was flavonoid-dominated
A. cina / *H. contortus*	n-Hexane extract produced 75–82.6% L3 activity at 1–2 mg/mL and 100% reduction in infected gerbils; Hc29 expression changed after exposure	Fractionation/biomarker lead; expression change is not direct target engagement
A. cina / *H. contortus* L3	Isolated cinic acid: LC50 0.13 mg/mL and LC90 0.40 mg/mL versus 3.30 mg/mL extract LC90	Defined sesquiterpene bridge; host exposure, selectivity, target, and transfer to *A. absinthium* unresolved
A. douglassiana / *T. cruzi*	Purified dehydroleucodine induced phosphatidylserine exposure and DNA fragmentation in epimastigotes and trypomastigotes	Compound-level STL phenotype; protein target and host selectivity unresolved
A. sieberi / *P. berghei*	Methanolic, ether, and chloroform-derived aerial-part extracts reduced murine parasitemia in the reported study	Crude, species-specific preparation; active constituent and human translation unresolved
A. khorassanica / *P. berghei*	Flowering-stage extract showed limited murine activity; chrysanthenone and cis-thujone were reported among GC-MS constituents	Composition markers are not causal proof; effect is preparation-, dose-, and parasite-specific
A. afra / *P. berghei*	Aqueous powder suspension did not show antimalarial activity in infected mice, unlike the *A. annua* comparator	Important negative in-vivo control for artemisinin-independent infusion hypotheses
A. vulgaris / *F. gigantica*	Extract reduced day-16 egg hatching to 3.33% at 5% and reached 66.67% adult mortality at 20% after 40 minutes	Comparative in-vitro extract phenotype; active compounds and mechanism unresolved
A. absinthium/*A. dracunculus* / root-knot nematodes	Oil activity diverged by species and composition: *A. absinthium* was weak while *A. dracunculus* reached 100% *M. incognita* mortality	Chemotype- and mixture-dependent comparison; docking does not establish causal constituent
A. absinthium / *Eimeria* spp.	Single 2 g/kg ethanolic extract dose in naturally infected goats underperformed toltrazuril and amprolium; oocysts persisted to day 28	Negative/limited veterinary translation control; crude chemistry unresolved
A. absinthium / chemotype controls	Flowers had higher oil content than leaves, and 11 ex-situ populations separated into sabinyl-acetate, mixed, and thujone chemotypes	Organ-, stage-, and population-specific chemistry; no direct efficacy or safety inference

The wormwood literature therefore supports a focused experimental program. First, voucher-linked *A. absinthium* accessions should be chemically profiled for thujones, camphor, cineole, epoxyocimenes, chrysanthenol, sesquiterpene lactones, and nonvolatile co-metabolites. Second, the same preparations should be tested against matched life stages of *H. contortus* and *T. spiralis*, with adult-worm motility, egg hatch, larval development, intracellular or tissue stages where relevant, and mammalian-cell counterscreens. Third, oils and extracts should be fractionated and recombined at measured ratios to distinguish single-compound activity from matrix effects. Finally, parasite-protein interaction work should begin from the observed *A. absinthium* chemistry and parasite phenotype; docking to a convenient protein without chemical provenance is not a target discovery result.

Wormwood also sharpens the safety boundary. Thujone is a neuroactive, preparation-dependent risk marker, and santonin is an additional historically important but toxic Artemisia anthelmintic marker that requires direct quantitation. Some of the strongest nematode evidence above came from thujone-free cultivated oils, whereas other wormwood chemotypes may contain thujones. Consequently, "vermifuge" cannot be used as a synonym for "safe," and veterinary efficacy cannot be transferred to human self-treatment. The Terpedia records preserve the distinction among traditional use, in-vitro nematode phenotype, in-vivo veterinary efficacy, quantified thujone/santonin chemistry, and human therapeutic evidence.

7. Safety and translation

Derivative safety can be triaged, but not declared, by integrated computational and experimental workflows. A machine-learning platform screened 374 artemisinin derivatives across cardiotoxicity, hepatotoxicity, reproductive toxicity, mutagenicity, and tumorigenicity models; seven candidates passed the reported filters, with cardiomyocyte and zebrafish experiments providing limited validation (Kadioglu et al. 2021). This is a useful model for prioritizing compounds and measuring selectivity, but a shortlist is not a clinical safety conclusion and cannot be transferred to crude *A. annua* or *A. absinthium* preparations. The reproductive-endocrine boundary is equally important: an *A. absinthium* oil dominated by cis-chrysanthenyl acetate and sabinyl acetate altered a pseudopregnancy rat endpoint at 25 mg/kg (Demirel et al. 2025). That result is neither an anthelmintic assay nor a human risk threshold, but it makes reproductive exposure a required counterscreen for wormwood oils proposed for internal use.

The safety mechanism of alpha-thujone must be kept separate from its historical anthelmintic reputation. A primary study examined alpha-thujone as a toxic absinthe/wormwood-oil constituent, showing GABA-A receptor modulation, convulsant neurotoxicity, and metabolic detoxification pathways (Höld et al. 2000). This supports exposure- and lot-specific toxicology; it does not establish parasite selectivity, a safe therapeutic dose, or modern anthelmintic efficacy. Wormwood-oil safety signals extend beyond thujone. In a human hepatic-stellate-cell model, *A. absinthium* essential oil at 0.4 microgram/microliter for 24 hours altered lipid metabolism and activated detoxification, stress, and pro-inflammatory responses (Barreto et al. 2024). This is a preparation- and concentration-specific cellular signal, not proof of clinical hepatotoxicity or a defined human risk threshold, but it supports including hepatic counterscreens and lot-specific chemical fingerprints in vermifuge studies.

Off-target mammalian pharmacology is another translation gate. Artemisinin directly activated recombinant and native TRPC3-containing calcium channels in heterologous and PC12-cell assays (Urban and Schaefer 2020). This result does not establish clinical toxicity, but it demonstrates why parasite selectivity should be tested against host ion channels and calcium signaling rather than inferred from parasite activity alone. Host microbiome metabolism adds a second exposure variable: seventeen human intestinal fungal species converted artemisinin mainly through isomerization and deoxygenation, and transplantation of one isolate into antibiotic-treated mice yielded multiple fecal artemisinin isomers (He et al. 2025). The metabolites were assessed for breast-cancer cytotoxicity, not antiparasitic efficacy, so the study supports a pharmacokinetic/biotransformation hypothesis rather than a new therapeutic claim.

Reproductive safety is a separate translational gate even for *A. annua* material. In pregnant Wistar rats, an ethanolic leaf extract containing 1.098% artemisinin produced limited maternal toxicity signals at 100-300 mg/kg in the reported assays, but the highest dose was associated with reduced litter size and fetal nonviability or malformation (Abolaji et al. 2013). This preparation- and dose-specific result does not define human pregnancy risk or a safe dose for teas, oils, isolated artemisinin, or wormwood, but it supports an explicit reproductive-toxicology gate in any antiparasitic translation program.

Thujone is neurotoxic and must be tracked as a quantified, preparation-specific analyte. Artemisinin-derived antimalarial treatment should not be conflated with crude *A. annua* tea, essential oil, or an unidentified *Artemisia* preparation. In-vitro antioxidant, antimicrobial, insecticidal, cytotoxic, or antiparasitic results do not establish a safe human dose or therapeutic efficacy.

Composition-aware toxicology is necessary because risk markers vary among accessions and processing methods. A comparative toxicological study of *A. annua* volatiles paired chemical profiles with pharmacological, microbiological, and toxicity assays, demonstrating that "*A. annua* oil" is not interchangeable with artemisinin or with a standardized extract (Radulović et al. 2013). Regional *A. afra* oils likewise differed in camphor-, cineole-, and thujone-associated composition, making provenance relevant to safety interpretation (Oyedeji et al. 2009). A safety table should therefore include not only activity and a generic cytotoxicity result but quantified hazardous constituents, route, exposure duration, metabolic activation, and relevant host or ecological controls.

Cross-species oil toxicology supplies a further calibration point. Iranian *A. armeniaca* and *A. incana* oils differed sharply: the former contained substantial non-terpene hydrocarbons with alpha-pinene and spathulenol among its major constituents, whereas the latter was dominated by oxygenated monoterpenes including camphor, 1,8-cineole, and both thujones; both oils showed toxicity in a brine-shrimp assay (Mojarrab et al. 2013). Brine-shrimp lethality is a general screening endpoint, not a mammalian therapeutic window, but the paired chemistry demonstrates why a species label cannot predict either parasite selectivity or host risk. An independent *A. princeps* oil study illustrates the need for a host-response control in wormwood research. Dietary essential oil, with or without vitamin E, altered LDL-oxidation readouts and cholesterol-related gene expression in HepG2 cells; higher treatment concentrations produced cytotoxicity (Chung et al. 2007). The study does not test helminths and does not identify a causal volatile, but it shows that a wormwood oil can have concentration-dependent mammalian effects unrelated to vermifuge activity. Matched parasite and hepatic-cell assays should therefore be treated as a minimum selectivity panel for candidate wormwood preparations.

A 13-week rat study of wormwood extract reported no obvious treatment-related changes in survival, body weight, clinical chemistry, organ weights, or histopathology at 0.125-2% extract and estimated a study-specific NOAEL at or above 1.27 g/kg/day in males and 2.06 g/kg/day in females (Muto et al. 2003). This is useful repeated-dose toxicology, but it cannot be converted into a human safe dose or generalized across thujone-rich chemotypes, essential oils, extraction methods, or parasite-treatment routes. The appropriate safety bridge remains quantified thujone and co-constituent exposure in the exact product under study.

Reproductive risk is also supported by a distinct purified-drug experiment. Oral artemisinin at 7, 35, or 70 mg/kg/day during either early or later gestation produced dose- and timing-dependent adverse pregnancy outcomes in Wistar rats, including post-implantation loss at higher doses (Boareto et al. 2008). This does not define human risk or establish that an *A. annua* tea has the same exposure, but it makes reproductive counterscreens mandatory when comparing artemisinin-rich malaria preparations with wormwood extracts or oils.

Thujone risk should not be reduced to a single-constituent narrative. In a comparative study of thujone-containing oils, multivariate analysis of rat behavior, brine-shrimp toxicity, antimicrobial activity, and detailed oil chemistry indicated that constituents beyond alpha- and beta-thujone contributed to observed effects (Radulović et al. 2017). The result strengthens the need for mixture-aware safety assessment of wormwood oils, while leaving human dose limits and parasite selectivity unresolved. It also explains why the same thujone percentage cannot be assumed to predict the biological behavior of every oil.

Analytical quality control is a prerequisite for that bridge. A validated LC-MS method reached a 0.22 ng/mL artemisinin detection limit and distinguished measurable *Artemisia* content from likely cross-contamination in material associated with retracted clinical claims (Lee et al. 2022). The lesson applies equally to wormwood vermifuge work: voucher identity, lot-specific GC-MS or LC-MS fingerprints, quantified thujones and lactones, and chain-of-custody records should precede biological comparison. A high-sensitivity assay improves attribution; it does not turn chemical detection into efficacy or safety evidence.

Preparation matrices also change exposure. Aqueous infusions extract a different chemical subset from hydrodistilled oils; solvent extracts can concentrate nonvolatile lactones and other electrophilic metabolites; and nanoformulations can alter stability, tissue distribution, and apparent potency. These are not minor formulation details. They determine which compounds reach a parasite, which host proteins are exposed, and whether an in-vitro concentration can plausibly be achieved safely in vivo.

The artemisinin matrix also has a drug-interaction boundary. In human hepatic models, all tested *Artemisia* teas inhibited CYP2B6 and CYP3A4 activity, whereas transcriptional induction was observed with artemisinin and a high-artemisinin tea but not with artemisinin-deficient teas; the authors interpreted the broad activity inhibition as likely due to other tea phytochemicals (Kane et al. 2022). A related human-microsome and rat study found that dried leaves increased artemisinin tissue exposure, consistent with inhibition of first-pass metabolism, with sex-dependent differences in downstream cytokine effects (Desrosiers et al. 2020). These data support measuring host CYP and transporter interactions alongside parasite assays. They do not identify a safe dose, validate an herbal regimen, or prove that a volatile terpene is responsible.

The regulatory boundary is preparation-specific. The EMA monograph requires the amount of thujone in covered *A. absinthium* products to be specified and sets a daily exposure limit of 6.0 mg (EMA monograph). For malaria, WHO recommends quality-assured artemisinin-based combination therapy and explicitly does not support promotion of *Artemisia* plant material such as teas, tablets, or capsules for prevention or treatment (WHO treatment guidance). These authorities address defined products and clinical policy; they do not establish the safety or efficacy of an arbitrary species, oil, extract, or terpene mixture.

The record-level safety and translation boundaries are archived in `safety-translation.csv`. This prevents a low-toxicity result in one assay or a traditional-use claim from being converted into a generalized therapeutic recommendation.

Artemisinin safety also has a regimen-dependent neurotoxicity boundary. A review of laboratory studies described relatively selective brainstem lesions, species differences in vulnerable nuclei, and a stronger association with sustained circulating drug or metabolite exposure than with a single peak; the mechanism and human translation remain unresolved (Genovese and Newman 2008). This does not negate the clinical value of quality-assured artemisinin-based therapies, but it does rule out treating plant-material dose, purified-drug dose, and repeated exposure as interchangeable variables in a terpene/protein or whole-plant translation experiment.

Two additional exposure records sharpen this boundary. Artemisinin and artesunate produced dose-dependent DNA damage in cultured human HepG2 cells at the tested concentrations, without significant changes in CASP3 or SOD1 expression (Aquino et al. 2013). Separately, repeated ingestion of an *A. vulgaris* infusion was associated with confirmed serum thujone concentrations of 27.7 and 24.1 microgram/mL on successive days in a poisoning case (Di Lorenzo et al. 2018). These records are not universal dose thresholds or evidence against clinically indicated artemisinin therapy; they show why route, duration, species, product chemistry, and exposure biomarkers must be retained when comparing malaria compounds with wormwood preparations.

A human-lymphocyte study provides an independent in-vitro genotoxicity signal. Artemisinin and artemether increased comet-assay DNA damage and micronucleus frequency at the tested concentrations, while artemisinin also increased necrotic cell death after 48 hours (Cardoso et al. 2019). The experiment does not establish clinical risk, therapeutic exposure, or whole-plant toxicity, but it strengthens the requirement for orthogonal genotoxicity, reproductive, and pharmacokinetic testing before translating artemisinin-rich or wormwood-derived candidates.

8. Testable hypotheses and research agenda

The new production and analytical records add two explicit tests to this agenda. Nutrient-by-symbiont effects should be modeled as interactions among soil, phosphate, fungal inoculation, genotype, and cultivation system rather than as a universal artemisinin increase. Likewise, chiral oil composition should be tested as a possible moderator of parasite phenotype and host toxicity, with enantiomer-resolved standards and matched exposure rather than achiral peak percentages alone.

We propose six hypotheses: (1) WGD- and lineage-specific TPS retention predicts chemical diversity only when copy number is paired with expression and metabolomics; (2) antiparasitic chemistry associated with artemisinin-like products tracks pathway completion more closely than total TPS count; (3) essential-oil activity is often a chemotype-specific mixture property; (4) secretory-tissue expression predicts accumulated compounds better than bulk leaf expression; (5) santonin/thujone abundance and nonvolatile STL profiles predict wormwood anthelmintic phenotypes only after preparation-specific fractionation and host-selectivity testing; and (6) cytotype- and diploidization-aware sampling will explain a measurable fraction of apparent Artemisia terpene-family expansion that is otherwise confounded by homeolog retention and annotation choice. Discriminating tests combine frozen assemblies, standardized annotation, orthology-aware gene trees, trichome transcriptomics, targeted metabolomics, enzyme assays, and fractionated parasite assays.

8.1 Minimum reporting standard for a future meta-analysis

Each observation should be stored as a specimen-level record with a stable taxon identifier, voucher or accession, tissue, developmental stage, locality and collection date, preparation, extraction yield, analytical platform, retention-index or spectral confirmation, compound identity confidence, abundance units, and assay metadata. A quantitative synthesis should not pool normalized peak percentages with absolute concentrations, purified compounds with oils, or promastigote-only endpoints with intracellular or in-vivo endpoints. This design makes heterogeneity a modeled variable rather than an unexplained nuisance.

8.2 Evidence-audit status of this review

The article's auditable evidence is distributed across five linked layers. `sources.json` registers 516 sources; `evidence-matrix.csv` contains 519 source-linked records spanning chemistry, pathway biology, genomes, chemotypes, reviews, and parasite assays; `antiparasitic-evidence.csv` retains preparation, parasite-stage, dose, endpoint, and host-control fields; `parasite-protein-interact` separates 18 direct, functional, computational, and unresolved-target records; and `claim-audit.csv` maps 324 central article claims to source identifiers and permitted inference boundaries. The package also archives 695 bounded title/abstract screening decisions from a 1,387-record priority queue and 177 compound/specimen extraction records. These counts describe the current versioned evidence state, not a completed systematic-review denominator.

The audit supports an extensive, source-traceable synthesis but does not justify calling the review exhaustive. Manual screening, full-text eligibility assessment, quantitative harmonization, broad multi-species species-tree sampling, and common reannotation remain open. Claims based on the 509-record matrix are source-linked and scope-qualified; claims based only on the unscreened queue are not used as findings. The exact package version, validation output, and SHA-256 manifest are stored in the Terpedia KB snapshot.

9. Conclusions

Artemisia terpene biology is best understood as a structured interaction between evolutionary history, gene-family innovation, tissue compartmentalization, ecological context, and analytical method. A publishable comparative review must therefore preserve specimen-level provenance and separate observed chemistry, direct enzyme evidence, annotation-supported hypotheses, mixture-level phenotypes, and unestablished claims. The Terpedia-linked dataset and protocol provide the basis for expanding this draft into a fully screened, quantitatively harmonized review.

Data and code availability

All source tables, figures, and code are versioned in this package. Large comparative-genomics inputs and outputs, accessions, retrieval metadata, manifests, and checksums are archived in the linked Terpedia KB study.